GEOG0030

2024-04-23

```
library(sf)## Warning: package 'sf' was built under R version 4.3.3## Linking to GEOS
3.11.2, GDAL 3.8.2, PROJ 9.3.1; sf_use_s2() is TRUElibrary(spData)## Warning: package
'spData' was built under R version 4.3.3## To access larger datasets in this package,
install the spDataLarge
## package with: `install.packages('spDataLarge',
## repos='https://nowosad.github.io/drat/', type='source')`library(spdep)## Warning:
package 'spdep' was built under R version 4.3.3library(tmap)## Warning: package 'tmap'
was built under R version 4.3.3## Breaking News: tmap 3.x is retiring. Please test v4,
e.g. with
## remotes::install_github('r-tmap/tmap')library(RColorBrewer)
library(spgwr)## Warning: package 'spgwr' was built under R version 4.3.3## Loading
required package: sp## Warning: package 'sp' was built under R version 4.3.3## NOTE: This
package does not constitute approval of GWR
## as a method of spatial analysis; see example(gwr)library(grid)
library(gridExtra)## Warning: package 'gridExtra' was built under R version
4.3.3library(spatstat)## Warning: package 'spatstat' was built under R version 4.3.3##
Loading required package: spatstat.data## Warning: package 'spatstat.data' was built
under R version 4.3.3## Loading required package: spatstat.geom## Warning: package
'spatstat.geom' was built under R version 4.3.3## spatstat.geom 3.2-9##
## Attaching package: 'spatstat.geom'## The following object is masked from
'package:grid':
##
##     as.mask## Loading required package: spatstat.random## Warning: package
'spatstat.random' was built under R version 4.3.3## spatstat.random 3.2-3## Loading
required package: spatstat.explore## Warning: package 'spatstat.explore' was built under
R version 4.3.3## Loading required package: nlme## spatstat.explore 3.2-7## Loading
required package: spatstat.model## Warning: package 'spatstat.model' was built under R
version 4.3.3## Loading required package: rpart## spatstat.model 3.2-11## Loading
required package: spatstat.linnet## Warning: package 'spatstat.linnet' was built under R
version 4.3.3## spatstat.linnet 3.1-5##
## spatstat 3.0-8
## For an introduction to spatstat, type 'beginner'library(spdep)
library(sp)
library(spatstat)
library(factoextra)## Warning: package 'factoextra' was built under R version 4.3.3##
Loading required package: ggplot2## Warning: package 'ggplot2' was built under R version
4.3.3## Welcome! Want to learn more? See two factoextra-related books at
https://goo.gl/ve3WBalibrary(dbscan)## Warning: package 'dbscan' was built under R
version 4.3.3##
## Attaching package: 'dbscan'## The following object is masked from 'package:stats':
##
##     as.dendrogramlibrary(SpatialKDE)## Warning: package 'SpatialKDE' was built under R
version 4.3.3library(osmdata)## Warning: package 'osmdata' was built under R version
```

```
4.3.3## Data (c) OpenStreetMap contributors, ODbL 1.0.
https://www.openstreetmap.org/copyrightlibrary(tmaptools)## Warning: package 'tmaptools'
was built under R version 4.3.3library(ggplot2)
library(plyr)## Warning: package 'plyr' was built under R version 4.3.3library(dplyr)##
Warning: package 'dplyr' was built under R version 4.3.3##
## Attaching package: 'dplyr'## The following objects are masked from 'package:plyr':
##
##     arrange, count, desc, failwith, id, mutate, rename, summarise,
##     summarize## The following object is masked from 'package:nlme':
##
##     collapse## The following object is masked from 'package:gridExtra':
##
##     combine## The following objects are masked from 'package:stats':
##
##     filter, lag## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, unionlibrary(lmtest)## Warning: package 'lmtest' was
built under R version 4.3.3## Loading required package: zoo## Warning: package 'zoo' was
built under R version 4.3.3##
## Attaching package: 'zoo'## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numericlibrary(terra)## Warning: package 'terra' was built under
R version 4.3.3## terra 1.7.71##
## Attaching package: 'terra'## The following object is masked from 'package:zoo':
##
##     time<-## The following objects are masked from 'package:spatstat.geom':
##
##     area, delaunay, is.empty, rescale, rotate, shift, where.max,
##     where.min## The following object is masked from 'package:grid':
##
##     depthlibrary(tmap)
library(viridis)## Warning: package 'viridis' was built under R version 4.3.3## Loading
required package: viridisLite## Warning: package 'viridisLite' was built under R version
4.3.3library(openair)## Warning: package 'openair' was built under R version
4.3.3library(gstat)## Warning: package 'gstat' was built under R version 4.3.3##
## Attaching package: 'gstat'## The following object is masked from
'package:spatstat.explore':
##
##     idwlibrary(sfheaders)## Warning: package 'sfheaders' was built under R version
4.3.3library(rmarkdown)## Warning: package 'rmarkdown' was built under R version
4.3.3#load census data
ethnicity_data <-read.csv("cleancensus20.1.csv")
housing_data <- read.csv("Housing.csv")
age_data <- read.csv("Age.csv")
Census_all <- merge(housing_data, age_data, by.x="Subzone.of.Residence",
by.y="Subzone.of.Residence")
Census_all <- merge(ethnicity_data, Census_all, by.x="Subzone.of.Residence",
by.y="Subzone.of.Residence")
```

```
Subzone.2019 <- st_read("MasterPlan2019SubzoneBoundaryNoSeaGEOJSON.geojson")## Reading
layer `MasterPlan2019SubzoneBoundaryNoSeaGEOJSON' from data source
##   `N:\Documents\GEOG0030\MasterPlan2019SubzoneBoundaryNoSeaGEOJSON.geojson'
##   using driver `GeoJSON'
## Simple feature collection with 332 features and 2 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY, XYZ
## Bounding box:  xmin: 103.6057 ymin: 1.158699 xmax: 104.0885 ymax: 1.470775
## z_range:       zmin: 0 zmax: 0
## Geodetic CRS:  WGS 84Subzone.2019$names <- sub(".*SUBZONE_N</th> <td>(.*?)</td>.*",
"\\1", Subzone.2019$Description)
Census_all$Subzone.of.Residence <- toupper(Census_all$Subzone.of.Residence)
Subzone.Census<- merge(Subzone.2019, Census_all, by.x="names",
by.y="Subzone.of.Residence")
Subzone.Census <- st_as_sf(Subzone.Census)

# data cleaning
colnames(Subzone.Census)##  [1] "names"                            "Name"
##  [3] "Description"                     "Planning.Area"
##  [5] "Total"                           "Males"
##  [7] "Females"                         "Total.Chinese"
##  [9] "Chinese.M"                       "Chinese.F"
## [11] "Total.Malay"                     "Malay.M"
## [13] "Malay.F"                         "Total.Indian"
## [15] "Indian.M"                        "Indian.F"
## [17] "Total.Other"                     "Other.M"
## [19] "Other.F"                         "Total.x"
## [21] "Total.HDB"                       "X1....2..Room.Flats1."
## [23] "X3.Room.Flats"                   "X4.Room.Flats"
## [25] "X5.Room...Executive.Flats"       "Condominiums...Other.Apartments"
## [27] "Landed.Properties"               "Others"
## [29] "Total.y"                         "X0...4"
## [31] "X5.Sep"                          "Oct.14"
## [33] "X15...19"                        "X20...24"
## [35] "X25...29"                        "X30...34"
## [37] "X35...39"                        "X40...44"
## [39] "X45...49"                        "X50...54"
## [41] "X55...59"                        "X60...64"
## [43] "X65...69"                        "X70...74"
## [45] "X75...79"                        "X80...84"
## [47] "X85...89"                        "X90...Over"
## [49] "geometry"Subzone.Census <- Subzone.Census %>% select(-Total.y, -Total.x)
# convert all -'s to 0s
cleaning <- c("Total", "Males", "Females",
              "Total.Chinese", "Chinese.M", "Chinese.F",
              "Total.Malay", "Malay.M", "Malay.F",
              "Total.Indian", "Indian.M", "Indian.F",
              "Total.Other", "Other.M", "Other.F",
              "Total.HDB","X1....2..Room.Flats1.", "X3.Room.Flats", "X4.Room.Flats",
```

```
"X5.Room...Executive.Flats",
              "Condominiums...Other.Apartments", "Landed.Properties", "Others",
              "X0...4", "X5.Sep", "Oct.14", "X15...19", "X20...24", "X25...29",
              "X30...34", "X35...39", "X40...44", "X45...49", "X50...54", "X55...59",
              "X60...64", "X65...69", "X70...74", "X75...79", "X80...84", "X85...89",
              "X90...Over")
Subzone.Census[cleaning] <- lapply(Subzone.Census[cleaning], function(x) {
  # Replace "-" with NA
  x <- gsub("-", NA, x)
  # Convert to numeric
  x <- as.numeric(x)
  # Replace NA with 0
  x[is.na(x)] <- 0
  return(x)
})## Warning in FUN(X[[i]], ...): NAs introduced by coercion## Warning in
`[<-.data.frame`(`*tmp*`, cleaning, value = list(Total = c(13920, :
## provided 43 variables to replace 42 variables#load electoral data
Electoral <-st_read("ELD2020.kml")## Reading layer `ELD2020' from data source
`N:\Documents\GEOG0030\ELD2020.kml' using driver `KML'
## Simple feature collection with 31 features and 2 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XYZ
## Bounding box:  xmin: 103.6057 ymin: 1.158762 xmax: 104.0885 ymax: 1.470783
## z_range:       zmin: 0 zmax: 0
## Geodetic CRS:  WGS 84Electoral <- st_as_sf(Electoral)
Turnout <-read.csv("GE2020.csv")
Result <-read.csv("Election data.csv")

# remove candidates, voters, rejected columns
Result$Candidate.s. <- NULL
Result$Voters <- NULL
Result$Rejected <- NULL
Result <- Result[complete.cases(Result$Party), ] # removes rows w missing values

Turnout$constituency <- toupper(Turnout$constituency)
# Turnout$constituency <- toupper(Turnout$?..constituency)
Result$Constituency <- toupper(Result$Constituency)
# Result$Constituency <- toupper(Result$?..Constituency)
colnames(Result)## [1] "Constituency"    "Seat"            "Party"           "Votes"
## [5] "Votes.."         "Sample.counts.." "Swing"           "Margins"#
Result$Constituency <- gsub("????"", "-", Result$Constituency) # the constituencies with
dashes aren't showing up - convert the dashes and re-run code
Result$Constituency <- gsub("–", "-", Result$Constituency)
Result.Turnout <- merge(Result, Turnout, by.x="Constituency", by.y="constituency")

Result.Turnout$no_of_registered_electors <- as.numeric(gsub("[^0-9.-]", "",
Result.Turnout$no_of_registered_electors))
Result.Turnout$no_of_registered_electors <-
as.numeric(Result.Turnout$no_of_registered_electors)
```

```
Result.Turnout$electors_per_seat <-
(Result.Turnout$no_of_registered_electors/Result.Turnout$Seat)
Result.Turnout$Swing <-as.numeric(Result.Turnout$Swing)## Warning: NAs introduced by
coercionResult.Turnout$Seat <-as.numeric(Result.Turnout$Seat)
Result.Turnout$Swing[is.na(Result.Turnout$Swing)] <- 0

# party distribution
Result.Turnout$Winner <- NA #creates new column
winning_rows <- !is.na(Result.Turnout$Margins)

# Assign 'Winner'
Result.Turnout$Winner[winning_rows] <- Result.Turnout$Party[winning_rows]
Result.Turnout$Winner##  [1] " Workers' Party"        " People's Action Party" " People's
Action Party"
##  [4] " People's Action Party" " People's Action Party" " People's Action Party"
##  [7] " People's Action Party" " People's Action Party" " People's Action Party"
## [10] " Workers' Party"        " People's Action Party" " People's Action Party"
## [13] " People's Action Party" " People's Action Party" " People's Action Party"
## [16] " People's Action Party" " People's Action Party" " People's Action Party"
## [19] " People's Action Party" " People's Action Party" " People's Action Party"
## [22] " People's Action Party" " People's Action Party" " People's Action Party"
## [25] " People's Action Party" " Workers' Party"        " People's Action Party"
## [28] " People's Action Party" " People's Action Party" " People's Action Party"
## [31] " People's Action Party"# Defining margins in relation to PAP
#WP <- Electoral %>%
#  filter(Winner == " Worker's Party")

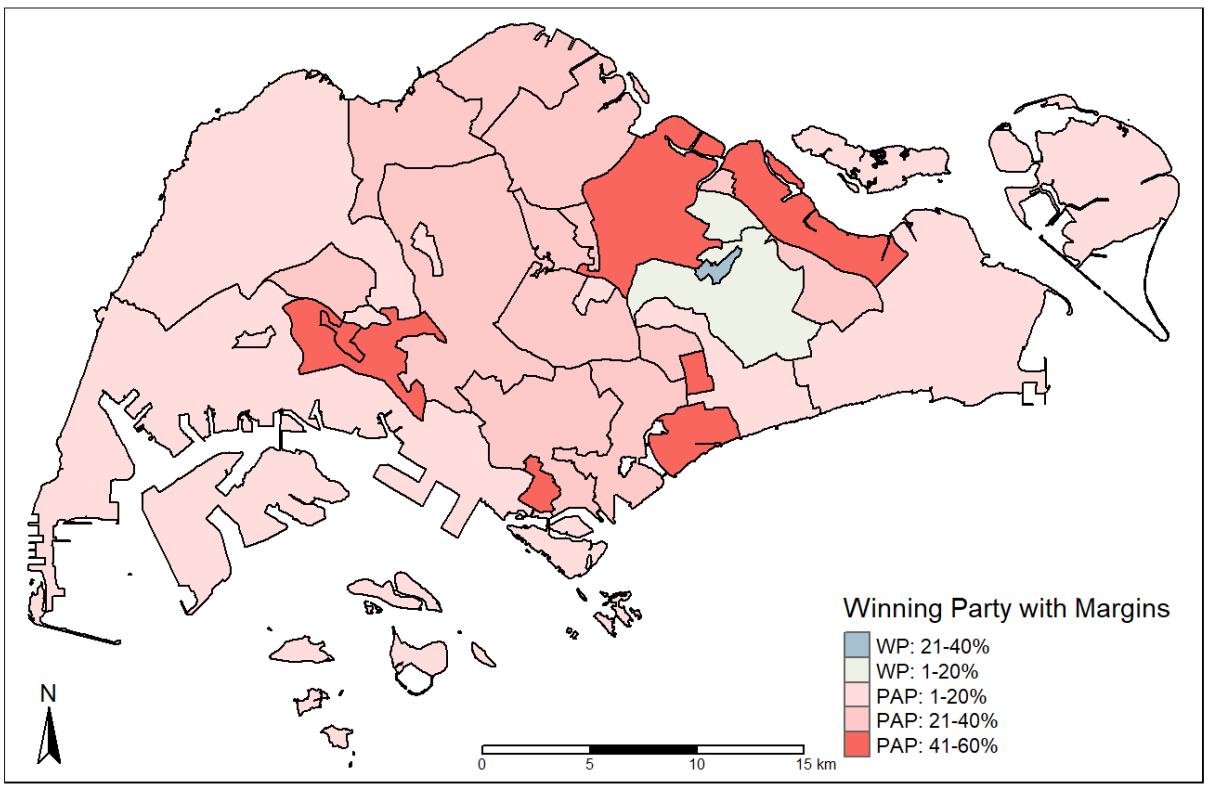
# Electoral <- Electoral %>%
#  mutate(Margins = ifelse(Party == "Workers' Party", -1 * Margins, Margins))
# commands not working - do manually
Result.Turnout[1, 8] <- -1 * Result.Turnout[1, 8]
Result.Turnout[10, 8] <- -1 * Result.Turnout[10, 8]
Result.Turnout[26, 8] <- -1 * Result.Turnout[26, 8]

Electoral <- merge(Result.Turnout, Electoral, by.x="Constituency", by.y="Name")
Electoral <- st_as_sf(Electoral)
Electoral <- st_make_valid(Electoral)

#form new object
#st_write(Electoral, dsn = "~/Electoral.2.geojson",
#         driver = "GeoJSON")

#Electoral <- st_read("Electoral.2.geojson")#mapping
pastel_RdBu <- colorRampPalette(c("#95b4cc", "#afc7d0", "#eef1e6", "#ffdddd", "#fec9c9",
"#f9665e"))
tm_shape(Electoral) +
  tm_polygons(
    col = "Margins",
```

```
    palette = pastel_RdBu(6),
    title = "Winning Party with Margins",
    labels = c("WP: 21-40%", "WP: 1-20%", "PAP: 1-20%", "PAP: 21-40%", "PAP: 41-60%")
  ) +
  #add border
  tm_shape(Electoral) +  # Separate tm_shape call for borders
  tm_borders(col = "black") +
  # add legend
  tm_layout(
    legend.position = c("right", "bottom")
  ) +
  # add scale bar to the bottom right
  tm_scale_bar(position = c("RIGHT", "BOTTOM")) +
  # add compass (north arrow) to the bottom left
  tm_compass(position = c("LEFT", "bottom"))## Variable(s) "Margins" contains positive
and negative values, so midpoint is set to 0. Set midpoint = NA to show the full spectrum
of the color palette.
```




```
# run spatial autocorrelation test
# global morans i
# identify neighbours
#ward_neighbours_queen <- Electoral |>
#  poly2nb(queen = T)
```

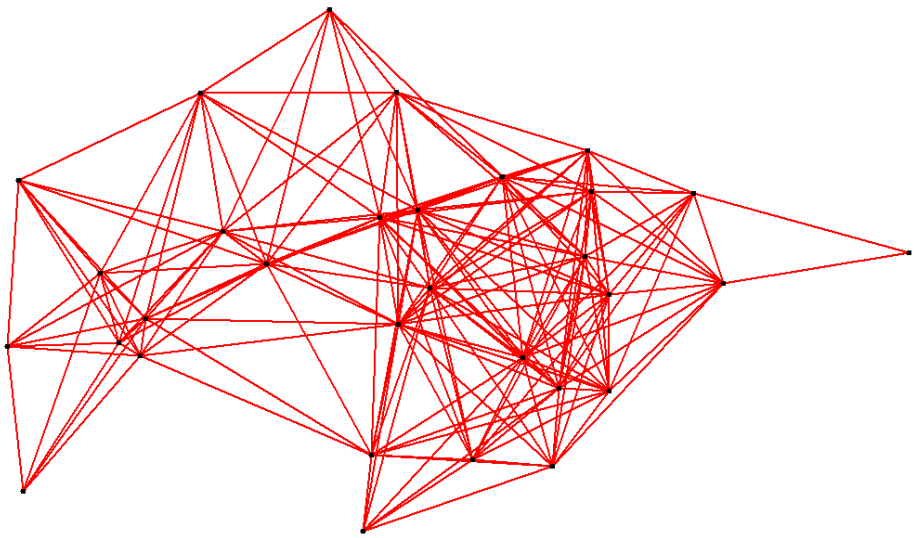
```
#ward_spatial_weights_queen <- ward_neighbours_queen |>
#  nb2listw(style = "W")


#plot(ward_neighbours_queen, st_geometry(ward_centroid), col = "red", pch = 20, cex =
0.5)

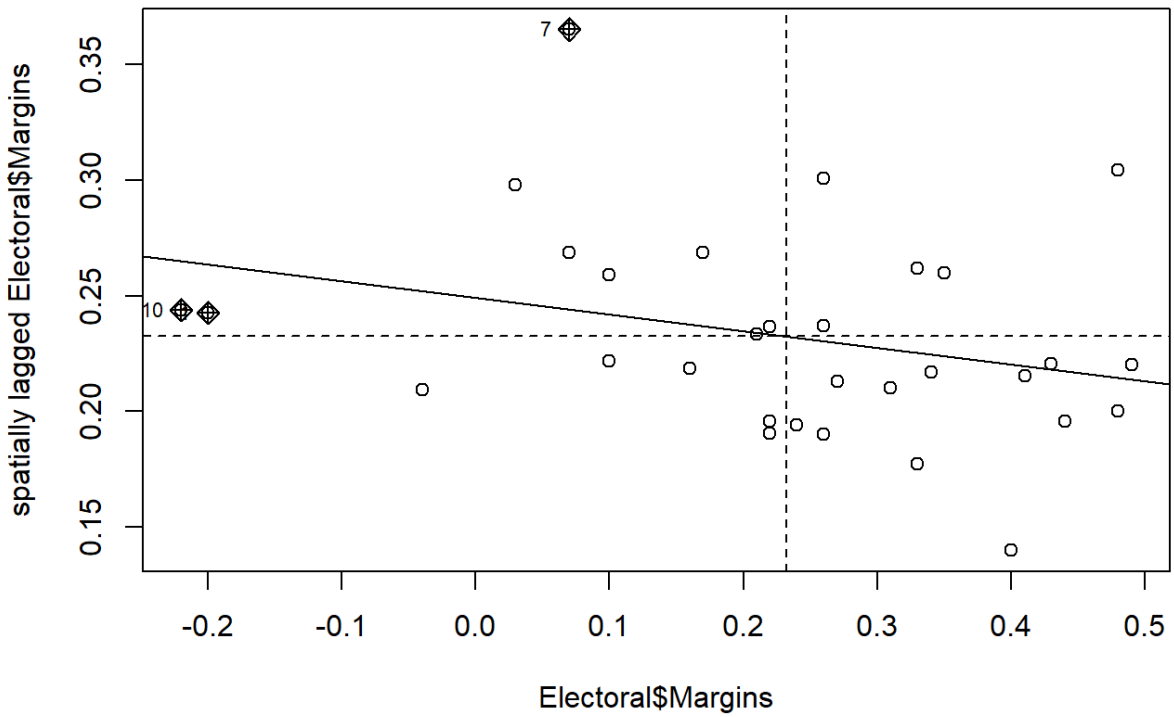
# empty neighbour sets found - manually edit

ward_neighbours_fd <- dnearneigh(st_geometry(st_centroid(Electoral)),
                                 0,10)## Warning: st_centroid assumes attributes are
constant over geometriesward_spatial_weights_fd <- ward_neighbours_fd |>
  nb2listw(style = "W", zero.policy = TRUE)

ward_centroid <- Electoral |>
  st_centroid()## Warning: st_centroid assumes attributes are constant over geometries#
plot
plot(ward_neighbours_fd, st_geometry(ward_centroid), col = "red", pch = 20, cex = 0.5)
```



```
#global moran's i
moran.plot(Electoral$Margins, listw = ward_spatial_weights_fd)
```




```
moran_fd <- Electoral |>
  pull(Margins) |>
  as.vector() |>
  moran.test(ward_spatial_weights_fd)

# inspect result
moran_fd #-0.07 = low negative (no pattern)##
##  Moran I test under randomisation
##
## data:  as.vector(pull(Electoral, Margins))
## weights: ward_spatial_weights_fd
##
## Moran I statistic standard deviate = -0.63259, p-value = 0.7365
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##      -0.071890815      -0.033333333       0.003715144#local moran's i
local_moran_fd <- Electoral |>
  pull(Margins) |>
  as.vector() |>
  localmoran(ward_spatial_weights_fd)

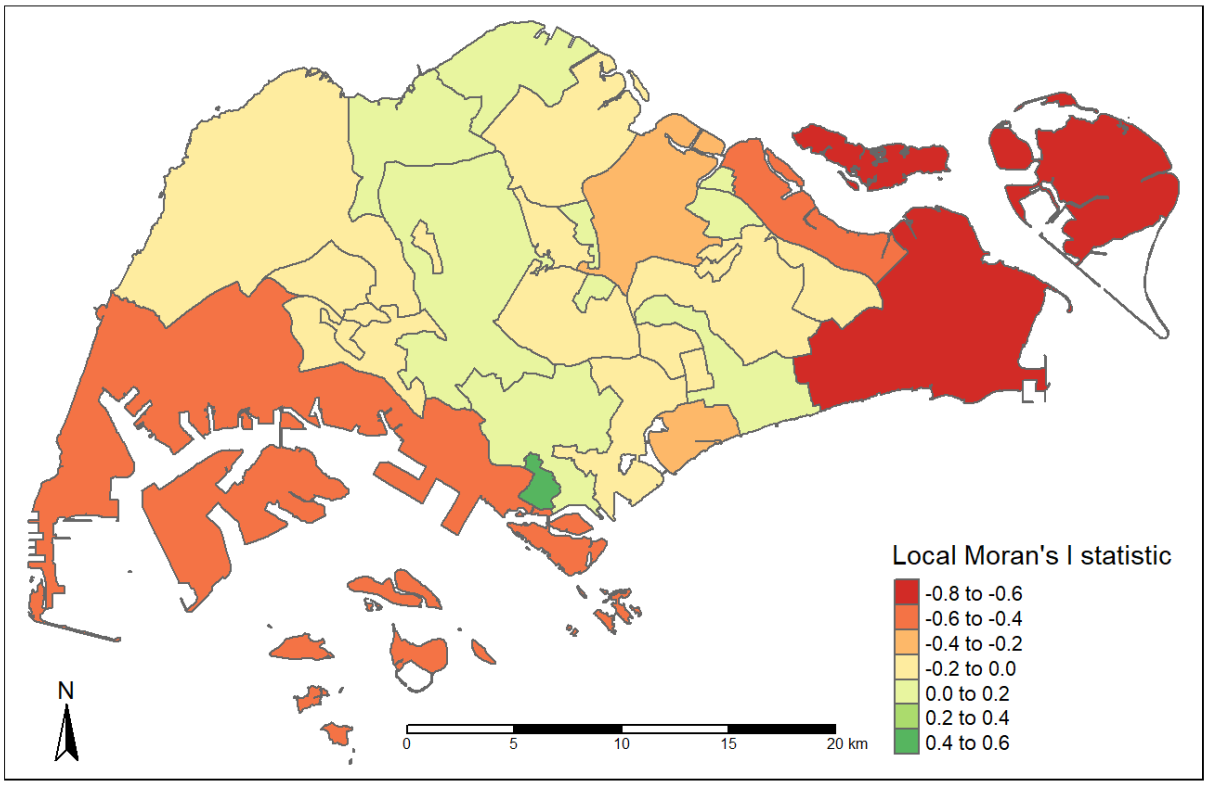
# inspect result
```

```
head(local_moran_fd)##            Ii         E.Ii      Var.Ii       Z.Ii Pr(z != E(Ii))
## 1 -0.14518832 -0.194796753 0.167668287  0.1211519      0.9035708
## 2 -0.23664387 -0.045200162 0.040366777 -0.9528599      0.3406610
## 3 -0.05055541 -0.012192989 0.006437481 -0.4781322      0.6325561
## 4 -0.11253648 -0.018174697 0.028612536 -0.5578507      0.5769464
## 5 -0.18582948 -0.027379656 0.032533215 -0.8784727      0.3796872
## 6 -0.07142066 -0.004005189 0.011726718 -0.6225458      0.5335830# coerce to dataframe
local_moran_fd <- as.data.frame(local_moran_fd)

# update the names for easier reference
names(local_moran_fd) <- c("LMI_Ii", "LMI_eIi", "LMI_varIi", "LMI_zIi", "LMI_sigP")

# join
Electoral <- Electoral |>
  mutate(local_moran_fd)

#mapping
tm_shape(Electoral) +
  tm_polygons(
    col = "LMI_Ii",
    style = "pretty",
    midpoint = 0,
    title = "Local Moran's I statistic"
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "bottom")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  )
```

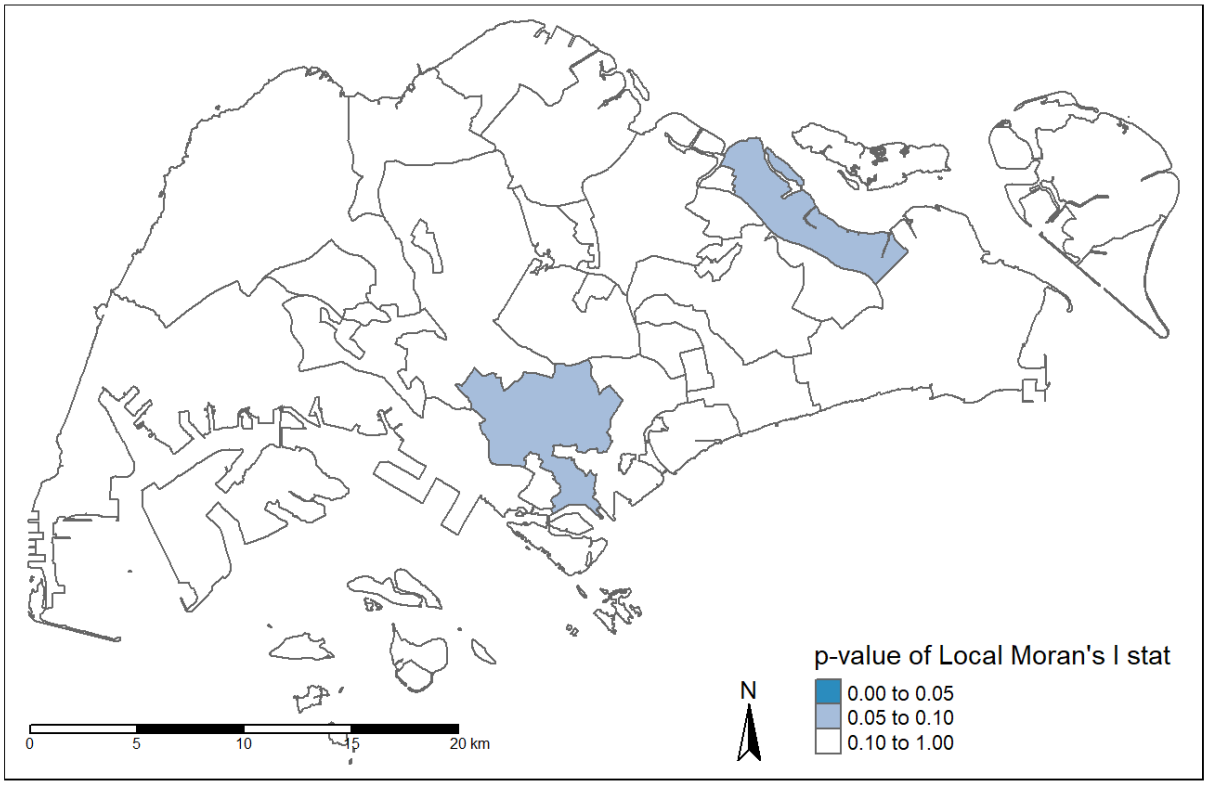



```
#statistically significant ares
# significance breaks
breaks <- c(0, 0.05, 0.1, 1)

# colour palette
colours <- c("#ffffff", "#a6bddb", "#2b8cbe")

# map the local Moran's I statistic / significance only
tm_shape(Electoral) +
  tm_polygons(
    col = "LMI_sigP",
    style = "fixed",
    breaks = breaks,
    palette = rev(colours),
    title = "p-value of Local Moran's I stat"
  ) +
  tm_compass(
    type = "arrow",
    position = c("right", "bottom")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
```

```
    position = c("left", "bottom")
  )
```




```
# lisa map
# significance breaks
Electoral <- Electoral |>
  mutate(obs_diff = (Margins - mean(Electoral$Margins)))

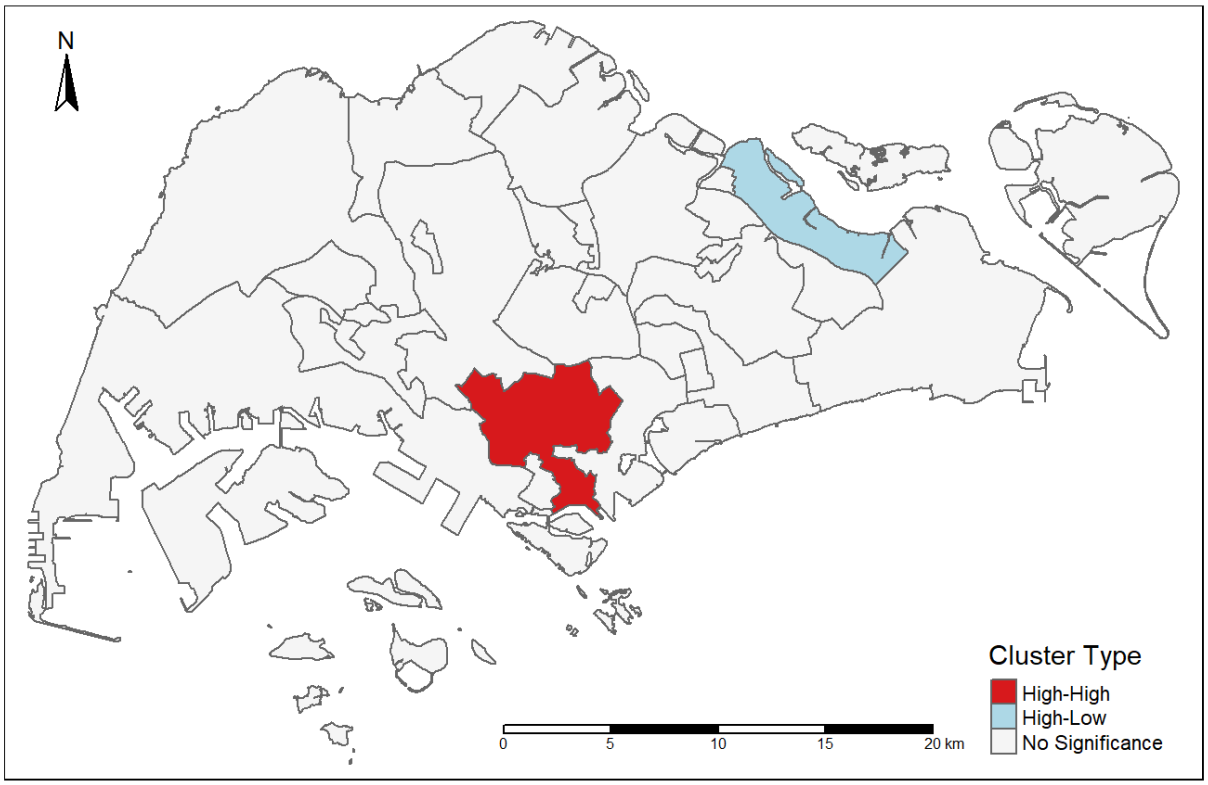
# compare local LMI value with mean LMI value
Electoral <- Electoral |>
  mutate(LMI_diff = (Electoral$LMI_Ii - mean(Electoral$LMI_Ii)))

# set significance threshold
signif <- 0.10

# generate column with cluster type, using values above
Electoral <- Electoral |>
  mutate(cluster_type = case_when(
    obs_diff > 0 & LMI_diff > 0 & LMI_sigP < signif ~ "High-High",
    obs_diff < 0 & LMI_diff < 0 & LMI_sigP < signif ~ "Low-Low",
    obs_diff < 0 & LMI_diff > 0 & LMI_sigP < signif ~ "Low-High",
    obs_diff > 0 & LMI_diff < 0 & LMI_sigP < signif ~ "High-Low",
    LMI_sigP > signif ~ "No Significance"
```

```
  ))

count(Electoral, cluster_type)## Simple feature collection with 3 features and 2 fields
## Geometry type: GEOMETRY
## Dimension:     XY
## Bounding box:  xmin: 103.6057 ymin: 1.158762 xmax: 104.0885 ymax: 1.470783
## Geodetic CRS:  WGS 84
##      cluster_type  n                       geometry
## 1       High-High  1 POLYGON ((103.8458 1.261627...
## 2        High-Low  1 MULTIPOLYGON (((103.898 1.4...
## 3 No Significance 29 MULTIPOLYGON (((103.6619 1....# create palette
pal <- c("#d7191c", "#ADD8E6", "#F5F5F5")
tm_shape(Electoral) +
  tm_polygons(
    col = "cluster_type",
    palette = pal,
    title = "Cluster Type"
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  )
```




```
# GI test
gi_distance <- Electoral |>
  pull(Margins) |>
  as.vector() |>
  localG(ward_spatial_weights_fd)

# join
Electoral <- Electoral |>
  mutate(MarginsGI = as.numeric(gi_distance))

# inspect
Electoral## Simple feature collection with 31 features and 24 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 103.6057 ymin: 1.158762 xmax: 104.0885 ymax: 1.470783
## Geodetic CRS:  WGS 84
## First 10 features:
##           Constituency Seat                  Party  Votes Votes..
## 1             ALJUNIED    5         Workers' Party  85815   59.95
## 2           ANG MO KIO    5  People's Action Party 124597   71.91
## 3     BISHAN-TOA PAYOH    4  People's Action Party  62983   67.23
## 4          BUKIT BATOK    1  People's Action Party  15500   54.80
## 5        BUKIT PANJANG    1  People's Action Party  18085   53.73
```

```
## 6        CHUA CHU KANG    4  People's Action Party  59554   58.64
## 7           EAST COAST    5  People's Action Party  61144   53.39
## 8  HOLLAND-BUKIT TIMAH    4  People's Action Party  71218   66.36
## 9       HONG KAH NORTH    1  People's Action Party  16347   60.99
## 10             HOUGANG    1         Workers' Party  15451   61.21
##    Sample.counts.. Swing Margins no_of_registered_electors no_of_rejected_votes
## 1               60  8.99   -0.20                    150821                 1582
## 2               72  6.72    0.44                    185261                 5016
## 3               67  6.36    0.34                    101220                 2049
## 4               57  6.43    0.10                     29948                  533
## 5               56 14.65    0.07                     35437                  586
## 6               59 18.25    0.17                    106632                 1410
## 7               54  7.34    0.07                    121644                 1393
## 8               68  0.24    0.33                    114973                 1999
## 9               63 13.77    0.22                     28046                  403
## 10              58  3.52   -0.22                     26432                  272
##    no_of_spoilt_ballot_papers Minority electors_per_seat                 Winner
## 1                         103        M          30164.20         Workers' Party
## 2                          90       IO          37052.20  People's Action Party
## 3                          58        M          25305.00  People's Action Party
## 4                          25                   29948.00  People's Action Party
## 5                          32                   35437.00  People's Action Party
## 6                          62        M          26658.00  People's Action Party
## 7                          98        M          24328.80  People's Action Party
## 8                         136       IO          28743.25  People's Action Party
## 9                          14                   28046.00  People's Action Party
## 10                         21                   26432.00         Workers' Party
##
Description
## 1                        <html xmlns:fo="http://www.w3.org/1999/XSL/Format"
xmlns:msxsl="urn:schemas-microsoft-com:xslt"> <head> <META http-equiv="Content-Type"
content="text/html"> <meta http-equiv="content-type" content="text/html; charset=UTF-8">
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<td>ED_DESC</td> <td>ALJUNIED</td> </tr> <tr> <td>ED_CODE</td> <td>AJ</td> </tr> </table>
</td> </tr> </table> </body> </html>
## 2                     <html xmlns:fo="http://www.w3.org/1999/XSL/Format"
xmlns:msxsl="urn:schemas-microsoft-com:xslt"> <head> <META http-equiv="Content-Type"
content="text/html"> <meta http-equiv="content-type" content="text/html; charset=UTF-8">
</head> <body style="margin:0px 0px 0px 0px;overflow:auto;background:#FFFFFF;"> <table
style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;width:100%;border-
collapse:collapse;padding:3px 3px 3px 3px"> <tr style="text-align:center;font-
weight:bold;background:#9CBCE2"> <td>ANG MO KIO</td> </tr> <tr> <td> <table style="font-
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padding:3px 3px 3px 3px"> <tr> <td>FID</td> <td>26</td> </tr> <tr bgcolor="#D4E4F3">
<td>ED_DESC</td> <td>ANG MO KIO</td> </tr> <tr> <td>ED_CODE</td> <td>AM</td> </tr>
```

</table> </td> </tr> </table> </body> </html>
3 <html xmlns:fo="http://www.w3.org/1999/XSL/Format" xmlns:msxsl="urn:schemas-microsoft-com:xslt"> <head> <META http-equiv="Content-Type" content="text/html"> <meta http-equiv="content-type" content="text/html; charset=UTF-8"> </head> <body style="margin:0px 0px 0px 0px;overflow:auto;background:#FFFFFF;"> <table style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;width:100%;border-collapse:collapse;padding:3px 3px 3px 3px"> <tr style="text-align:center;font-weight:bold;background:#9CBCE2"> <td>BISHAN-TOA PAYOH</td> </tr> <tr> <td> <table style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;width:100%;border-spacing:0px; padding:3px 3px 3px 3px"> <tr> <td>FID</td> <td>13</td> </tr> <tr bgcolor="#D4E4F3"> <td>ED_DESC</td> <td>BISHAN-TOA PAYOH</td> </tr> <tr> <td>ED_CODE</td> <td>BS</td> </tr> </table> </td> </tr> </table> </body> </html>
4 <html xmlns:fo="http://www.w3.org/1999/XSL/Format" xmlns:msxsl="urn:schemas-microsoft-com:xslt"> <head> <META http-equiv="Content-Type" content="text/html"> <meta http-equiv="content-type" content="text/html; charset=UTF-8"> </head> <body style="margin:0px 0px 0px 0px;overflow:auto;background:#FFFFFF;"> <table style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;width:100%;border-collapse:collapse;padding:3px 3px 3px 3px"> <tr style="text-align:center;font-weight:bold;background:#9CBCE2"> <td>BUKIT BATOK</td> </tr> <tr> <td> <table style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;width:100%;border-spacing:0px; padding:3px 3px 3px 3px"> <tr> <td>FID</td> <td>8</td> </tr> <tr bgcolor="#D4E4F3"> <td>ED_DESC</td> <td>BUKIT BATOK</td> </tr> <tr> <td>ED_CODE</td> <td>BK</td> </tr> </table> </td> </tr> </table> </body> </html>
5 <html xmlns:fo="http://www.w3.org/1999/XSL/Format" xmlns:msxsl="urn:schemas-microsoft-com:xslt"> <head> <META http-equiv="Content-Type" content="text/html"> <meta http-equiv="content-type" content="text/html; charset=UTF-8"> </head> <body style="margin:0px 0px 0px 0px;overflow:auto;background:#FFFFFF;"> <table style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;width:100%;border-collapse:collapse;padding:3px 3px 3px 3px"> <tr style="text-align:center;font-weight:bold;background:#9CBCE2"> <td>BUKIT PANJANG</td> </tr> <tr> <td> <table style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;width:100%;border-spacing:0px; padding:3px 3px 3px 3px"> <tr> <td>FID</td> <td>18</td> </tr> <tr bgcolor="#D4E4F3"> <td>ED_DESC</td> <td>BUKIT PANJANG</td> </tr> <tr> <td>ED_CODE</td> <td>BP</td> </tr> </table> </td> </tr> </table> </body> </html>
6 <html xmlns:fo="http://www.w3.org/1999/XSL/Format" xmlns:msxsl="urn:schemas-microsoft-com:xslt"> <head> <META http-equiv="Content-Type" content="text/html"> <meta http-equiv="content-type" content="text/html; charset=UTF-8"> </head> <body style="margin:0px 0px 0px 0px;overflow:auto;background:#FFFFFF;"> <table style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;width:100%;border-collapse:collapse;padding:3px 3px 3px 3px"> <tr style="text-align:center;font-weight:bold;background:#9CBCE2"> <td>CHUA CHU KANG</td> </tr> <tr> <td> <table style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;width:100%;border-spacing:0px; padding:3px 3px 3px 3px"> <tr> <td>FID</td> <td>28</td> </tr> <tr bgcolor="#D4E4F3"> <td>ED_DESC</td> <td>CHUA CHU KANG</td> </tr> <tr> <td>ED_CODE</td> <td>CK</td> </tr> </table> </td> </tr> </table> </body> </html>
7 <html xmlns:fo="http://www.w3.org/1999/XSL/Format" xmlns:msxsl="urn:schemas-microsoft-com:xslt"> <head> <META http-equiv="Content-Type" content="text/html"> <meta http-equiv="content-type" content="text/html; charset=UTF-8"> </head> <body style="margin:0px 0px 0px 0px;overflow:auto;background:#FFFFFF;"> <table style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;width:100%;border-

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collapse:collapse;padding:3px 3px 3px 3px"> <tr style="text-align:center;font-
weight:bold;background:#9CBCE2"> <td>EAST COAST</td> </tr> <tr> <td> <table style="font-
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</table> </td> </tr> </table> </body> </html>
## 8  <html xmlns:fo="http://www.w3.org/1999/XSL/Format" xmlns:msxsl="urn:schemas-
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collapse:collapse;padding:3px 3px 3px 3px"> <tr style="text-align:center;font-
weight:bold;background:#9CBCE2"> <td>HOLLAND-BUKIT TIMAH</td> </tr> <tr> <td> <table
style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;width:100%;border-
spacing:0px; padding:3px 3px 3px 3px"> <tr> <td>FID</td> <td>23</td> </tr> <tr
bgcolor="#D4E4F3"> <td>ED_DESC</td> <td>HOLLAND-BUKIT TIMAH</td> </tr> <tr>
<td>ED_CODE</td> <td>HT</td> </tr> </table> </td> </tr> </table> </body> </html>
## 9            <html xmlns:fo="http://www.w3.org/1999/XSL/Format"
xmlns:msxsl="urn:schemas-microsoft-com:xslt"> <head> <META http-equiv="Content-Type"
content="text/html"> <meta http-equiv="content-type" content="text/html; charset=UTF-8">
</head> <body style="margin:0px 0px 0px 0px;overflow:auto;background:#FFFFFF;"> <table
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collapse:collapse;padding:3px 3px 3px 3px"> <tr style="text-align:center;font-
weight:bold;background:#9CBCE2"> <td>HONG KAH NORTH</td> </tr> <tr> <td> <table
style="font-family:Arial,Verdana,Times;font-size:12px;text-align:left;width:100%;border-
spacing:0px; padding:3px 3px 3px 3px"> <tr> <td>FID</td> <td>15</td> </tr> <tr
bgcolor="#D4E4F3"> <td>ED_DESC</td> <td>HONG KAH NORTH</td> </tr> <tr> <td>ED_CODE</td>
<td>HN</td> </tr> </table> </td> </tr> </table> </body> </html>
## 10                        <html xmlns:fo="http://www.w3.org/1999/XSL/Format"
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content="text/html"> <meta http-equiv="content-type" content="text/html; charset=UTF-8">
</head> <body style="margin:0px 0px 0px 0px;overflow:auto;background:#FFFFFF;"> <table
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weight:bold;background:#9CBCE2"> <td>HOUGANG</td> </tr> <tr> <td> <table style="font-
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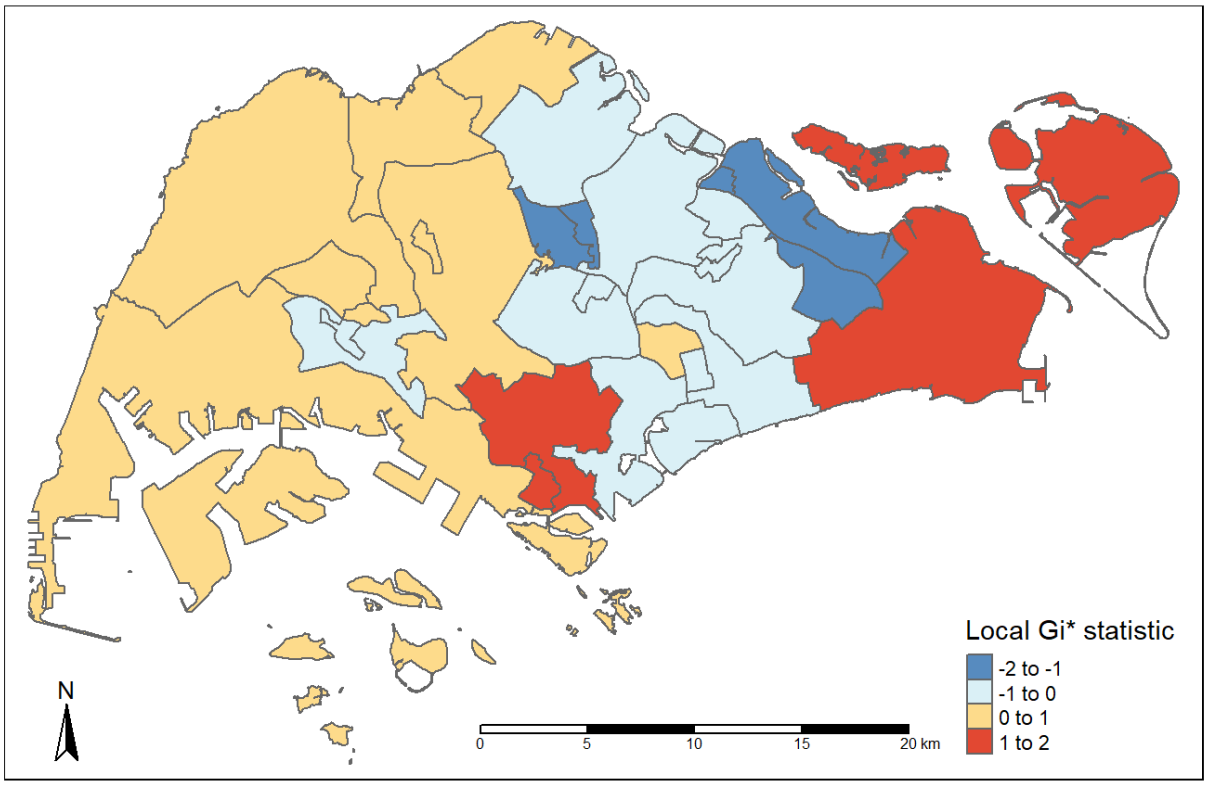
padding:3px 3px 3px 3px"> <tr> <td>FID</td> <td>14</td> </tr> <tr bgcolor="#D4E4F3">
<td>ED_DESC</td> <td>HOUGANG</td> </tr> <tr> <td>ED_CODE</td> <td>HG</td> </tr> </table>
</td> </tr> </table> </body> </html>
##                          geometry       LMI_Ii       LMI_eIi    LMI_varIi
## 1  MULTIPOLYGON (((103.9175 1.... -0.145188319 -0.1947967527 0.1676682867
## 2  MULTIPOLYGON (((103.8974 1.... -0.236643870 -0.0452001617 0.0403667769
## 3  MULTIPOLYGON (((103.8244 1.... -0.050555407 -0.0121929887 0.0064374813
## 4  MULTIPOLYGON (((103.7484 1.... -0.112536478 -0.0181746968 0.0286125359
## 5  MULTIPOLYGON (((103.7758 1.... -0.185829483 -0.0273796564 0.0325332151
## 6  MULTIPOLYGON (((103.6619 1.... -0.071420656 -0.0040051890 0.0117267180
## 7  MULTIPOLYGON (((103.9201 1.... -0.674946708 -0.0273796564 0.3985318856
## 8  MULTIPOLYGON (((103.8089 1....  0.092348811 -0.0100407863 0.0106254841
## 9  MULTIPOLYGON (((103.7484 1.... -0.001768785 -0.0001487387 0.0003709368
## 10 MULTIPOLYGON (((103.8855 1.... -0.167246710 -0.2132538249 0.1569290041
```

```
##        LMI_zIi  LMI_sigP    obs_diff      LMI_diff    cluster_type   MarginsGI
## 1   0.12115185 0.9035708 -0.43193548 -0.0732975043 No Significance -0.12115185
## 2  -0.95285992 0.3406610  0.20806452 -0.1647530549 No Significance -0.95285992
## 3  -0.47813220 0.6325561  0.10806452  0.0213354077 No Significance -0.47813220
## 4  -0.55785067 0.5769464 -0.13193548 -0.0406456637 No Significance  0.55785067
## 5  -0.87847271 0.3796872 -0.16193548 -0.1139386685 No Significance  0.87847271
## 6  -0.62254580 0.5335830 -0.06193548  0.0004701591 No Significance  0.62254580
## 7  -1.02577759 0.3049964 -0.16193548 -0.6030558935 No Significance  1.02577759
## 8   0.99330238 0.3205626  0.09806452  0.1642396261 No Significance  0.99330238
## 9  -0.08411576 0.9329644 -0.01193548  0.0701220302 No Significance  0.08411576
## 10  0.11613775 0.9075434 -0.45193548 -0.0953558948 No Significance -0.11613775#plot
local GI clusters
tm_shape(Electoral) +
  tm_polygons(
    col = "MarginsGI",
    style = "pretty",
    midpoint = 0,
    title = "Local Gi* statistic",
    palette = "-RdYlBu"
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "bottom")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  )
```




```
# thiessen polygon: assign boundaries of areas closest to each unique point.
# Compute centroids of the polygons in your census data
Subzone.Census <- st_make_valid(Subzone.Census)
centroids <- st_centroid(Subzone.Census)## Warning: st_centroid assumes attributes are
constant over geometriescentroid_coords <- st_coordinates(centroids)

# Create a new spatial dataframe with the centroids and the same attributes as your
original census data
centroid_data <- st_as_sf(
  cbind(centroids, centroid_coords),
  coords = c("X", "Y"),  # Assuming the coordinates are named X and Y
  crs = st_crs(Subzone.Census)
)
# Check if CRS is in longitude/latitude
is_longlat <- st_is_longlat(centroid_data)

st_thiessen_point <- function(points) {

  # input check
  if (!all(st_geometry_type(points) == "POINT")) {
    stop("Input not POINT geometries")
  }
```

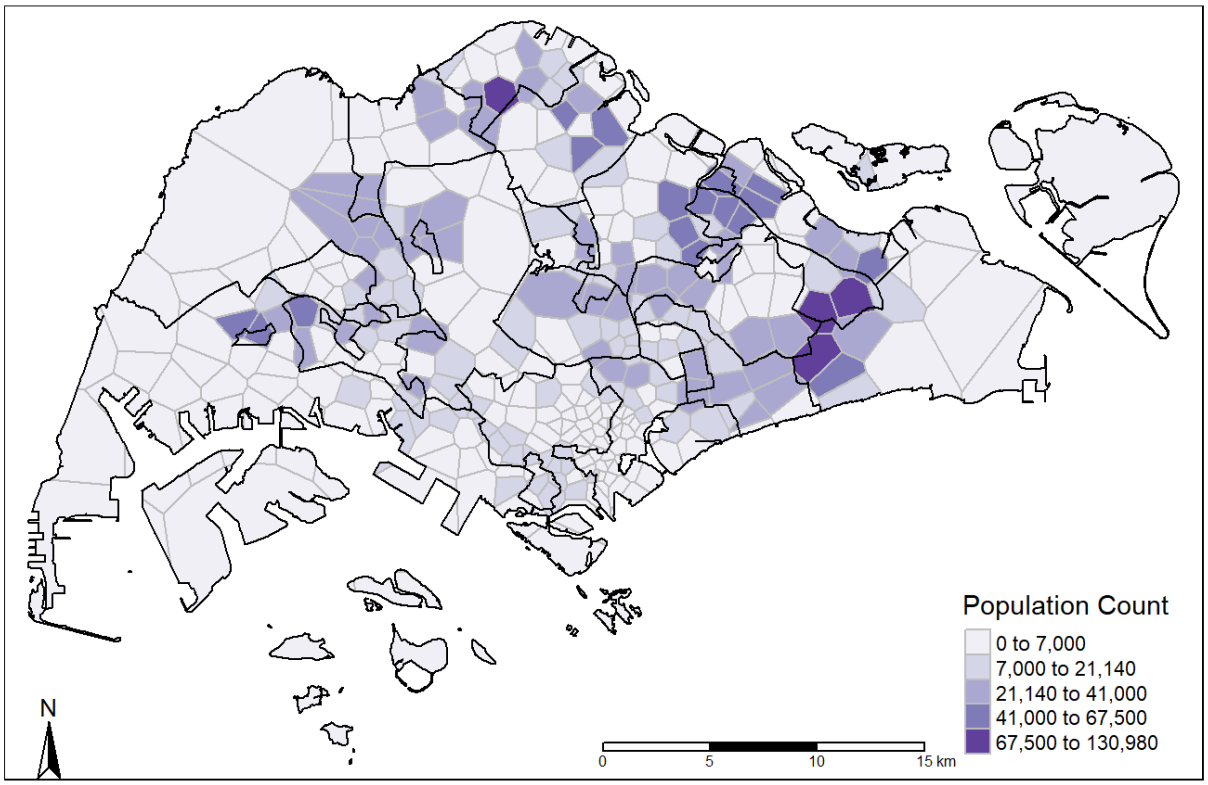
```
  # to multipoint
  g = st_combine(st_geometry(points))

  # to thiessen
  v = st_voronoi(g)
  v = st_collection_extract(v)

  # return
  return(v[unlist(st_intersects(points, v))])
}

# call function
census_thiessen <- st_thiessen_point(centroid_data)## Warning in st_voronoi.sfc(g):
st_voronoi does not correctly triangulate
## longitude/latitude data# force thiessen polygons to measurement sites dataframe
Subzone.Census <- st_make_valid(Subzone.Census)
Electoral <- st_make_valid(Electoral)

Census_thiessen_df <- Subzone.Census |>
  st_set_geometry(census_thiessen) |>
  st_intersection(Electoral)## Warning: attribute variables are assumed to be spatially
constant throughout
## all geometries#visualisation
tm_shape(Census_thiessen_df) +
  tm_borders(col = "grey") +
  tm_polygons(
    col = "Total",
    palette = "Purples",
    title = "Population Count",
    style = "jenks",
    n = 5
  ) +
  # add legend
  tm_layout(
    legend.position = c("right", "bottom")
  ) +
  # add scale bar to the bottom right
  tm_scale_bar(position = c("RIGHT", "BOTTOM")) +
  # add compass (north arrow) to the bottom left
  tm_compass(position = c("LEFT", "BOTTOM")) +
  tm_shape(Electoral) +  # Separate tm_shape call for borders
  tm_borders(col = "black")## Warning: One tm layer group has duplicated layer types,
which are omitted. To
## draw multiple layers of the same type, use multiple layer groups (i.e. specify
## tm_shape prior to each of them).
```




```
#calculate area of each polygon
Census_thiessen_df$area <- st_area(Census_thiessen_df)
colnames(Census_thiessen_df)##  [1] "names"                                "Name"
##  [3] "Description"                      "Planning.Area"
##  [5] "Total"                            "Males"
##  [7] "Females"                          "Total.Chinese"
##  [9] "Chinese.M"                        "Chinese.F"
## [11] "Total.Malay"                      "Malay.M"
## [13] "Malay.F"                          "Total.Indian"
## [15] "Indian.M"                         "Indian.F"
## [17] "Total.Other"                      "Other.M"
## [19] "Other.F"                          "Total.HDB"
## [21] "X1....2..Room.Flats1."            "X3.Room.Flats"
## [23] "X4.Room.Flats"                    "X5.Room...Executive.Flats"
## [25] "Condominiums...Other.Apartments" "Landed.Properties"
## [27] "Others"                           "X0...4"
## [29] "X5.Sep"                           "Oct.14"
## [31] "X15...19"                         "X20...24"
## [33] "X25...29"                         "X30...34"
## [35] "X35...39"                         "X40...44"
## [37] "X45...49"                         "X50...54"
## [39] "X55...59"                         "X60...64"
## [41] "X65...69"                         "X70...74"
```

```
## [43] "X75...79"                        "X80...84"
## [45] "X85...89"                        "X90...Over"
## [47] "Constituency"                    "Seat"
## [49] "Party"                           "Votes"
## [51] "Votes.."                         "Sample.counts.."
## [53] "Swing"                           "Margins"
## [55] "no_of_registered_electors"       "no_of_rejected_votes"
## [57] "no_of_spoilt_ballot_papers"      "Minority"
## [59] "electors_per_seat"               "Winner"
## [61] "Description.1"                   "LMI_Ii"
## [63] "LMI_eIi"                         "LMI_varIi"
## [65] "LMI_zIi"                         "LMI_sigP"
## [67] "obs_diff"                        "LMI_diff"
## [69] "cluster_type"                    "MarginsGI"
## [71] "geometry"                        "area"variables <- c("Total", "Males",
"Females", "Total.Chinese", "Chinese.M", "Chinese.F", "Total.Malay", "Malay.M",
"Malay.F", "Total.Indian", "Indian.M", "Indian.F", "Total.Other", "Other.M", "Other.F",
              "electors_per_seat")
for (col in variables) {
  Census_thiessen_df[[col]] <- as.numeric(unclass(Census_thiessen_df[[col]]))
}#proportion of area respecitve to its corresponding subzone
area_sub <- aggregate(Census_thiessen_df["area"], by = list(Census_thiessen_df$names),
FUN = sum)  %>%
  rename(sub = Group.1, area_sub = area) %>%
  st_drop_geometry(area_sub)
area_elec <- aggregate(Census_thiessen_df["area"], by =
list(Census_thiessen_df$Constituency), FUN = sum)  %>%
  rename(sub = Group.1, area_elec = area) %>%
  st_drop_geometry(area_elec)

Area <- merge(Census_thiessen_df, area_sub, by.x = "names", by.y = "sub", all.x = TRUE)
Area <- merge(Area, area_elec, by.x = "Constituency", by.y = "sub", all.x = TRUE)

variables.c <- c("Total", "Males", "Females", "Total.Chinese", "Chinese.M", "Chinese.F",
"Total.Malay", "Malay.M", "Malay.F", "Total.Indian", "Indian.M", "Indian.F",
"Total.Other", "Other.M", "Other.F",
                "Total.HDB", "X1....2..Room.Flats1.", "X3.Room.Flats", "X4.Room.Flats",
"X5.Room...Executive.Flats", "Condominiums...Other.Apartments", "Landed.Properties",
"Others",
                "X0...4",                          "X5.Sep",
"Oct.14",
                "X15...19",                        "X20...24",
"X25...29",
                "X30...34",                        "X35...39",
"X40...44",
                "X45...49",                        "X50...54",
"X55...59",
                "X60...64",                        "X65...69",
"X70...74",
```

```
                "X75...79",                        "X80...84",
"X85...89",
                "X90...Over")
for (col in variables.c) {
  Area[[col]] <- Area[[col]] * (Area$area)/(Area$area_sub)
}

variables <- c("Total", "Males", "Females", "Total.Chinese", "Chinese.M", "Chinese.F",
"Total.Malay", "Malay.M", "Malay.F", "Total.Indian", "Indian.M", "Indian.F",
"Total.Other", "Other.M", "Other.F",
               "Votes", "Votes..", "Sample.counts..", "Swing", "Margins",
"no_of_registered_electors", "no_of_rejected_votes", "no_of_spoilt_ballot_papers",
"electors_per_seat")
for (col in variables) {
  Area[[col]] <- as.numeric(unclass(Area[[col]]))
}

#form new object
# st_write(Area, dsn = "~/Area.2.geojson",
#         driver = "GeoJSON")

# Area <- st_read("Area.2.geojson")

# analysis using polygons
# geography of race and electoral data
# population per proportion of area times electoral area/area
Area$Chinese.per <- Area$Total.Chinese / Area$Total *100
Area$Chinese.per <- gsub("NaN", 0, Area$Chinese.per)
Area$Chinese.per <- as.numeric(Area$Chinese.per)
Area$Malay.per <- Area$Total.Malay / Area$Total * 100
Area$Malay.per <- gsub("NaN", 0, Area$Malay.per)
Area$Malay.per <- as.numeric(Area$Malay.per)
Area$Indian.per <- Area$Total.Indian / Area$Total *100
Area$Indian.per <- gsub("NaN", 0, Area$Indian.per)
Area$Indian.per <- as.numeric(Area$Indian.per)
Area$Other.per <- Area$Total.Other / Area$Total * 100
Area$Other.per <- gsub("NaN", 0, Area$Other.per)
Area$Other.per <- as.numeric(Area$Other.per)# spatial autocorrelation of race using
pseudo map
# identify neighbours
# queen <- Area |>
#  poly2nb(queen = T)

# spatialweights_queen <- queen |>
#  nb2listw(style = "W")

# empty neighbour sets found = use distance based
```

```
nb2 <- dnearneigh(st_centroid(Area), 0, 8)## Warning: st_centroid assumes attributes are
constant over geometrieslist2 <- nb2listw(nb2, style = "S")
list2## Characteristics of weights list object:
## Neighbour list object:
## Number of regions: 626
## Number of nonzero links: 120302
## Percentage nonzero weights: 30.699
## Average number of links: 192.1757
##
## Weights style: S
## Weights constants summary:
##     n     nn  S0       S1       S2
## S 626 391876 626 6.733699 2657.344moran.test(Area$Chinese.per, list2) #8.55e-02 v
low##
##  Moran I test under randomisation
##
## data:  Area$Chinese.per
## weights: list2
##
## Moran I statistic standard deviate = 25.748, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##      8.556471e-02     -1.600000e-03      1.145987e-05moran.test(Area$Malay.per, list2)
#-8.99e-2 vvv low##
##  Moran I test under randomisation
##
## data:  Area$Malay.per
## weights: list2
##
## Moran I statistic standard deviate = 27.075, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##      8.997887e-02     -1.600000e-03      1.144065e-05moran.test(Area$Indian.per,
list2) #-2.62e-2##
##  Moran I test under randomisation
##
## data:  Area$Indian.per
## weights: list2
##
## Moran I statistic standard deviate = 11.249, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##      3.621640e-02     -1.600000e-03      1.130205e-05moran.test(Area$Other.per, list2)
#1.22e-1##
##  Moran I test under randomisation
##
## data:  Area$Other.per
```

```
## weights: list2
##
## Moran I statistic standard deviate = 36.979, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
##      1.222977e-01     -1.600000e-03      1.122566e-05poly_centroid <-
st_centroid(Area)## Warning: st_centroid assumes attributes are constant over
geometries#clusters of each race
#chinese
# GI test
gi_distance_C <- Area |>
  pull(Chinese.per) |>
  as.vector() |>
  localG(list2)

# join
Area.c.p <- Area |>
  mutate(GI_C = as.numeric(gi_distance_C))

# inspect
Area.c.p## Simple feature collection with 626 features and 78 fields
## Geometry type: GEOMETRY
## Dimension:     XY
## Bounding box:  xmin: 103.6057 ymin: 1.158762 xmax: 104.0885 ymax: 1.470783
## Geodetic CRS:  WGS 84
## First 10 features:
##    Constituency                names    Name
## 1      ALJUNIED     PAYA LEBAR NORTH kml_222
## 2      ALJUNIED          KAMPONG UBI  kml_99
## 3      ALJUNIED         HOUGANG EAST kml_223
## 4      ALJUNIED      BEDOK RESERVOIR kml_215
## 5      ALJUNIED     UPPER PAYA LEBAR kml_201
## 6      ALJUNIED DEFU INDUSTRIAL PARK kml_158
## 7      ALJUNIED        LORONG AH SOO kml_186
## 8      ALJUNIED            KEMBANGAN kml_149
## 9      ALJUNIED         LORONG HALUS kml_224
## 10     ALJUNIED                 PLAB kml_325
##
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##    Planning.Area        Total      Males    Females Total.Chinese  Chinese.M
## 1     Paya Lebar     9.378953     0.0000     0.0000         0.000     0.0000
## 2        Geylang  1673.990373   835.5766   838.4138      1150.514   571.7103
## 3        Hougang 12369.812551  6011.1730  6358.6396     10126.171  4934.0264
## 4          Bedok 25419.294287 12455.1548 12964.1395     19111.876  9321.4059
## 5      Serangoon  1975.444245   952.6965  1022.7477      1730.265   832.4419
## 6        Hougang    20.000000     0.0000     0.0000         0.000     0.0000
## 7        Hougang 31785.192085 15017.3231 16767.8690     26612.231 12489.8496
## 8          Bedok  3685.201349  1816.0389  1870.1462      2626.665  1292.6734
## 9        Hougang     0.000000     0.0000     0.0000         0.000     0.0000
## 10    Paya Lebar     0.000000     0.0000     0.0000         0.000     0.0000
##     Chinese.F Total.Malay    Malay.M    Malay.F Total.Indian   Indian.M
```

```
## 1      0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
## 2    578.8035   331.96080  167.39904  163.14313     146.1195   76.60634
## 3   5192.1444   997.72565  481.48949  516.23616    1002.6895  501.34473
## 4   9790.4702  3263.49008 1606.79481 1656.69527    2215.5804 1137.73049
## 5    897.8231    59.54353   29.18801   30.35553     145.9400   72.38625
## 6      0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
## 7  14112.5466  2252.10674 1121.13611 1130.97063    2320.9484 1150.63969
## 8   1333.9917   577.47282  282.34191  296.11468     349.2382  179.04609
## 9      0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
## 10     0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
##      Indian.F Total.Other   Other.M   Other.F     Total.HDB
## 1     0.00000     0.00000   0.00000   0.00000     0.000 [1]
## 2    68.09452    46.81499  18.44227  26.95408  1655.548 [1]
## 3   501.34473   243.22665  94.31237 148.91428 11382.015 [1]
## 4  1077.84993   828.34763 389.22359 439.12405 17405.280 [1]
## 5    73.55378    39.69569  18.68032  19.84784  1058.941 [1]
## 6     0.00000     0.00000   0.00000   0.00000     0.000 [1]
## 7  1170.30874   609.74069 255.69771 354.04298 23681.542 [1]
## 8   170.19216   131.82514  61.97749  69.84765  2312.843 [1]
## 9     0.00000     0.00000   0.00000   0.00000     0.000 [1]
## 10    0.00000     0.00000   0.00000   0.00000     0.000 [1]
##    X1....2..Room.Flats1. X3.Room.Flats X4.Room.Flats X5.Room...Executive.Flats
## 1            0.00000 [1]    0.0000 [1]    0.0000 [1]                0.0000 [1]
## 2            0.00000 [1]  136.1890 [1] 1239.8878 [1]              280.8899 [1]
## 3            0.00000 [1]  789.2457 [1] 6651.5043 [1]             3941.2645 [1]
## 4            0.00000 [1] 2834.3461 [1] 8303.4365 [1]             6257.5177 [1]
## 5           38.52817 [1]  171.6255 [1]  555.7396 [1]              294.2151 [1]
## 6            0.00000 [1]    0.0000 [1]    0.0000 [1]                0.0000 [1]
## 7            0.00000 [1] 7572.5860 [1] 9441.1462 [1]             6667.8095 [1]
## 8          312.83877 [1]  708.3142 [1]  618.7912 [1]              672.8985 [1]
## 9            0.00000 [1]    0.0000 [1]    0.0000 [1]                0.0000 [1]
## 10           0.00000 [1]    0.0000 [1]    0.0000 [1]                0.0000 [1]
##    Condominiums...Other.Apartments Landed.Properties         Others
## 1                       0.0000 [1]       0.00000 [1]   0.000000 [1]
## 2                       0.0000 [1]       0.00000 [1]  18.442267 [1]
## 3                     982.8342 [1]       0.00000 [1]   9.927618 [1]
## 4                    7914.2129 [1]      49.90046 [1]  39.920368 [1]
## 5                     573.2524 [1]     331.57575 [1]  11.675202 [1]
## 6                       0.0000 [1]       0.00000 [1]   0.000000 [1]
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## 8                     495.8199 [1]     793.90216 [1]  83.620426 [1]
## 9                       0.0000 [1]       0.00000 [1]   0.000000 [1]
## 10                      0.0000 [1]       0.00000 [1]   0.000000 [1]
##            X0...4         X5.Sep         Oct.14       X15...19      X20...24
## 1     0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
## 2    56.74544 [1]   65.25725 [1]   73.76907 [1]   79.44361 [1]  106.3977 [1]
## 3   675.07805 [1]  610.54853 [1]  516.23616 [1]  541.05520 [1]  729.6800 [1]
## 4  1097.81012 [1] 1267.47168 [1] 1397.21288 [1] 1506.99389 [1] 1487.0337 [1]
## 5    68.88369 [1]   91.06658 [1]   86.39650 [1]   91.06658 [1]  105.0768 [1]
## 6     0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
```

```
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## 8   138.71153 [1]  133.79268 [1]  145.59792 [1]  175.11101 [1]  198.7215 [1]
## 9     0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
## 10    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
##         X25...29      X30...34      X35...39      X40...44      X45...49
## 1     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]
## 2   147.5381 [1]  143.2822 [1]  113.4909 [1]  113.4909 [1]  106.3977 [1]
## 3  1047.3637 [1] 1136.7123 [1] 1072.1828 [1]  848.8114 [1]  814.0647 [1]
## 4  1596.8147 [1] 1826.3568 [1] 1896.2175 [1] 2085.8392 [1] 2105.7994 [1]
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## 6     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]
## 7  2232.4377 [1] 2143.9269 [1] 1976.7400 [1] 2143.9269 [1] 2448.7973 [1]
## 8   257.7477 [1]  274.4718 [1]  247.9100 [1]  225.2833 [1]  247.9100 [1]
## 9     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]
## 10    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]
##         X50...54      X55...59      X60...64      X65...69       X70...74
## 1     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.00000 [1]
## 2   106.3977 [1]  143.2822 [1]  156.0500 [1]  119.1654 [1]   73.76907 [1]
## 3   799.1733 [1] 1027.5085 [1]  928.2323 [1]  680.0419 [1]  456.67045 [1]
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## 7  2370.1211 [1] 2537.3080 [1] 2655.3224 [1] 2261.9413 [1] 1780.04943 [1]
## 8   257.7477 [1]  289.2283 [1]  318.7414 [1]  270.5367 [1]  210.52672 [1]
## 9     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.00000 [1]
## 10    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.00000 [1]
##         X75...79      X80...84       X85...89     X90...Over Seat
## 1    0.00000 [1]   0.00000 [1]   0.000000 [1]   0.000000 [1]    5
## 2   36.88453 [1]  18.44227 [1]   8.511815 [1]   5.674544 [1]    5
## 3  238.26284 [1] 138.98666 [1]  74.457138 [1]  39.710474 [1]    5
## 4  469.06432 [1] 309.38285 [1] 159.681472 [1]  79.840736 [1]    5
## 5   51.37089 [1]  36.19313 [1]  19.847844 [1]  10.507682 [1]    5
## 6    0.00000 [1]   0.00000 [1]   0.000000 [1]   0.000000 [1]    5
## 7  963.78367 [1] 737.58954 [1] 403.215617 [1] 177.021491 [1]    5
## 8  122.97121 [1]  84.60420 [1]  52.139795 [1]  34.431940 [1]    5
## 9    0.00000 [1]   0.00000 [1]   0.000000 [1]   0.000000 [1]    5
## 10   0.00000 [1]   0.00000 [1]   0.000000 [1]   0.000000 [1]    5
##              Party Votes Votes.. Sample.counts.. Swing Margins
## 1   Workers' Party 85815   59.95              60  8.99    -0.2
## 2   Workers' Party 85815   59.95              60  8.99    -0.2
## 3   Workers' Party 85815   59.95              60  8.99    -0.2
## 4   Workers' Party 85815   59.95              60  8.99    -0.2
## 5   Workers' Party 85815   59.95              60  8.99    -0.2
## 6   Workers' Party 85815   59.95              60  8.99    -0.2
## 7   Workers' Party 85815   59.95              60  8.99    -0.2
## 8   Workers' Party 85815   59.95              60  8.99    -0.2
## 9   Workers' Party 85815   59.95              60  8.99    -0.2
## 10  Workers' Party 85815   59.95              60  8.99    -0.2
##    no_of_registered_electors no_of_rejected_votes no_of_spoilt_ballot_papers
## 1                     150821                 1582                        103
```

```
## 2                     150821                 1582                          103
## 3                     150821                 1582                          103
## 4                     150821                 1582                          103
## 5                     150821                 1582                          103
## 6                     150821                 1582                          103
## 7                     150821                 1582                          103
## 8                     150821                 1582                          103
## 9                     150821                 1582                          103
## 10                    150821                 1582                          103
##    Minority electors_per_seat          Winner
## 1         M           30164.2  Workers' Party
## 2         M           30164.2  Workers' Party
## 3         M           30164.2  Workers' Party
## 4         M           30164.2  Workers' Party
## 5         M           30164.2  Workers' Party
## 6         M           30164.2  Workers' Party
## 7         M           30164.2  Workers' Party
## 8         M           30164.2  Workers' Party
## 9         M           30164.2  Workers' Party
## 10        M           30164.2  Workers' Party
##
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</td> </tr> </table> </body> </html>
##        LMI_Ii    LMI_eIi LMI_varIi   LMI_zIi  LMI_sigP   obs_diff   LMI_diff
## 1  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 2  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 3  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 4  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 5  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 6  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 7  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 8  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 9  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 10 -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
##       cluster_type  MarginsGI            area        area_sub      area_elec
## 1  No Significance -0.1211519  876035.5 [m^2] 2802131.9 [m^2] 29321692 [m^2]
## 2  No Significance -0.1211519  340984.5 [m^2] 2403608.6 [m^2] 29321692 [m^2]
## 3  No Significance -0.1211519  633123.4 [m^2] 1275478.9 [m^2] 29321692 [m^2]
## 4  No Significance -0.1211519 2787574.5 [m^2] 2793135.0 [m^2] 29321692 [m^2]
## 5  No Significance -0.1211519  128680.9 [m^2] 1102172.9 [m^2] 29321692 [m^2]
## 6  No Significance -0.1211519  958617.3 [m^2]  958617.3 [m^2] 29321692 [m^2]
```

```
## 7  No Significance -0.1211519 1561918.5 [m^2] 1588198.8 [m^2] 29321692 [m^2]
## 8  No Significance -0.1211519  269395.8 [m^2] 2738403.2 [m^2] 29321692 [m^2]
## 9  No Significance -0.1211519 1053946.3 [m^2] 1164289.4 [m^2] 29321692 [m^2]
## 10 No Significance -0.1211519 2825932.6 [m^2] 2825932.6 [m^2] 29321692 [m^2]
##                          geometry Chinese.per Malay.per Indian.per Other.per
## 1  POLYGON ((103.9193 1.379211...     0.00000  0.000000   0.000000  0.000000
## 2  MULTIPOLYGON (((103.9058 1....    68.72881 19.830508   8.728814  2.796610
## 3  POLYGON ((103.9042 1.376471...    81.86196  8.065811   8.105939  1.966292
## 4  POLYGON ((103.9343 1.338343...    75.18649 12.838634   8.716137  3.258736
## 5  POLYGON ((103.8826 1.351, 1...    87.58865  3.014184   7.387707  2.009456
## 6  POLYGON ((103.8954 1.343897...     0.00000  0.000000   0.000000  0.000000
## 7  POLYGON ((103.8845 1.350097...    83.72525  7.085396   7.301980  1.918317
## 8  POLYGON ((103.9253 1.32934,...    71.27603 15.670048   9.476775  3.577149
## 9  POLYGON ((103.9011 1.370104...     0.00000  0.000000   0.000000  0.000000
## 10 POLYGON ((103.9077 1.346787...     0.00000  0.000000   0.000000  0.000000
##        GI_C
## 1  3.346438
## 2  4.060033
## 3  3.380584
## 4  2.956619
## 5  4.052181
## 6  4.539248
## 7  4.169717
## 8  3.349416
## 9  3.709835
## 10 4.206839#plot local GI clusters
C.con <- Electoral |>
  filter(Minority == "")

non.C.con <- Electoral |>
  filter(!Minority == "")

C.spot <- tm_shape(Area.c.p) +
  tm_polygons(
    col = "GI_C",
    style = "pretty",
    midpoint = 0,
    title = "Local Gi* Chinese",
    palette = "-RdYlBu",
    border.col = "grey", border.alpha = 0.2,
  )+
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  ) +
```

```
  tm_shape(C.con) +
  tm_borders(col = "black", lwd = 1.5) +
  tm_shape(non.C.con) +
  tm_fill(
    palette = "grey",
    alpha = 0.5
  )

#malay
# GI test
gi_distance_M <- Area |>
  pull(Malay.per) |>
  as.vector() |>
  localG(list2)

# join
Area.M.p <- Area |>
  mutate(GI_M = as.numeric(gi_distance_M))

# inspect
Area.M.p## Simple feature collection with 626 features and 78 fields
## Geometry type: GEOMETRY
## Dimension:     XY
## Bounding box:  xmin: 103.6057 ymin: 1.158762 xmax: 104.0885 ymax: 1.470783
## Geodetic CRS:  WGS 84
## First 10 features:
##    Constituency                names    Name
## 1      ALJUNIED     PAYA LEBAR NORTH kml_222
## 2      ALJUNIED          KAMPONG UBI  kml_99
## 3      ALJUNIED         HOUGANG EAST kml_223
## 4      ALJUNIED      BEDOK RESERVOIR kml_215
## 5      ALJUNIED     UPPER PAYA LEBAR kml_201
## 6      ALJUNIED DEFU INDUSTRIAL PARK kml_158
## 7      ALJUNIED        LORONG AH SOO kml_186
## 8      ALJUNIED            KEMBANGAN kml_149
## 9      ALJUNIED         LORONG HALUS kml_224
## 10     ALJUNIED                 PLAB kml_325
##
Description
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<th>CA_IND</th> <td>N</td> </tr><tr bgcolor="#E3E3F3"> <th>PLN_AREA_N</th> <td>PAYA
LEBAR</td> </tr><tr bgcolor=""> <th>PLN_AREA_C</th> <td>PL</td> </tr><tr
bgcolor="#E3E3F3"> <th>REGION_N</th> <td>EAST REGION</td> </tr><tr bgcolor="">
<th>REGION_C</th> <td>ER</td> </tr><tr bgcolor="#E3E3F3"> <th>INC_CRC</th>
<td>100D316E8AEA9465</td> </tr><tr bgcolor=""> <th>FMEL_UPD_D</th>
<td>20191223152313</td> </tr></table></center>
```

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<td>GEYLANG</td> </tr><tr bgcolor=""> <th>PLN_AREA_C</th> <td>GL</td> </tr><tr
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<th>REGION_C</th> <td>CR</td> </tr><tr bgcolor="#E3E3F3"> <th>INC_CRC</th>
<td>418A24B07122EF05</td> </tr><tr bgcolor=""> <th>FMEL_UPD_D</th>
<td>20191223152313</td> </tr></table></center>
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<th>CA_IND</th> <td>N</td> </tr><tr bgcolor="#E3E3F3"> <th>PLN_AREA_N</th>
<td>HOUGANG</td> </tr><tr bgcolor=""> <th>PLN_AREA_C</th> <td>HG</td> </tr><tr
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<th>REGION_C</th> <td>NER</td> </tr><tr bgcolor="#E3E3F3"> <th>INC_CRC</th>
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<th>REGION_N</th> <td>EAST REGION</td> </tr><tr bgcolor=""> <th>REGION_C</th> <td>ER</td>
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## 1     Paya Lebar     9.378953     0.0000     0.0000         0.000     0.0000
## 2        Geylang  1673.990373   835.5766   838.4138      1150.514   571.7103
## 3        Hougang 12369.812551  6011.1730  6358.6396     10126.171  4934.0264
## 4          Bedok 25419.294287 12455.1548 12964.1395     19111.876  9321.4059
## 5      Serangoon  1975.444245   952.6965  1022.7477      1730.265   832.4419
## 6        Hougang    20.000000     0.0000     0.0000         0.000     0.0000
## 7        Hougang 31785.192085 15017.3231 16767.8690     26612.231 12489.8496
## 8          Bedok  3685.201349  1816.0389  1870.1462      2626.665  1292.6734
## 9        Hougang     0.000000     0.0000     0.0000         0.000     0.0000
## 10    Paya Lebar     0.000000     0.0000     0.0000         0.000     0.0000
##     Chinese.F Total.Malay    Malay.M    Malay.F Total.Indian   Indian.M
## 1      0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
```

```
## 2    578.8035   331.96080  167.39904  163.14313     146.1195   76.60634
## 3   5192.1444   997.72565  481.48949  516.23616    1002.6895  501.34473
## 4   9790.4702  3263.49008 1606.79481 1656.69527    2215.5804 1137.73049
## 5    897.8231    59.54353   29.18801   30.35553     145.9400   72.38625
## 6      0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
## 7  14112.5466  2252.10674 1121.13611 1130.97063    2320.9484 1150.63969
## 8   1333.9917   577.47282  282.34191  296.11468     349.2382  179.04609
## 9      0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
## 10     0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
##      Indian.F Total.Other   Other.M   Other.F     Total.HDB
## 1     0.00000     0.00000   0.00000   0.00000     0.000 [1]
## 2    68.09452    46.81499  18.44227  26.95408  1655.548 [1]
## 3   501.34473   243.22665  94.31237 148.91428 11382.015 [1]
## 4  1077.84993   828.34763 389.22359 439.12405 17405.280 [1]
## 5    73.55378    39.69569  18.68032  19.84784  1058.941 [1]
## 6     0.00000     0.00000   0.00000   0.00000     0.000 [1]
## 7  1170.30874   609.74069 255.69771 354.04298 23681.542 [1]
## 8   170.19216   131.82514  61.97749  69.84765  2312.843 [1]
## 9     0.00000     0.00000   0.00000   0.00000     0.000 [1]
## 10    0.00000     0.00000   0.00000   0.00000     0.000 [1]
##    X1....2..Room.Flats1. X3.Room.Flats X4.Room.Flats X5.Room...Executive.Flats
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## 3            0.00000 [1]  789.2457 [1] 6651.5043 [1]             3941.2645 [1]
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## 5           38.52817 [1]  171.6255 [1]  555.7396 [1]              294.2151 [1]
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## 7            0.00000 [1] 7572.5860 [1] 9441.1462 [1]             6667.8095 [1]
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## 10           0.00000 [1]    0.0000 [1]    0.0000 [1]                0.0000 [1]
##    Condominiums...Other.Apartments Landed.Properties         Others
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## 2                       0.0000 [1]       0.00000 [1]  18.442267 [1]
## 3                     982.8342 [1]       0.00000 [1]   9.927618 [1]
## 4                    7914.2129 [1]      49.90046 [1]  39.920368 [1]
## 5                     573.2524 [1]     331.57575 [1]  11.675202 [1]
## 6                       0.0000 [1]       0.00000 [1]   0.000000 [1]
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##            X0...4         X5.Sep         Oct.14       X15...19      X20...24
## 1     0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
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## 10    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
##         X25...29      X30...34      X35...39      X40...44      X45...49
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##         X50...54      X55...59      X60...64      X65...69       X70...74
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## 8   257.7477 [1]  289.2283 [1]  318.7414 [1]  270.5367 [1]  210.52672 [1]
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## 10    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.00000 [1]
##         X75...79      X80...84       X85...89     X90...Over Seat
## 1    0.00000 [1]   0.00000 [1]   0.000000 [1]   0.000000 [1]    5
## 2   36.88453 [1]  18.44227 [1]   8.511815 [1]   5.674544 [1]    5
## 3  238.26284 [1] 138.98666 [1]  74.457138 [1]  39.710474 [1]    5
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##              Party Votes Votes.. Sample.counts.. Swing Margins
## 1   Workers' Party 85815   59.95              60  8.99    -0.2
## 2   Workers' Party 85815   59.95              60  8.99    -0.2
## 3   Workers' Party 85815   59.95              60  8.99    -0.2
## 4   Workers' Party 85815   59.95              60  8.99    -0.2
## 5   Workers' Party 85815   59.95              60  8.99    -0.2
## 6   Workers' Party 85815   59.95              60  8.99    -0.2
## 7   Workers' Party 85815   59.95              60  8.99    -0.2
## 8   Workers' Party 85815   59.95              60  8.99    -0.2
## 9   Workers' Party 85815   59.95              60  8.99    -0.2
## 10  Workers' Party 85815   59.95              60  8.99    -0.2
##    no_of_registered_electors no_of_rejected_votes no_of_spoilt_ballot_papers
## 1                     150821                 1582                        103
## 2                     150821                 1582                        103
```

```
## 3                     150821                 1582                        103
## 4                     150821                 1582                        103
## 5                     150821                 1582                        103
## 6                     150821                 1582                        103
## 7                     150821                 1582                        103
## 8                     150821                 1582                        103
## 9                     150821                 1582                        103
## 10                    150821                 1582                        103
##    Minority electors_per_seat          Winner
## 1         M           30164.2  Workers' Party
## 2         M           30164.2  Workers' Party
## 3         M           30164.2  Workers' Party
## 4         M           30164.2  Workers' Party
## 5         M           30164.2  Workers' Party
## 6         M           30164.2  Workers' Party
## 7         M           30164.2  Workers' Party
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## 10        M           30164.2  Workers' Party
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## 2  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 3  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
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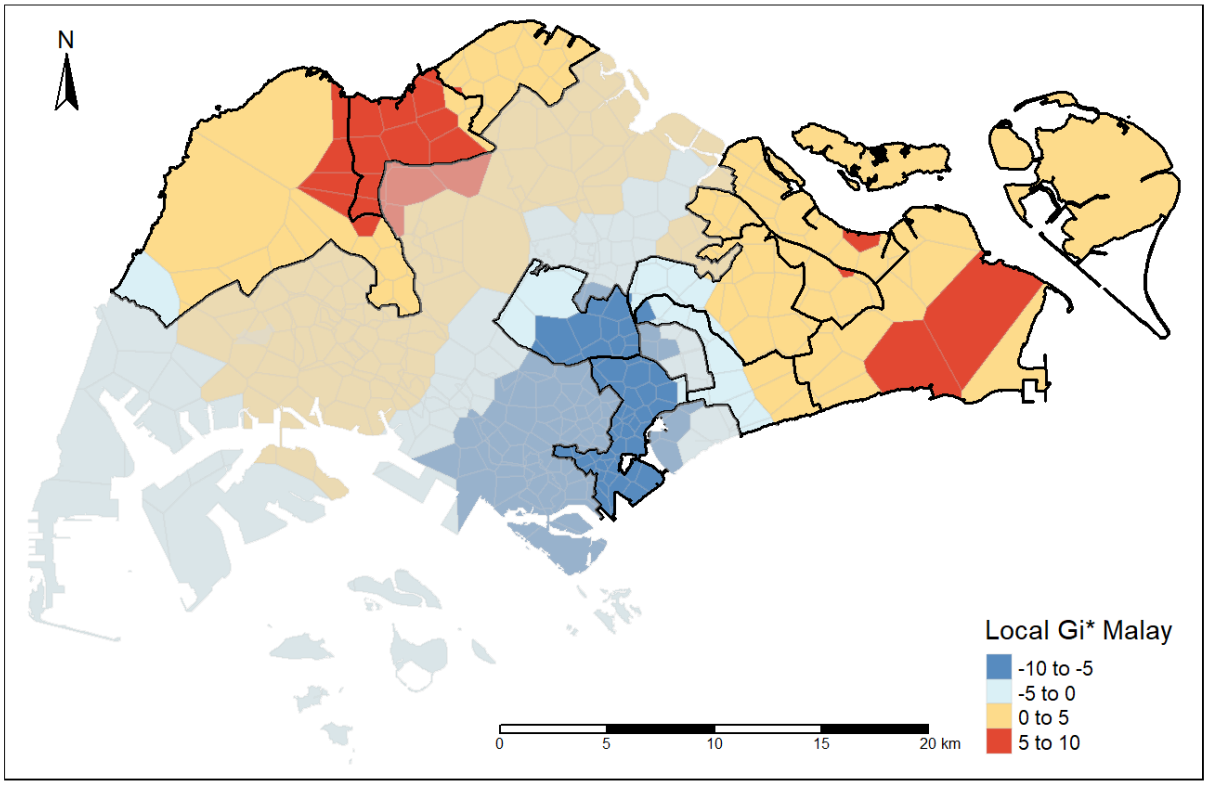
## 9  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 10 -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
##       cluster_type  MarginsGI            area        area_sub      area_elec
## 1  No Significance -0.1211519  876035.5 [m^2] 2802131.9 [m^2] 29321692 [m^2]
## 2  No Significance -0.1211519  340984.5 [m^2] 2403608.6 [m^2] 29321692 [m^2]
## 3  No Significance -0.1211519  633123.4 [m^2] 1275478.9 [m^2] 29321692 [m^2]
## 4  No Significance -0.1211519 2787574.5 [m^2] 2793135.0 [m^2] 29321692 [m^2]
## 5  No Significance -0.1211519  128680.9 [m^2] 1102172.9 [m^2] 29321692 [m^2]
## 6  No Significance -0.1211519  958617.3 [m^2]  958617.3 [m^2] 29321692 [m^2]
## 7  No Significance -0.1211519 1561918.5 [m^2] 1588198.8 [m^2] 29321692 [m^2]
```

```
## 8  No Significance -0.1211519  269395.8 [m^2] 2738403.2 [m^2] 29321692 [m^2]
## 9  No Significance -0.1211519 1053946.3 [m^2] 1164289.4 [m^2] 29321692 [m^2]
## 10 No Significance -0.1211519 2825932.6 [m^2] 2825932.6 [m^2] 29321692 [m^2]
##                          geometry Chinese.per Malay.per Indian.per Other.per
## 1  POLYGON ((103.9193 1.379211...     0.00000  0.000000   0.000000  0.000000
## 2  MULTIPOLYGON (((103.9058 1....    68.72881 19.830508   8.728814  2.796610
## 3  POLYGON ((103.9042 1.376471...    81.86196  8.065811   8.105939  1.966292
## 4  POLYGON ((103.9343 1.338343...    75.18649 12.838634   8.716137  3.258736
## 5  POLYGON ((103.8826 1.351, 1...    87.58865  3.014184   7.387707  2.009456
## 6  POLYGON ((103.8954 1.343897...     0.00000  0.000000   0.000000  0.000000
## 7  POLYGON ((103.8845 1.350097...    83.72525  7.085396   7.301980  1.918317
## 8  POLYGON ((103.9253 1.32934,...    71.27603 15.670048   9.476775  3.577149
## 9  POLYGON ((103.9011 1.370104...     0.00000  0.000000   0.000000  0.000000
## 10 POLYGON ((103.9077 1.346787...     0.00000  0.000000   0.000000  0.000000
##          GI_M
## 1   3.3630344
## 2  -0.2819378
## 3   2.4473933
## 4   2.5267969
## 5  -2.0040516
## 6   0.6558877
## 7  -0.3424804
## 8   1.4482405
## 9   2.1012975
## 10  2.3102831#hotspot
non.M.con <- Electoral |>
  filter(!Minority == "M")
M.con <- Electoral |>
  filter(Minority == "M")

M.spot <- tm_shape(Area.M.p) +
  tm_polygons(
    col = "GI_M",
    style = "pretty",
    midpoint = 0,
    title = "Local Gi* Malay",
    palette = "-RdYlBu",
    border.col = "grey", border.alpha = 0.2,
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  ) +
  tm_shape(M.con) +
  tm_borders(col = "black", lwd = 1.5) +
```

```
  tm_shape(non.M.con) +
  tm_fill(
    palette = "grey",
    alpha = 0.5
  )

tmap_options(check.and.fix = TRUE)
M.spot
```




```
# indian
# GI test
gi_distance_I <- Area |>
  pull(Indian.per) |>
  as.vector() |>
  localG(list2)

# join
Area.I.p <- Area |>
  mutate(GI_I = as.numeric(gi_distance_I))

# inspect
Area.I.p## Simple feature collection with 626 features and 78 fields
## Geometry type: GEOMETRY
```

```
## Dimension:     XY
## Bounding box:  xmin: 103.6057 ymin: 1.158762 xmax: 104.0885 ymax: 1.470783
## Geodetic CRS:  WGS 84
## First 10 features:
##    Constituency                names    Name
## 1      ALJUNIED     PAYA LEBAR NORTH kml_222
## 2      ALJUNIED          KAMPONG UBI  kml_99
## 3      ALJUNIED         HOUGANG EAST kml_223
## 4      ALJUNIED      BEDOK RESERVOIR kml_215
## 5      ALJUNIED     UPPER PAYA LEBAR kml_201
## 6      ALJUNIED DEFU INDUSTRIAL PARK kml_158
## 7      ALJUNIED        LORONG AH SOO kml_186
## 8      ALJUNIED            KEMBANGAN kml_149
## 9      ALJUNIED         LORONG HALUS kml_224
## 10     ALJUNIED                 PLAB kml_325
##
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##    Planning.Area        Total      Males    Females Total.Chinese  Chinese.M
## 1     Paya Lebar     9.378953     0.0000     0.0000         0.000     0.0000
## 2        Geylang  1673.990373   835.5766   838.4138      1150.514   571.7103
## 3        Hougang 12369.812551  6011.1730  6358.6396     10126.171  4934.0264
## 4          Bedok 25419.294287 12455.1548 12964.1395     19111.876  9321.4059
## 5      Serangoon  1975.444245   952.6965  1022.7477      1730.265   832.4419
## 6        Hougang    20.000000     0.0000     0.0000         0.000     0.0000
## 7        Hougang 31785.192085 15017.3231 16767.8690     26612.231 12489.8496
## 8          Bedok  3685.201349  1816.0389  1870.1462      2626.665  1292.6734
## 9        Hougang     0.000000     0.0000     0.0000         0.000     0.0000
## 10    Paya Lebar     0.000000     0.0000     0.0000         0.000     0.0000
##     Chinese.F Total.Malay    Malay.M    Malay.F Total.Indian   Indian.M
## 1      0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
## 2    578.8035   331.96080  167.39904  163.14313     146.1195   76.60634
## 3   5192.1444   997.72565  481.48949  516.23616    1002.6895  501.34473
## 4   9790.4702  3263.49008 1606.79481 1656.69527    2215.5804 1137.73049
## 5    897.8231    59.54353   29.18801   30.35553     145.9400   72.38625
## 6      0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
## 7  14112.5466  2252.10674 1121.13611 1130.97063    2320.9484 1150.63969
## 8   1333.9917   577.47282  282.34191  296.11468     349.2382  179.04609
## 9      0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
## 10     0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
##      Indian.F Total.Other   Other.M   Other.F     Total.HDB
## 1     0.00000     0.00000   0.00000   0.00000     0.000 [1]
## 2    68.09452    46.81499  18.44227  26.95408  1655.548 [1]
## 3   501.34473   243.22665  94.31237 148.91428 11382.015 [1]
## 4  1077.84993   828.34763 389.22359 439.12405 17405.280 [1]
## 5    73.55378    39.69569  18.68032  19.84784  1058.941 [1]
## 6     0.00000     0.00000   0.00000   0.00000     0.000 [1]
## 7  1170.30874   609.74069 255.69771 354.04298 23681.542 [1]
## 8   170.19216   131.82514  61.97749  69.84765  2312.843 [1]
## 9     0.00000     0.00000   0.00000   0.00000     0.000 [1]
## 10    0.00000     0.00000   0.00000   0.00000     0.000 [1]
##    X1....2..Room.Flats1. X3.Room.Flats X4.Room.Flats X5.Room...Executive.Flats
## 1            0.00000 [1]    0.0000 [1]    0.0000 [1]                0.0000 [1]
## 2            0.00000 [1]  136.1890 [1] 1239.8878 [1]              280.8899 [1]
```

```
## 3            0.00000 [1]  789.2457 [1] 6651.5043 [1]             3941.2645 [1]
## 4            0.00000 [1] 2834.3461 [1] 8303.4365 [1]             6257.5177 [1]
## 5           38.52817 [1]  171.6255 [1]  555.7396 [1]              294.2151 [1]
## 6            0.00000 [1]    0.0000 [1]    0.0000 [1]                0.0000 [1]
## 7            0.00000 [1] 7572.5860 [1] 9441.1462 [1]             6667.8095 [1]
## 8          312.83877 [1]  708.3142 [1]  618.7912 [1]              672.8985 [1]
## 9            0.00000 [1]    0.0000 [1]    0.0000 [1]                0.0000 [1]
## 10           0.00000 [1]    0.0000 [1]    0.0000 [1]                0.0000 [1]
##    Condominiums...Other.Apartments Landed.Properties         Others
## 1                       0.0000 [1]       0.00000 [1]   0.000000 [1]
## 2                       0.0000 [1]       0.00000 [1]  18.442267 [1]
## 3                     982.8342 [1]       0.00000 [1]   9.927618 [1]
## 4                    7914.2129 [1]      49.90046 [1]  39.920368 [1]
## 5                     573.2524 [1]     331.57575 [1]  11.675202 [1]
## 6                       0.0000 [1]       0.00000 [1]   0.000000 [1]
## 7                    4228.8467 [1]    3668.27867 [1] 216.359600 [1]
## 8                     495.8199 [1]     793.90216 [1]  83.620426 [1]
## 9                       0.0000 [1]       0.00000 [1]   0.000000 [1]
## 10                      0.0000 [1]       0.00000 [1]   0.000000 [1]
##            X0...4         X5.Sep         Oct.14       X15...19      X20...24
## 1     0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
## 2    56.74544 [1]   65.25725 [1]   73.76907 [1]   79.44361 [1]  106.3977 [1]
## 3   675.07805 [1]  610.54853 [1]  516.23616 [1]  541.05520 [1]  729.6800 [1]
## 4  1097.81012 [1] 1267.47168 [1] 1397.21288 [1] 1506.99389 [1] 1487.0337 [1]
## 5    68.88369 [1]   91.06658 [1]   86.39650 [1]   91.06658 [1]  105.0768 [1]
## 6     0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
## 7   934.28009 [1] 1121.13611 [1] 1366.99929 [1] 1662.03511 [1] 1878.3947 [1]
## 8   138.71153 [1]  133.79268 [1]  145.59792 [1]  175.11101 [1]  198.7215 [1]
## 9     0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
## 10    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
##         X25...29      X30...34      X35...39      X40...44      X45...49
## 1     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]
## 2   147.5381 [1]  143.2822 [1]  113.4909 [1]  113.4909 [1]  106.3977 [1]
## 3  1047.3637 [1] 1136.7123 [1] 1072.1828 [1]  848.8114 [1]  814.0647 [1]
## 4  1596.8147 [1] 1826.3568 [1] 1896.2175 [1] 2085.8392 [1] 2105.7994 [1]
## 5   166.9554 [1]  166.9554 [1]  141.2699 [1]  141.2699 [1]  135.4323 [1]
## 6     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]
## 7  2232.4377 [1] 2143.9269 [1] 1976.7400 [1] 2143.9269 [1] 2448.7973 [1]
## 8   257.7477 [1]  274.4718 [1]  247.9100 [1]  225.2833 [1]  247.9100 [1]
## 9     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]
## 10    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]
##         X50...54      X55...59      X60...64      X65...69       X70...74
## 1     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.00000 [1]
## 2   106.3977 [1]  143.2822 [1]  156.0500 [1]  119.1654 [1]   73.76907 [1]
## 3   799.1733 [1] 1027.5085 [1]  928.2323 [1]  680.0419 [1]  456.67045 [1]
## 4  1916.1777 [1] 1836.3369 [1] 1796.4166 [1] 1546.9143 [1] 1047.90966 [1]
## 5   129.5947 [1]  141.2699 [1]  162.2853 [1]  128.4272 [1]  102.74178 [1]
## 6     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.00000 [1]
## 7  2370.1211 [1] 2537.3080 [1] 2655.3224 [1] 2261.9413 [1] 1780.04943 [1]
## 8   257.7477 [1]  289.2283 [1]  318.7414 [1]  270.5367 [1]  210.52672 [1]
```

```
## 9     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.00000 [1]
## 10    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.00000 [1]
##         X75...79      X80...84       X85...89     X90...Over Seat
## 1    0.00000 [1]   0.00000 [1]   0.000000 [1]   0.000000 [1]    5
## 2   36.88453 [1]  18.44227 [1]   8.511815 [1]   5.674544 [1]    5
## 3  238.26284 [1] 138.98666 [1]  74.457138 [1]  39.710474 [1]    5
## 4  469.06432 [1] 309.38285 [1] 159.681472 [1]  79.840736 [1]    5
## 5   51.37089 [1]  36.19313 [1]  19.847844 [1]  10.507682 [1]    5
## 6    0.00000 [1]   0.00000 [1]   0.000000 [1]   0.000000 [1]    5
## 7  963.78367 [1] 737.58954 [1] 403.215617 [1] 177.021491 [1]    5
## 8  122.97121 [1]  84.60420 [1]  52.139795 [1]  34.431940 [1]    5
## 9    0.00000 [1]   0.00000 [1]   0.000000 [1]   0.000000 [1]    5
## 10   0.00000 [1]   0.00000 [1]   0.000000 [1]   0.000000 [1]    5
##              Party Votes Votes.. Sample.counts.. Swing Margins
## 1   Workers' Party 85815   59.95              60  8.99    -0.2
## 2   Workers' Party 85815   59.95              60  8.99    -0.2
## 3   Workers' Party 85815   59.95              60  8.99    -0.2
## 4   Workers' Party 85815   59.95              60  8.99    -0.2
## 5   Workers' Party 85815   59.95              60  8.99    -0.2
## 6   Workers' Party 85815   59.95              60  8.99    -0.2
## 7   Workers' Party 85815   59.95              60  8.99    -0.2
## 8   Workers' Party 85815   59.95              60  8.99    -0.2
## 9   Workers' Party 85815   59.95              60  8.99    -0.2
## 10  Workers' Party 85815   59.95              60  8.99    -0.2
##    no_of_registered_electors no_of_rejected_votes no_of_spoilt_ballot_papers
## 1                     150821                 1582                        103
## 2                     150821                 1582                        103
## 3                     150821                 1582                        103
## 4                     150821                 1582                        103
## 5                     150821                 1582                        103
## 6                     150821                 1582                        103
## 7                     150821                 1582                        103
## 8                     150821                 1582                        103
## 9                     150821                 1582                        103
## 10                    150821                 1582                        103
##    Minority electors_per_seat          Winner
## 1         M           30164.2  Workers' Party
## 2         M           30164.2  Workers' Party
## 3         M           30164.2  Workers' Party
## 4         M           30164.2  Workers' Party
## 5         M           30164.2  Workers' Party
## 6         M           30164.2  Workers' Party
## 7         M           30164.2  Workers' Party
## 8         M           30164.2  Workers' Party
## 9         M           30164.2  Workers' Party
## 10        M           30164.2  Workers' Party
##
Description.1
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microsoft-com:xslt"> <head> <META http-equiv="Content-Type" content="text/html"> <meta
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</td> </tr> </table> </body> </html>
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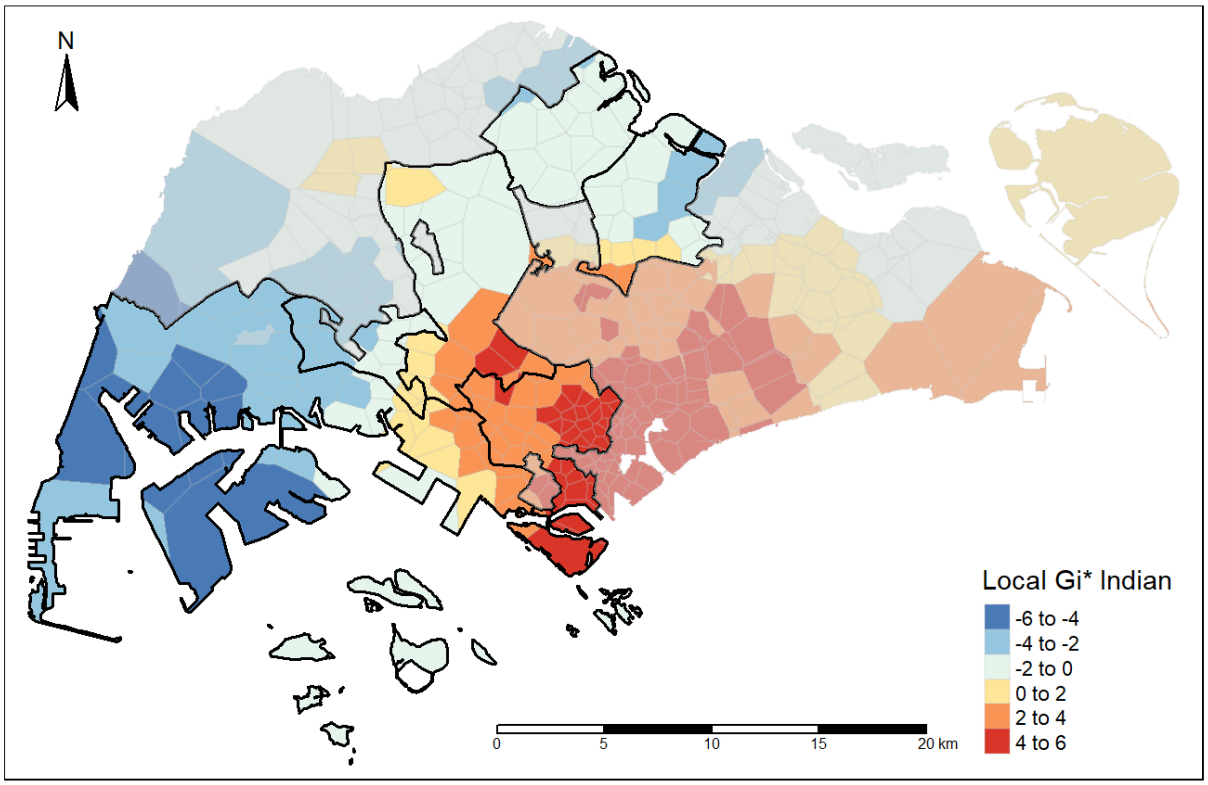
## 2  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 3  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 4  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 5  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 6  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 7  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 8  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 9  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 10 -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
##       cluster_type  MarginsGI            area        area_sub      area_elec
## 1  No Significance -0.1211519  876035.5 [m^2] 2802131.9 [m^2] 29321692 [m^2]
## 2  No Significance -0.1211519  340984.5 [m^2] 2403608.6 [m^2] 29321692 [m^2]
## 3  No Significance -0.1211519  633123.4 [m^2] 1275478.9 [m^2] 29321692 [m^2]
## 4  No Significance -0.1211519 2787574.5 [m^2] 2793135.0 [m^2] 29321692 [m^2]
## 5  No Significance -0.1211519  128680.9 [m^2] 1102172.9 [m^2] 29321692 [m^2]
## 6  No Significance -0.1211519  958617.3 [m^2]  958617.3 [m^2] 29321692 [m^2]
## 7  No Significance -0.1211519 1561918.5 [m^2] 1588198.8 [m^2] 29321692 [m^2]
## 8  No Significance -0.1211519  269395.8 [m^2] 2738403.2 [m^2] 29321692 [m^2]
## 9  No Significance -0.1211519 1053946.3 [m^2] 1164289.4 [m^2] 29321692 [m^2]
## 10 No Significance -0.1211519 2825932.6 [m^2] 2825932.6 [m^2] 29321692 [m^2]
##                          geometry Chinese.per Malay.per Indian.per Other.per
## 1  POLYGON ((103.9193 1.379211...     0.00000  0.000000   0.000000  0.000000
## 2  MULTIPOLYGON (((103.9058 1....    68.72881 19.830508   8.728814  2.796610
## 3  POLYGON ((103.9042 1.376471...    81.86196  8.065811   8.105939  1.966292
## 4  POLYGON ((103.9343 1.338343...    75.18649 12.838634   8.716137  3.258736
## 5  POLYGON ((103.8826 1.351, 1...    87.58865  3.014184   7.387707  2.009456
## 6  POLYGON ((103.8954 1.343897...     0.00000  0.000000   0.000000  0.000000
## 7  POLYGON ((103.8845 1.350097...    83.72525  7.085396   7.301980  1.918317
## 8  POLYGON ((103.9253 1.32934,...    71.27603 15.670048   9.476775  3.577149
## 9  POLYGON ((103.9011 1.370104...     0.00000  0.000000   0.000000  0.000000
## 10 POLYGON ((103.9077 1.346787...     0.00000  0.000000   0.000000  0.000000
##          GI_I
## 1   0.9996142
## 2   4.3909291
## 3  -0.8502159
## 4   2.3445171
## 5   3.5554370
## 6   4.3226592
## 7   3.4395274
## 8   4.4787116
```

```
## 9   0.7881194
## 10  2.2686324#plot local GI clusters
I.con <- Electoral |>
  filter(Minority == "IO")

non.I.con <- Electoral |>
  filter(!Minority == "IO")

I.spot <- tm_shape(Area.I.p) +
  tm_polygons(
    col = "GI_I",
    style = "pretty",
    midpoint = 0,
    title = "Local Gi* Indian",
    palette = "-RdYlBu",
    border.col = "grey", border.alpha = 0.2,
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  ) +
  tm_shape(I.con) +
  tm_borders(col = "black", lwd = 1.5) +
  tm_shape(non.I.con) +
  tm_fill(
    palette = "grey",
    alpha = 0.5
  )

tmap_options(check.and.fix = TRUE)
I.spot
```




```
#others
# GI test
gi_distance_O <- Area |>
  pull(Other.per) |>
  as.vector() |>
  localG(list2)

# join
Area.O.p <- Area |>
  mutate(GI_O = as.numeric(gi_distance_O))

# inspect
Area.O.p## Simple feature collection with 626 features and 78 fields
## Geometry type: GEOMETRY
## Dimension:     XY
## Bounding box:  xmin: 103.6057 ymin: 1.158762 xmax: 104.0885 ymax: 1.470783
## Geodetic CRS:  WGS 84
## First 10 features:
##    Constituency                names    Name
## 1      ALJUNIED     PAYA LEBAR NORTH kml_222
## 2      ALJUNIED          KAMPONG UBI  kml_99
## 3      ALJUNIED         HOUGANG EAST kml_223
## 4      ALJUNIED      BEDOK RESERVOIR kml_215
```

```
## 5      ALJUNIED     UPPER PAYA LEBAR kml_201
## 6      ALJUNIED DEFU INDUSTRIAL PARK kml_158
## 7      ALJUNIED        LORONG AH SOO kml_186
## 8      ALJUNIED            KEMBANGAN kml_149
## 9      ALJUNIED         LORONG HALUS kml_224
## 10     ALJUNIED                 PLAB kml_325
##
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```
##    Planning.Area        Total      Males    Females Total.Chinese  Chinese.M
## 1     Paya Lebar     9.378953     0.0000     0.0000         0.000     0.0000
## 2        Geylang  1673.990373   835.5766   838.4138      1150.514   571.7103
## 3        Hougang 12369.812551  6011.1730  6358.6396     10126.171  4934.0264
## 4          Bedok 25419.294287 12455.1548 12964.1395     19111.876  9321.4059
## 5      Serangoon  1975.444245   952.6965  1022.7477      1730.265   832.4419
## 6        Hougang    20.000000     0.0000     0.0000         0.000     0.0000
## 7        Hougang 31785.192085 15017.3231 16767.8690     26612.231 12489.8496
## 8          Bedok  3685.201349  1816.0389  1870.1462      2626.665  1292.6734
## 9        Hougang     0.000000     0.0000     0.0000         0.000     0.0000
## 10    Paya Lebar     0.000000     0.0000     0.0000         0.000     0.0000
##     Chinese.F Total.Malay    Malay.M    Malay.F Total.Indian   Indian.M
## 1      0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
## 2    578.8035   331.96080  167.39904  163.14313     146.1195   76.60634
## 3   5192.1444   997.72565  481.48949  516.23616    1002.6895  501.34473
## 4   9790.4702  3263.49008 1606.79481 1656.69527    2215.5804 1137.73049
## 5    897.8231    59.54353   29.18801   30.35553     145.9400   72.38625
## 6      0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
## 7  14112.5466  2252.10674 1121.13611 1130.97063    2320.9484 1150.63969
## 8   1333.9917   577.47282  282.34191  296.11468     349.2382  179.04609
## 9      0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
## 10     0.0000     0.00000    0.00000    0.00000       0.0000    0.00000
##      Indian.F Total.Other   Other.M   Other.F     Total.HDB
## 1     0.00000     0.00000   0.00000   0.00000     0.000 [1]
## 2    68.09452    46.81499  18.44227  26.95408  1655.548 [1]
## 3   501.34473   243.22665  94.31237 148.91428 11382.015 [1]
## 4  1077.84993   828.34763 389.22359 439.12405 17405.280 [1]
## 5    73.55378    39.69569  18.68032  19.84784  1058.941 [1]
## 6     0.00000     0.00000   0.00000   0.00000     0.000 [1]
## 7  1170.30874   609.74069 255.69771 354.04298 23681.542 [1]
## 8   170.19216   131.82514  61.97749  69.84765  2312.843 [1]
## 9     0.00000     0.00000   0.00000   0.00000     0.000 [1]
## 10    0.00000     0.00000   0.00000   0.00000     0.000 [1]
##    X1....2..Room.Flats1. X3.Room.Flats X4.Room.Flats X5.Room...Executive.Flats
## 1            0.00000 [1]    0.0000 [1]    0.0000 [1]                0.0000 [1]
## 2            0.00000 [1]  136.1890 [1] 1239.8878 [1]              280.8899 [1]
## 3            0.00000 [1]  789.2457 [1] 6651.5043 [1]             3941.2645 [1]
## 4            0.00000 [1] 2834.3461 [1] 8303.4365 [1]             6257.5177 [1]
## 5           38.52817 [1]  171.6255 [1]  555.7396 [1]              294.2151 [1]
## 6            0.00000 [1]    0.0000 [1]    0.0000 [1]                0.0000 [1]
## 7            0.00000 [1] 7572.5860 [1] 9441.1462 [1]             6667.8095 [1]
## 8          312.83877 [1]  708.3142 [1]  618.7912 [1]              672.8985 [1]
## 9            0.00000 [1]    0.0000 [1]    0.0000 [1]                0.0000 [1]
## 10           0.00000 [1]    0.0000 [1]    0.0000 [1]                0.0000 [1]
##    Condominiums...Other.Apartments Landed.Properties         Others
```

```
## 1                      0.0000 [1]       0.00000 [1]   0.000000 [1]
## 2                      0.0000 [1]       0.00000 [1]  18.442267 [1]
## 3                    982.8342 [1]       0.00000 [1]   9.927618 [1]
## 4                   7914.2129 [1]      49.90046 [1]  39.920368 [1]
## 5                    573.2524 [1]     331.57575 [1]  11.675202 [1]
## 6                      0.0000 [1]       0.00000 [1]   0.000000 [1]
## 7                   4228.8467 [1]    3668.27867 [1] 216.359600 [1]
## 8                    495.8199 [1]     793.90216 [1]  83.620426 [1]
## 9                      0.0000 [1]       0.00000 [1]   0.000000 [1]
## 10                     0.0000 [1]       0.00000 [1]   0.000000 [1]
##            X0...4         X5.Sep         Oct.14       X15...19      X20...24
## 1     0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
## 2    56.74544 [1]   65.25725 [1]   73.76907 [1]   79.44361 [1]  106.3977 [1]
## 3   675.07805 [1]  610.54853 [1]  516.23616 [1]  541.05520 [1]  729.6800 [1]
## 4  1097.81012 [1] 1267.47168 [1] 1397.21288 [1] 1506.99389 [1] 1487.0337 [1]
## 5    68.88369 [1]   91.06658 [1]   86.39650 [1]   91.06658 [1]  105.0768 [1]
## 6     0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
## 7   934.28009 [1] 1121.13611 [1] 1366.99929 [1] 1662.03511 [1] 1878.3947 [1]
## 8   138.71153 [1]  133.79268 [1]  145.59792 [1]  175.11101 [1]  198.7215 [1]
## 9     0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
## 10    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.00000 [1]    0.0000 [1]
##         X25...29      X30...34      X35...39      X40...44      X45...49
## 1     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]
## 2   147.5381 [1]  143.2822 [1]  113.4909 [1]  113.4909 [1]  106.3977 [1]
## 3  1047.3637 [1] 1136.7123 [1] 1072.1828 [1]  848.8114 [1]  814.0647 [1]
## 4  1596.8147 [1] 1826.3568 [1] 1896.2175 [1] 2085.8392 [1] 2105.7994 [1]
## 5   166.9554 [1]  166.9554 [1]  141.2699 [1]  141.2699 [1]  135.4323 [1]
## 6     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]
## 7  2232.4377 [1] 2143.9269 [1] 1976.7400 [1] 2143.9269 [1] 2448.7973 [1]
## 8   257.7477 [1]  274.4718 [1]  247.9100 [1]  225.2833 [1]  247.9100 [1]
## 9     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]
## 10    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]
##         X50...54      X55...59      X60...64      X65...69       X70...74
## 1     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.00000 [1]
## 2   106.3977 [1]  143.2822 [1]  156.0500 [1]  119.1654 [1]   73.76907 [1]
## 3   799.1733 [1] 1027.5085 [1]  928.2323 [1]  680.0419 [1]  456.67045 [1]
## 4  1916.1777 [1] 1836.3369 [1] 1796.4166 [1] 1546.9143 [1] 1047.90966 [1]
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## 6     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.00000 [1]
## 7  2370.1211 [1] 2537.3080 [1] 2655.3224 [1] 2261.9413 [1] 1780.04943 [1]
## 8   257.7477 [1]  289.2283 [1]  318.7414 [1]  270.5367 [1]  210.52672 [1]
## 9     0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.00000 [1]
## 10    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.0000 [1]    0.00000 [1]
##         X75...79      X80...84       X85...89     X90...Over Seat
## 1    0.00000 [1]   0.00000 [1]   0.000000 [1]   0.000000 [1]    5
## 2   36.88453 [1]  18.44227 [1]   8.511815 [1]   5.674544 [1]    5
## 3  238.26284 [1] 138.98666 [1]  74.457138 [1]  39.710474 [1]    5
## 4  469.06432 [1] 309.38285 [1] 159.681472 [1]  79.840736 [1]    5
## 5   51.37089 [1]  36.19313 [1]  19.847844 [1]  10.507682 [1]    5
## 6    0.00000 [1]   0.00000 [1]   0.000000 [1]   0.000000 [1]    5
```

```
## 7  963.78367 [1] 737.58954 [1] 403.215617 [1] 177.021491 [1]    5
## 8  122.97121 [1]  84.60420 [1]  52.139795 [1]  34.431940 [1]    5
## 9    0.00000 [1]   0.00000 [1]   0.000000 [1]   0.000000 [1]    5
## 10   0.00000 [1]   0.00000 [1]   0.000000 [1]   0.000000 [1]    5
##              Party Votes Votes.. Sample.counts.. Swing Margins
## 1   Workers' Party 85815   59.95              60  8.99    -0.2
## 2   Workers' Party 85815   59.95              60  8.99    -0.2
## 3   Workers' Party 85815   59.95              60  8.99    -0.2
## 4   Workers' Party 85815   59.95              60  8.99    -0.2
## 5   Workers' Party 85815   59.95              60  8.99    -0.2
## 6   Workers' Party 85815   59.95              60  8.99    -0.2
## 7   Workers' Party 85815   59.95              60  8.99    -0.2
## 8   Workers' Party 85815   59.95              60  8.99    -0.2
## 9   Workers' Party 85815   59.95              60  8.99    -0.2
## 10  Workers' Party 85815   59.95              60  8.99    -0.2
##    no_of_registered_electors no_of_rejected_votes no_of_spoilt_ballot_papers
## 1                     150821                 1582                        103
## 2                     150821                 1582                        103
## 3                     150821                 1582                        103
## 4                     150821                 1582                        103
## 5                     150821                 1582                        103
## 6                     150821                 1582                        103
## 7                     150821                 1582                        103
## 8                     150821                 1582                        103
## 9                     150821                 1582                        103
## 10                    150821                 1582                        103
##    Minority electors_per_seat          Winner
## 1         M           30164.2  Workers' Party
## 2         M           30164.2  Workers' Party
## 3         M           30164.2  Workers' Party
## 4         M           30164.2  Workers' Party
## 5         M           30164.2  Workers' Party
## 6         M           30164.2  Workers' Party
## 7         M           30164.2  Workers' Party
## 8         M           30164.2  Workers' Party
## 9         M           30164.2  Workers' Party
## 10        M           30164.2  Workers' Party
##
Description.1
## 1  <html xmlns:fo="http://www.w3.org/1999/XSL/Format" xmlns:msxsl="urn:schemas-
microsoft-com:xslt"> <head> <META http-equiv="Content-Type" content="text/html"> <meta
http-equiv="content-type" content="text/html; charset=UTF-8"> </head> <body
style="margin:0px 0px 0px 0px;overflow:auto;background:#FFFFFF;"> <table style="font-
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collapse:collapse;padding:3px 3px 3px 3px"> <tr style="text-align:center;font-
weight:bold;background:#9CBCE2"> <td>ALJUNIED</td> </tr> <tr> <td> <table style="font-
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padding:3px 3px 3px 3px"> <tr> <td>FID</td> <td>17</td> </tr> <tr bgcolor="#D4E4F3">
<td>ED_DESC</td> <td>ALJUNIED</td> </tr> <tr> <td>ED_CODE</td> <td>AJ</td> </tr> </table>
</td> </tr> </table> </body> </html>
```

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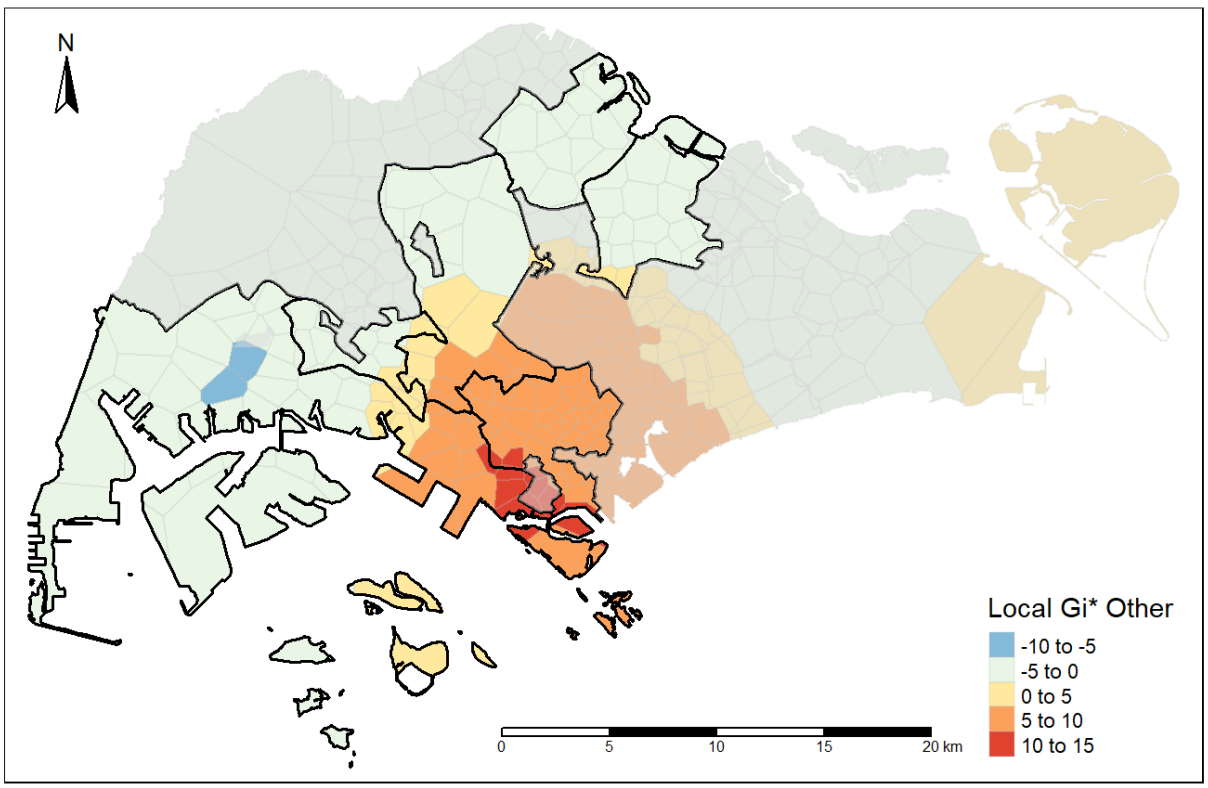
LMI_Ii LMI_eIi LMI_varIi LMI_zIi LMI_sigP obs_diff LMI_diff

```
## 1  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 2  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 3  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 4  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 5  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 6  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 7  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 8  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 9  -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
## 10 -0.1451883 -0.1947968 0.1676683 0.1211519 0.9035708 -0.4319355 -0.0732975
##       cluster_type  MarginsGI            area        area_sub      area_elec
## 1  No Significance -0.1211519  876035.5 [m^2] 2802131.9 [m^2] 29321692 [m^2]
## 2  No Significance -0.1211519  340984.5 [m^2] 2403608.6 [m^2] 29321692 [m^2]
## 3  No Significance -0.1211519  633123.4 [m^2] 1275478.9 [m^2] 29321692 [m^2]
## 4  No Significance -0.1211519 2787574.5 [m^2] 2793135.0 [m^2] 29321692 [m^2]
## 5  No Significance -0.1211519  128680.9 [m^2] 1102172.9 [m^2] 29321692 [m^2]
## 6  No Significance -0.1211519  958617.3 [m^2]  958617.3 [m^2] 29321692 [m^2]
## 7  No Significance -0.1211519 1561918.5 [m^2] 1588198.8 [m^2] 29321692 [m^2]
## 8  No Significance -0.1211519  269395.8 [m^2] 2738403.2 [m^2] 29321692 [m^2]
## 9  No Significance -0.1211519 1053946.3 [m^2] 1164289.4 [m^2] 29321692 [m^2]
## 10 No Significance -0.1211519 2825932.6 [m^2] 2825932.6 [m^2] 29321692 [m^2]
##                          geometry Chinese.per Malay.per Indian.per Other.per
## 1  POLYGON ((103.9193 1.379211...     0.00000  0.000000   0.000000  0.000000
## 2  MULTIPOLYGON (((103.9058 1....    68.72881 19.830508   8.728814  2.796610
## 3  POLYGON ((103.9042 1.376471...    81.86196  8.065811   8.105939  1.966292
## 4  POLYGON ((103.9343 1.338343...    75.18649 12.838634   8.716137  3.258736
## 5  POLYGON ((103.8826 1.351, 1...    87.58865  3.014184   7.387707  2.009456
## 6  POLYGON ((103.8954 1.343897...     0.00000  0.000000   0.000000  0.000000
## 7  POLYGON ((103.8845 1.350097...    83.72525  7.085396   7.301980  1.918317
## 8  POLYGON ((103.9253 1.32934,...    71.27603 15.670048   9.476775  3.577149
## 9  POLYGON ((103.9011 1.370104...     0.00000  0.000000   0.000000  0.000000
## 10 POLYGON ((103.9077 1.346787...     0.00000  0.000000   0.000000  0.000000
##          GI_O
## 1  -2.5235734
## 2   0.4925907
## 3  -3.9926749
## 4  -2.9833212
## 5   2.0369954
## 6  -1.6993554
## 7  -1.6058960
## 8  -2.8055342
## 9  -3.5931188
## 10 -3.0521515#plot local GI clusters
O.spot <- tm_shape(Area.O.p) +
  tm_polygons(
    col = "GI_O",
    style = "pretty",
    midpoint = 0,
    title = "Local Gi* Other",
    palette = "-RdYlBu",
```

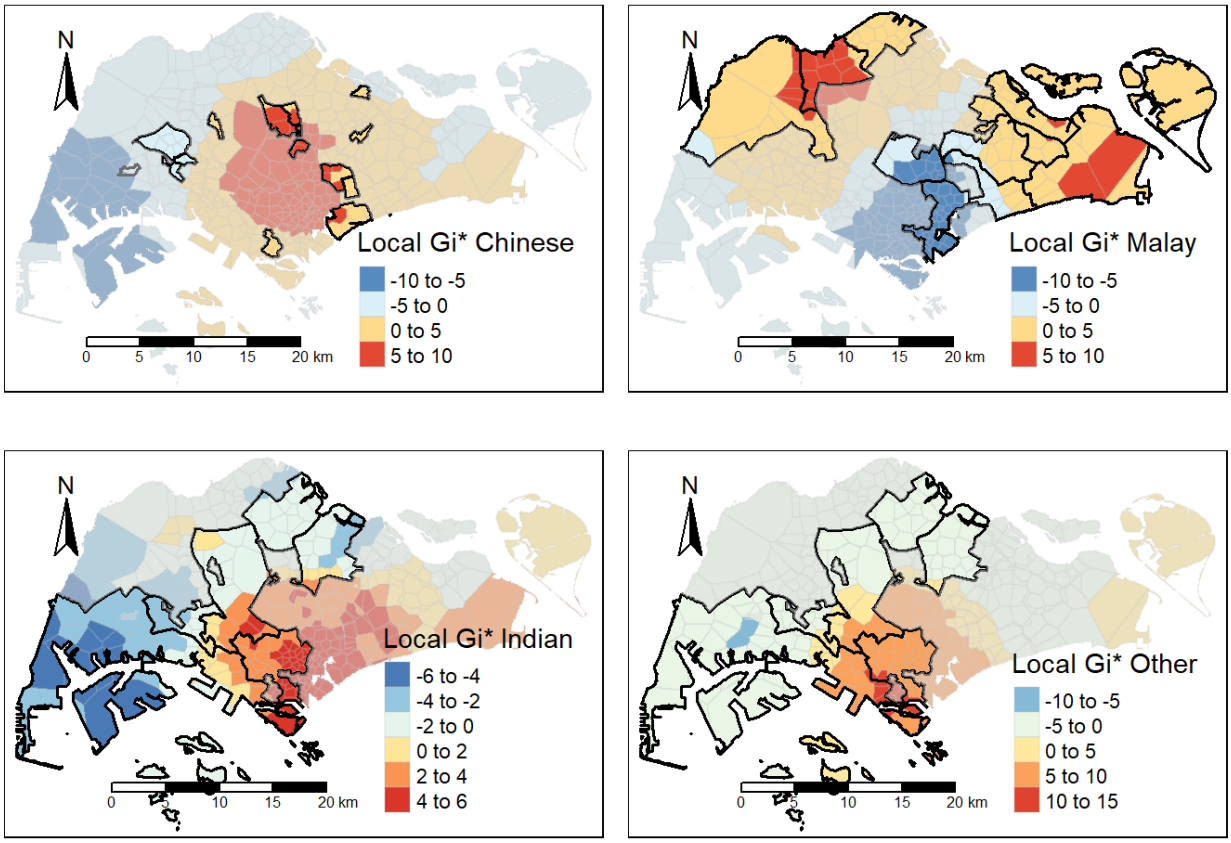
```
    border.col = "grey", border.alpha = 0.2,
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  ) +
  tm_shape(I.con) +
  tm_borders(col = "black", lwd = 1.5) +
  tm_shape(non.I.con) +
  tm_fill(
    palette = "grey",
    alpha = 0.5
  )

tmap_options(check.and.fix = TRUE)
O.spot
```




```
# hotspot of pop comp
grid.newpage()
```

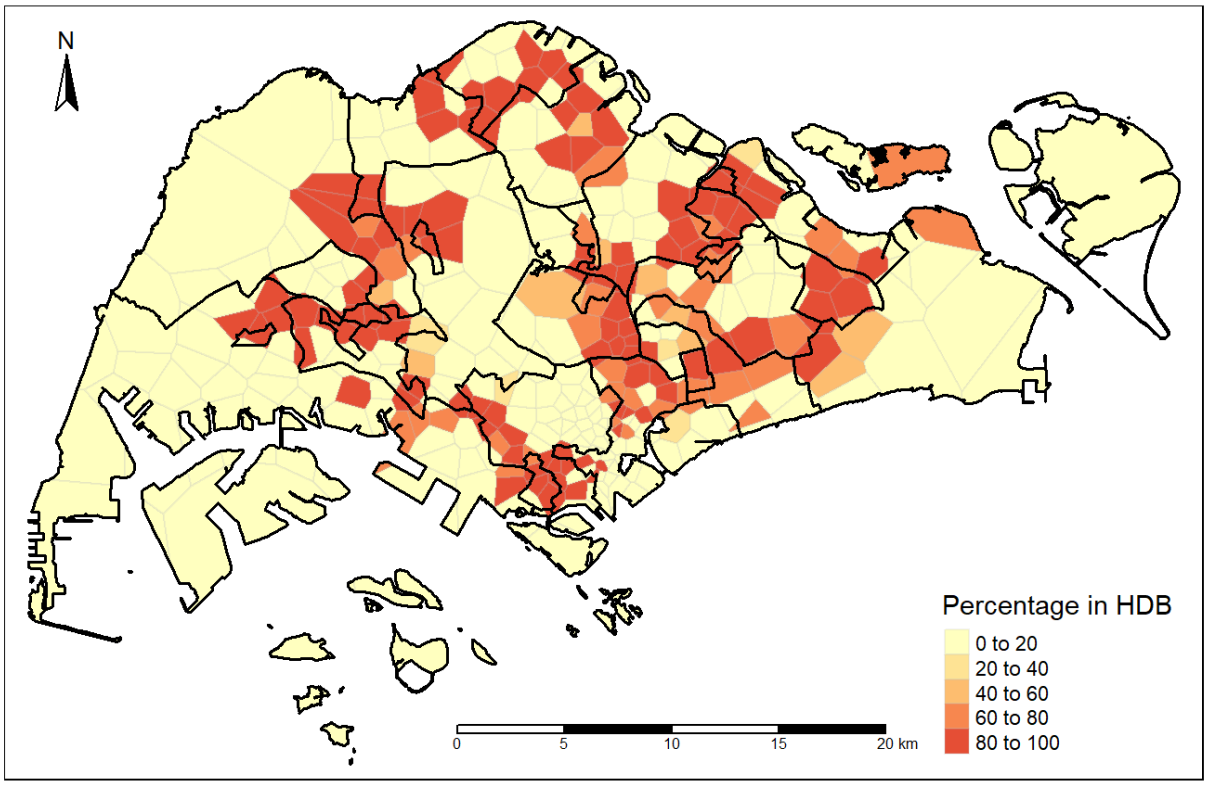
```
pushViewport(viewport(layout=grid.layout(2,2)))
print(I.spot, vp=viewport(layout.pos.col = 1, layout.pos.row =2))
print(O.spot, vp=viewport(layout.pos.col = 2, layout.pos.row =2))
print(C.spot, vp=viewport(layout.pos.col = 1, layout.pos.row =1))
print(M.spot, vp=viewport(layout.pos.col = 2, layout.pos.row =1))
```



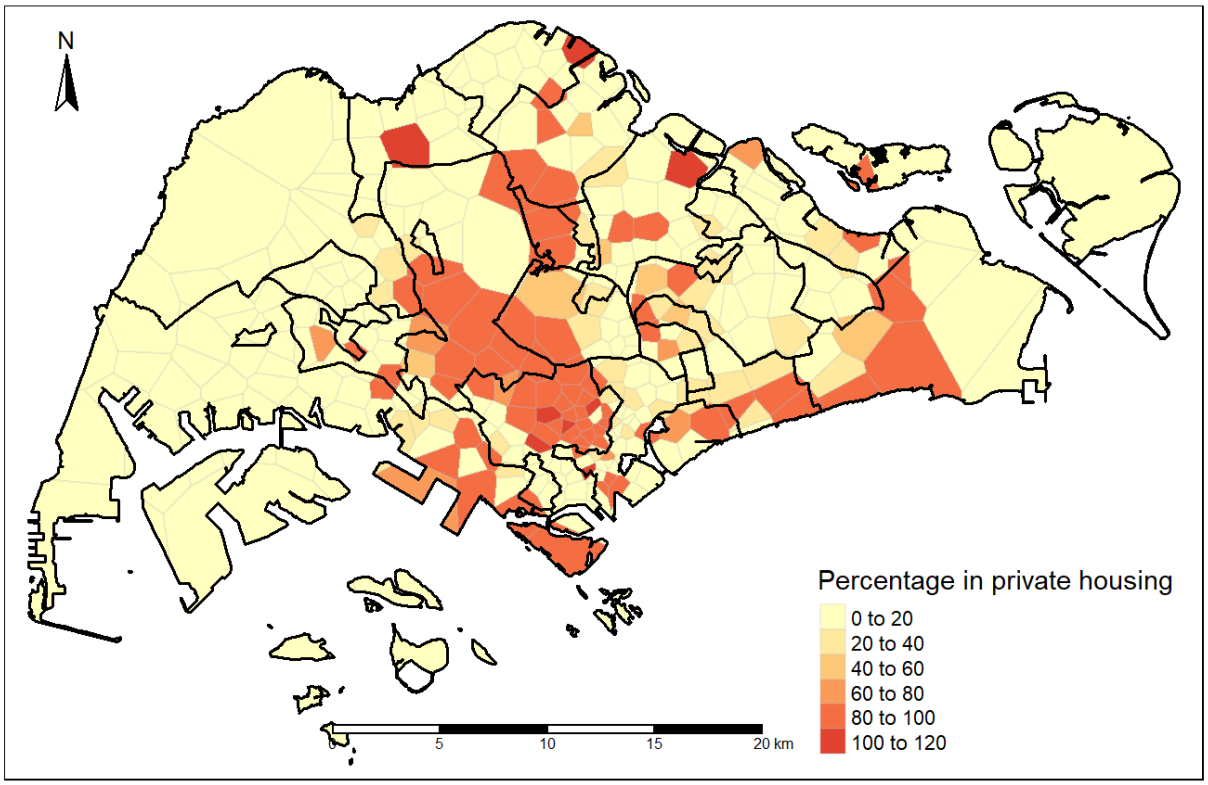

```
#GWR
# prop public and private housing
Area$HDB.per <- Area$Total.HDB / Area$Total *100
Area$HDB.per <- gsub("NaN", 0, Area$HDB.per)
Area$HDB.per <- as.numeric(Area$HDB.per)
Area$priv.per <- (Area$Condominiums...Other.Apartments + Area$Landed.Properties) /
Area$Total *100
Area$priv.per <- gsub("NaN", 0, Area$priv.per)
Area$priv.per <- as.numeric(Area$priv.per)

# prop by age group
Area$"0-19" <- (Area$X0...4 + Area$X5.Sep + Area$Oct.14 + Area$X15...19) / Area$Total
*100
Area$"0-19" <- gsub("NaN", 0, Area$"0-19")
Area$"0-19" <- as.numeric(Area$"0-19")
Area$"20-39" <- (Area$X20...24 + Area$X25...29 + Area$X30...34 + Area$X35...39) /
Area$Total *100
```

```r
Area$"20-39" <- gsub("NaN", 0, Area$"20-39")
Area$"20-39" <- as.numeric(Area$"20-39")
Area$"40-64" <- (Area$X40...44 + Area$X45...49 + Area$X50...54 + Area$X55...59 +
Area$X60...64) / Area$Total *100
Area$"40-64" <- gsub("NaN", 0, Area$"40-64")
Area$"40-64" <- as.numeric(Area$"40-64")
Area$Elderly <- (Area$X65...69 + Area$X70...74 + Area$X75...79 + Area$X80...84 +
Area$X85...89 + Area$X90...Over) /Area$Total*100
Area$Elderly <- gsub("NaN", 0, Area$Elderly)
Area$Elderly <- as.numeric(Area$Elderly)

#mapping
tm_shape(Area) +
  tm_polygons(
    col = "HDB.per",
    style = "pretty",
    midpoint = 0,
    title = "Percentage in HDB",
    palette = "-RdYlBu",
    border.col = "grey", border.alpha = 0.2,
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  ) +
  tm_shape(Electoral) +
  tm_borders(col = "black", lwd = 1.5)
```

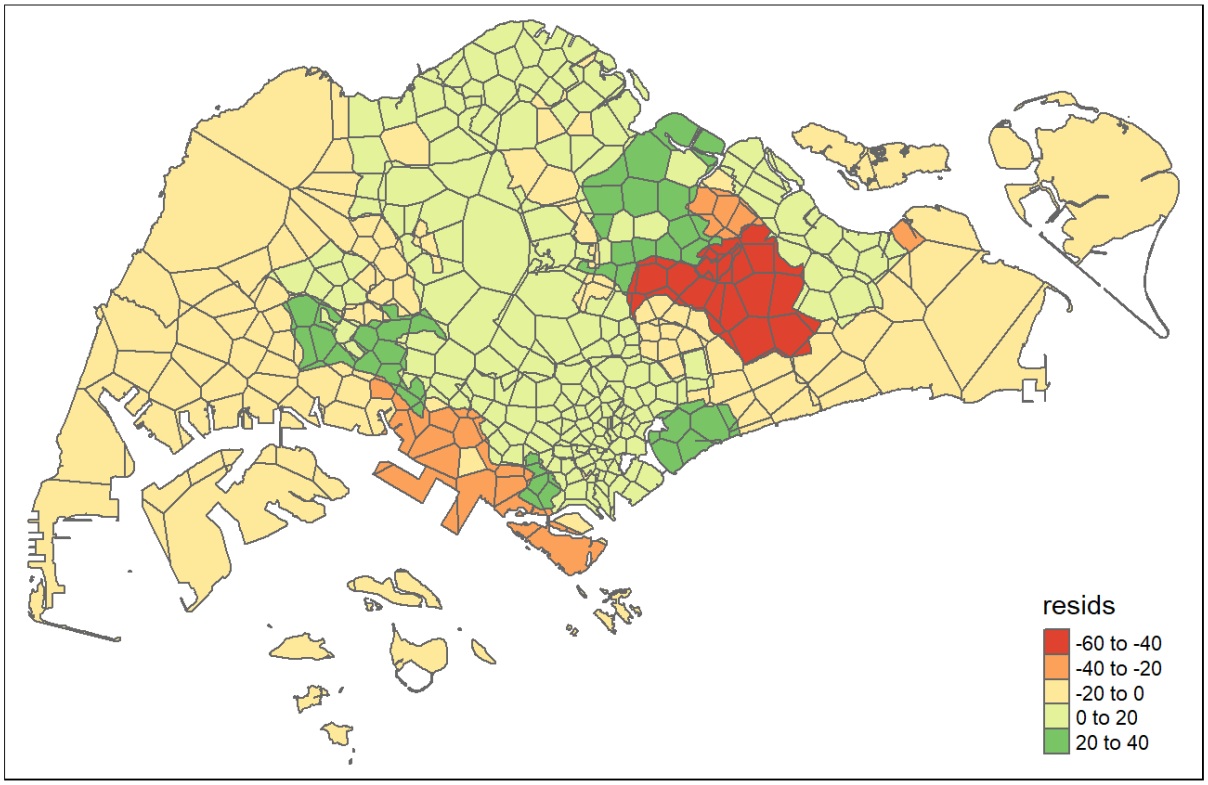



```
tm_shape(Area) +
  tm_polygons(
    col = "priv.per",
    style = "pretty",
    midpoint = 0,
    title = "Percentage in private housing",
    palette = "-RdYlBu",
    border.col = "grey", border.alpha = 0.2,
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  ) +
  tm_shape(Electoral) +
  tm_borders(col = "black", lwd = 1.5)
```




```
#reformat
Area$Margins <- Area$Margins*100
Poly.SP <- as(Area, "Spatial")
# linear model of margins, hdb, elderly, chinese
model <- lm(Area$Margins ~ Area$HDB.per + Area$Elderly + Area$Chinese.per)
summary(model)##
## Call:
## lm(formula = Area$Margins ~ Area$HDB.per + Area$Elderly + Area$Chinese.per)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -47.346  -8.799   2.418  10.516  27.672
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)      21.32792    1.37135  15.553   <2e-16 ***
## Area$HDB.per     -0.02640    0.01867  -1.414    0.158
## Area$Elderly      0.07942    0.10977   0.724    0.470
## Area$Chinese.per  0.03482    0.03218   1.082    0.280
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.37 on 622 degrees of freedom
```

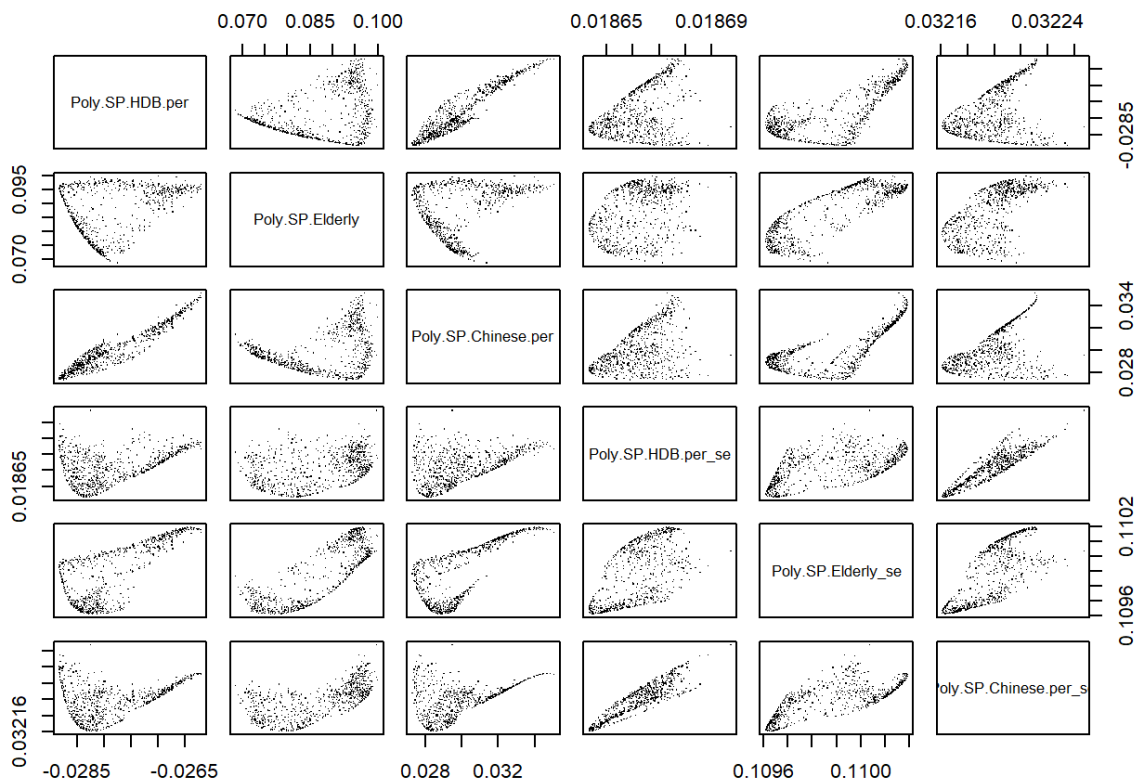
```
## Multiple R-squared:  0.008246,   Adjusted R-squared:  0.003462
## F-statistic: 1.724 on 3 and 622 DF,  p-value: 0.1608#plot residuals
resids <- residuals(model)
map.resids <- Area
map.resids <- cbind(Area, resids)
qtm(map.resids, fill = "resids")## Variable(s) "resids" contains positive and negative
values, so midpoint is set to 0. Set midpoint = NA to show the full spectrum of the color
palette.
```




```
# mots -ve resid in Wp wards, west cost

# try to combine variables into one model
formula <- Poly.SP$Margins ~ Poly.SP$HDB.per + Poly.SP$Elderly + Poly.SP$Chinese.per
GWRbandwidth <- gwr.sel(formula, data = Poly.SP, adapt = T)## Adaptive q: 0.381966 CV
score: 183291.9
## Adaptive q: 0.618034 CV score: 186650.3
## Adaptive q: 0.236068 CV score: 177554.6
## Adaptive q: 0.145898 CV score: 171482.6
## Adaptive q: 0.09016994 CV score: 165161.9
## Adaptive q: 0.05572809 CV score: 158197.9
## Adaptive q: 0.03444185 CV score: 151082.6
## Adaptive q: 0.02128624 CV score: 146510.7
## Adaptive q: 0.01315562 CV score: 149983.9
```

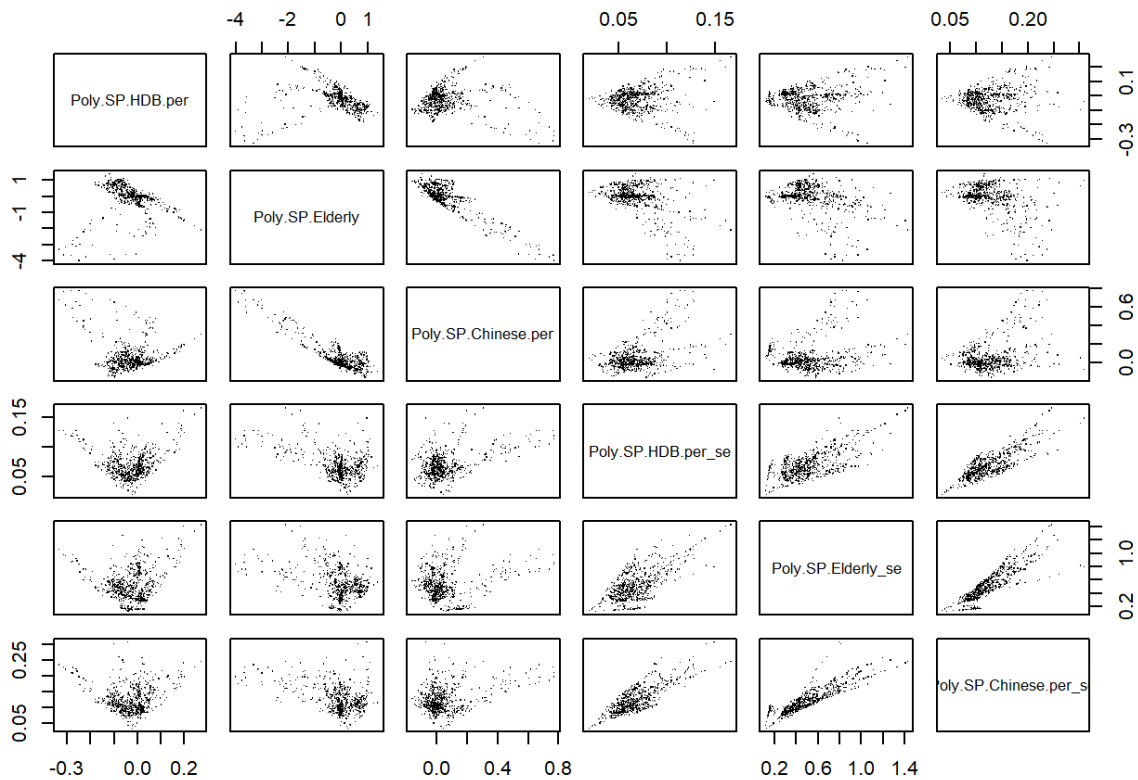
```
## Adaptive q: 0.02308962 CV score: 146785.8
## Adaptive q: 0.02088081 CV score: 146450.9
## Adaptive q: 0.01793005 CV score: 146750.3
## Adaptive q: 0.02008924 CV score: 146445.6
## Adaptive q: 0.02045651 CV score: 146434.7
## Adaptive q: 0.02041582 CV score: 146435.1
## Adaptive q: 0.02053337 CV score: 146434.3
## Adaptive q: 0.02066608 CV score: 146435.2
## Adaptive q: 0.02057406 CV score: 146434.4
## Adaptive q: 0.02053337 CV score: 146434.3for (i in c(1, GWRbandwidth)) {
  gwr.model = gwr(formula, data = Poly.SP, adapt = i, hatmatrix = TRUE, se.fit = TRUE)

  # print the results of the model
  print(gwr.model)

  results.gwr <- as.data.frame(gwr.model$SDF)
  names(results.gwr)

  print(round(cor(results.gwr[, c(3, 4, 5, 7, 8, 9)], use = "complete.obs"), 2))
  pairs(results.gwr[, c(3, 4, 5, 7, 8, 9)], pch = ".")
  #multicollinearity present between Elderly, Chinese
}## Call:
## gwr(formula = formula, data = Poly.SP, adapt = i, hatmatrix = TRUE,
##     se.fit = TRUE)
## Kernel function: gwr.Gauss
## Adaptive quantile: 1 (about 626 of 626 data points)
## Summary of GWR coefficient estimates at data points:
##                          Min.   1st Qu.    Median   3rd Qu.      Max.  Global
## X.Intercept.        21.196199 21.552308 21.737905 21.824854 21.961203 21.3279
## Poly.SP.HDB.per     -0.028830 -0.028403 -0.028063 -0.027186 -0.026202 -0.0264
## Poly.SP.Elderly      0.068580  0.079477  0.089868  0.095284  0.099987  0.0794
## Poly.SP.Chinese.per  0.027298  0.028493  0.029515  0.031248  0.035068  0.0348
## Number of data points: 626
## Effective number of parameters (residual: 2traceS - traceS'S): 4.641734
## Effective degrees of freedom (residual: 2traceS - traceS'S): 621.3583
## Sigma (residual: 2traceS - traceS'S): 17.34256
## Effective number of parameters (model: traceS): 4.330605
## Effective degrees of freedom (model: traceS): 621.6694
## Sigma (model: traceS): 17.33822
## Sigma (ML): 17.27815
## AICc (GWR p. 61, eq 2.33; p. 96, eq. 4.21): 5354.783
## AIC (GWR p. 96, eq. 4.22): 5348.344
## Residual sum of squares: 186882.5
## Quasi-global R2: 0.01274206
##                        Poly.SP.HDB.per Poly.SP.Elderly Poly.SP.Chinese.per
## Poly.SP.HDB.per                   1.00            0.35                0.95
## Poly.SP.Elderly                   0.35            1.00                0.32
## Poly.SP.Chinese.per               0.95            0.32                1.00
## Poly.SP.HDB.per_se                0.40            0.36                0.44
```

```
## Poly.SP.Elderly_se                0.76            0.82                0.73
## Poly.SP.Chinese.per_se            0.34            0.64                0.42
##                        Poly.SP.HDB.per_se Poly.SP.Elderly_se
## Poly.SP.HDB.per                      0.40               0.76
## Poly.SP.Elderly                      0.36               0.82
## Poly.SP.Chinese.per                  0.44               0.73
## Poly.SP.HDB.per_se                   1.00               0.59
## Poly.SP.Elderly_se                   0.59               1.00
## Poly.SP.Chinese.per_se               0.90               0.74
##                        Poly.SP.Chinese.per_se
## Poly.SP.HDB.per                          0.34
## Poly.SP.Elderly                          0.64
## Poly.SP.Chinese.per                      0.42
## Poly.SP.HDB.per_se                       0.90
## Poly.SP.Elderly_se                       0.74
## Poly.SP.Chinese.per_se                   1.00
```



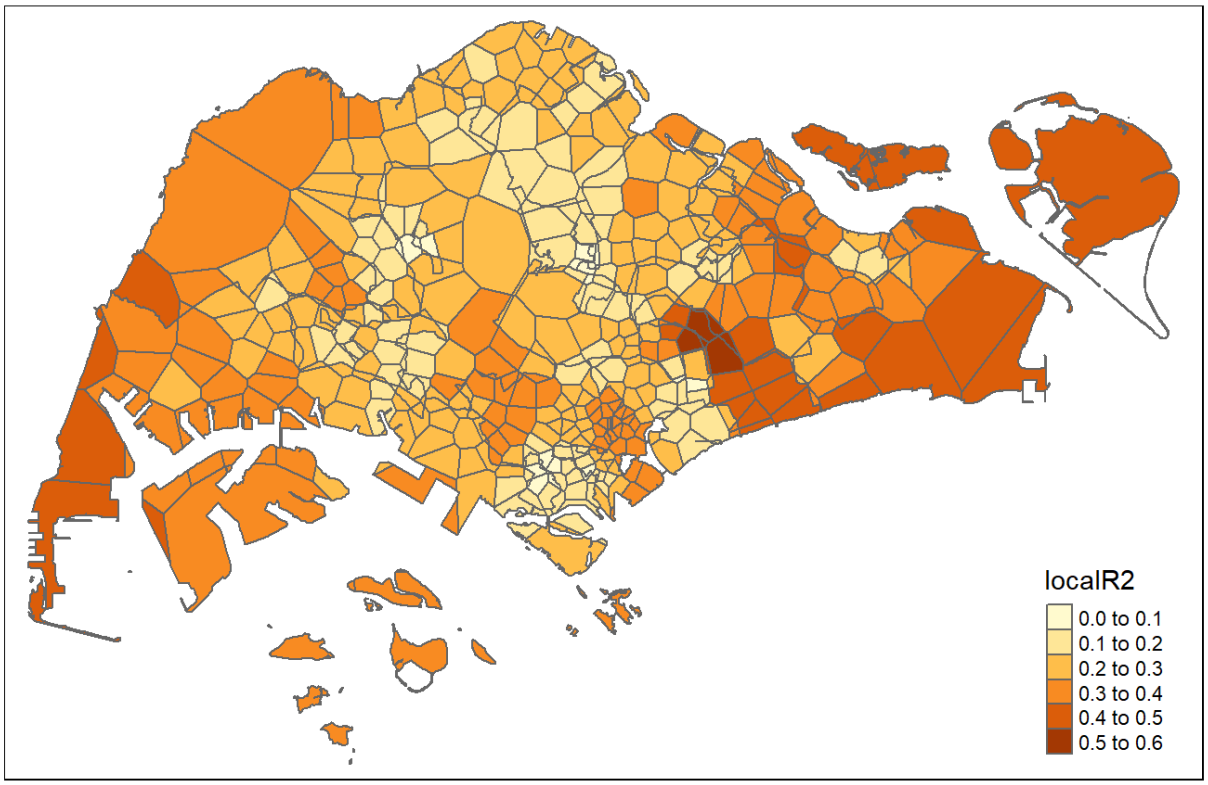

```
## Call:
## gwr(formula = formula, data = Poly.SP, adapt = i, hatmatrix = TRUE,
##     se.fit = TRUE)
## Kernel function: gwr.Gauss
## Adaptive quantile: 0.02053337 (about 12 of 626 data points)
## Summary of GWR coefficient estimates at data points:
##                            Min.     1st Qu.      Median     3rd Qu.        Max.
```

```
## X.Intercept.        -16.1439723  17.6756043  27.2847943  30.0132588  36.8746249
## Poly.SP.HDB.per      -0.3334784  -0.0703405  -0.0197780   0.0165259   0.2720896
## Poly.SP.Elderly      -3.9896839  -0.2223414   0.0122755   0.3509553   1.3865440
## Poly.SP.Chinese.per  -0.1609246  -0.0256332   0.0065956   0.0684295   0.7715789
##                       Global
## X.Intercept.        21.3279
## Poly.SP.HDB.per     -0.0264
## Poly.SP.Elderly      0.0794
## Poly.SP.Chinese.per  0.0348
## Number of data points: 626
## Effective number of parameters (residual: 2traceS - traceS'S): 103.0036
## Effective degrees of freedom (residual: 2traceS - traceS'S): 522.9964
## Sigma (residual: 2traceS - traceS'S): 14.71735
## Effective number of parameters (model: traceS): 74.82339
## Effective degrees of freedom (model: traceS): 551.1766
## Sigma (model: traceS): 14.33618
## Sigma (ML): 13.45215
## AICc (GWR p. 61, eq 2.33; p. 96, eq. 4.21): 5203.493
## AIC (GWR p. 96, eq. 4.22): 5105.456
## Residual sum of squares: 113281.2
## Quasi-global R2: 0.4015612
##                        Poly.SP.HDB.per Poly.SP.Elderly Poly.SP.Chinese.per
## Poly.SP.HDB.per                   1.00           -0.07               -0.17
## Poly.SP.Elderly                  -0.07            1.00               -0.87
## Poly.SP.Chinese.per              -0.17           -0.87                1.00
## Poly.SP.HDB.per_se                0.09           -0.49                0.46
## Poly.SP.Elderly_se                0.13           -0.37                0.22
## Poly.SP.Chinese.per_se            0.03           -0.43                0.37
##                        Poly.SP.HDB.per_se Poly.SP.Elderly_se
## Poly.SP.HDB.per                      0.09               0.13
## Poly.SP.Elderly                     -0.49              -0.37
## Poly.SP.Chinese.per                  0.46               0.22
## Poly.SP.HDB.per_se                   1.00               0.68
## Poly.SP.Elderly_se                   0.68               1.00
## Poly.SP.Chinese.per_se               0.82               0.88
##                        Poly.SP.Chinese.per_se
## Poly.SP.HDB.per                          0.03
## Poly.SP.Elderly                         -0.43
## Poly.SP.Chinese.per                      0.37
## Poly.SP.HDB.per_se                       0.82
## Poly.SP.Elderly_se                       0.88
## Poly.SP.Chinese.per_se                   1.00
```



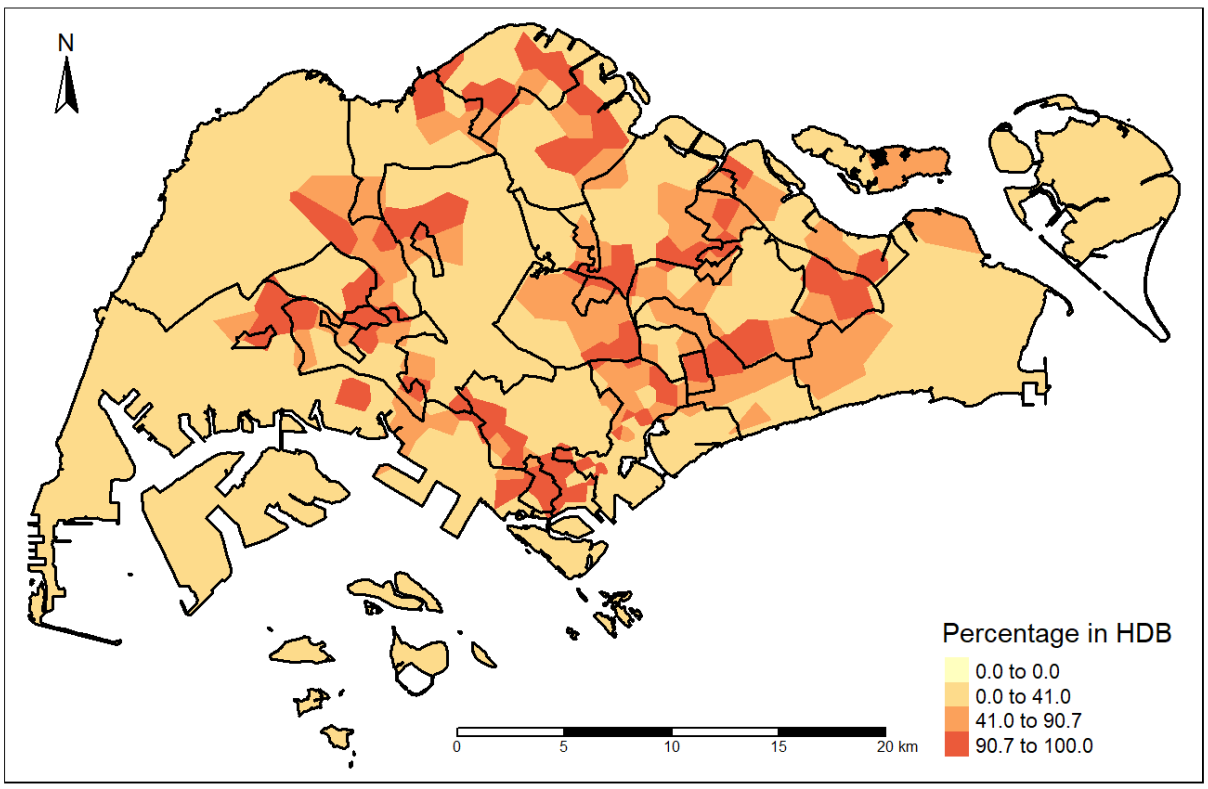

```
#separate regression for each
formulaHDB <- Poly.SP$Margins ~ Poly.SP$HDB.per
HDBbandwidth <- gwr.sel(formulaHDB, data = Poly.SP, adapt = T)## Adaptive q: 0.381966 CV
score: 184180.7
## Adaptive q: 0.618034 CV score: 187539.8
## Adaptive q: 0.236068 CV score: 178560.2
## Adaptive q: 0.145898 CV score: 172123
## Adaptive q: 0.09016994 CV score: 164387
## Adaptive q: 0.05572809 CV score: 155195.4
## Adaptive q: 0.03444185 CV score: 145415.6
## Adaptive q: 0.02128624 CV score: 135895.6
## Adaptive q: 0.01315562 CV score: 131288.5
## Adaptive q: 0.008130619 CV score: 132698.6
## Adaptive q: 0.01282176 CV score: 131211.1
## Adaptive q: 0.01192821 CV score: 131408.4
## Adaptive q: 0.01267454 CV score: 131215.5
## Adaptive q: 0.01277526 CV score: 131203.7
## Adaptive q: 0.01273457 CV score: 131208.1
## Adaptive q: 0.01277526 CV score: 131203.7gwr.HDB = gwr(Poly.SP$Margins ~
Poly.SP$HDB.per, data = Poly.SP, adapt=HDBbandwidth, hatmatrix=TRUE, se.fit=TRUE)
gwr.HDB## Call:
## gwr(formula = Poly.SP$Margins ~ Poly.SP$HDB.per, data = Poly.SP,
##     adapt = HDBbandwidth, hatmatrix = TRUE, se.fit = TRUE)
```

```
## Kernel function: gwr.Gauss
## Adaptive quantile: 0.01277526 (about 7 of 626 data points)
## Summary of GWR coefficient estimates at data points:
##                        Min.     1st Qu.      Median     3rd Qu.        Max.
## X.Intercept.    -16.8779654  19.6382504  27.3930054  29.7649265  41.3517644
## Poly.SP.HDB.per  -0.1523023  -0.0336478  -0.0017871   0.0279638   0.1630624
##                  Global
## X.Intercept.    23.3675
## Poly.SP.HDB.per -0.0042
## Number of data points: 626
## Effective number of parameters (residual: 2traceS - traceS'S): 95.05229
## Effective degrees of freedom (residual: 2traceS - traceS'S): 530.9477
## Sigma (residual: 2traceS - traceS'S): 13.95541
## Effective number of parameters (model: traceS): 67.27162
## Effective degrees of freedom (model: traceS): 558.7284
## Sigma (model: traceS): 13.60404
## Sigma (ML): 12.85231
## AICc (GWR p. 61, eq 2.33; p. 96, eq. 4.21): 5127.055
## AIC (GWR p. 96, eq. 4.22): 5040.794
## Residual sum of squares: 103403.9
## Quasi-global R2: 0.4537408resultsHDB <-as.data.frame(gwr.HDB$SDF)
gwr.HDBmap <- Area
gwr.HDBmap <- cbind(gwr.HDBmap, as.matrix(resultsHDB))
qtm(gwr.HDBmap, fill = "localR2")
```



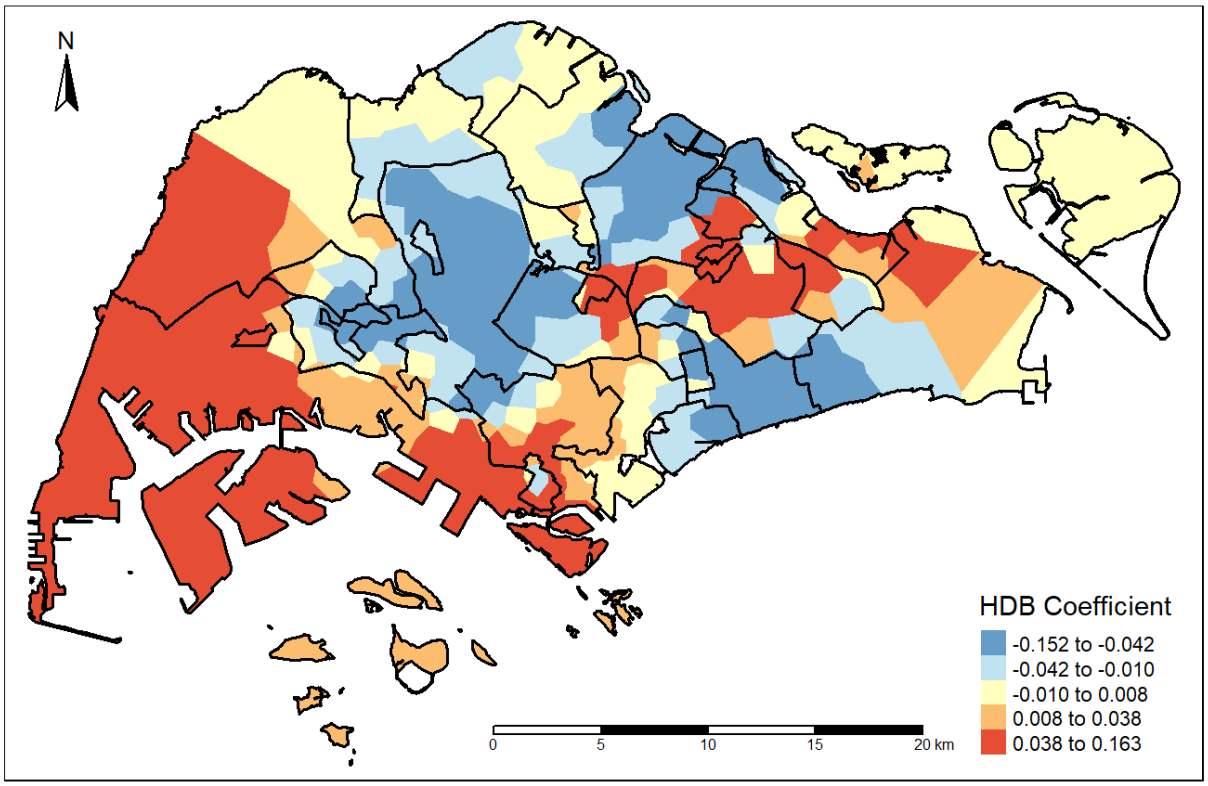

```
forHDB <- Poly.SP$Margins ~ Poly.SP$HDB.per
GWRbandwidthHDB <- gwr.sel(forHDB, data = Poly.SP, adapt = T)## Adaptive q: 0.381966 CV
score: 184180.7
## Adaptive q: 0.618034 CV score: 187539.8
## Adaptive q: 0.236068 CV score: 178560.2
## Adaptive q: 0.145898 CV score: 172123
## Adaptive q: 0.09016994 CV score: 164387
## Adaptive q: 0.05572809 CV score: 155195.4
## Adaptive q: 0.03444185 CV score: 145415.6
## Adaptive q: 0.02128624 CV score: 135895.6
## Adaptive q: 0.01315562 CV score: 131288.5
## Adaptive q: 0.008130619 CV score: 132698.6
## Adaptive q: 0.01282176 CV score: 131211.1
## Adaptive q: 0.01192821 CV score: 131408.4
## Adaptive q: 0.01267454 CV score: 131215.5
## Adaptive q: 0.01277526 CV score: 131203.7
## Adaptive q: 0.01273457 CV score: 131208.1
## Adaptive q: 0.01277526 CV score: 131203.7mapHDB <- tm_shape(gwr.HDBmap) +
  tm_fill("HDB.per", n = 4, style = "quantile",
    midpoint = 0,
    title = "Percentage in HDB",
    palette = "-RdYlBu",
    border.col = "grey", border.alpha = 0.2,
```

```
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  ) +
  tm_shape(Electoral) +
  tm_borders(col = "black", lwd = 1.5)
mapHDB
```

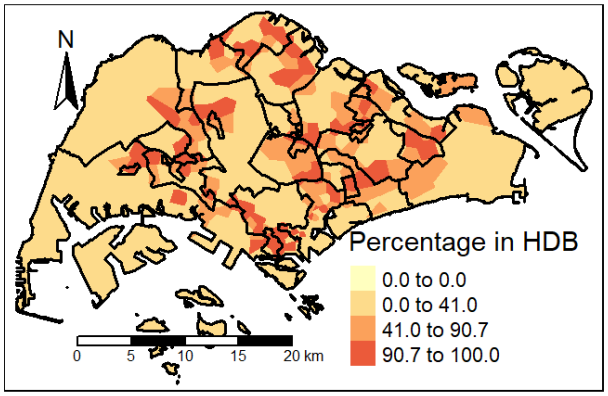


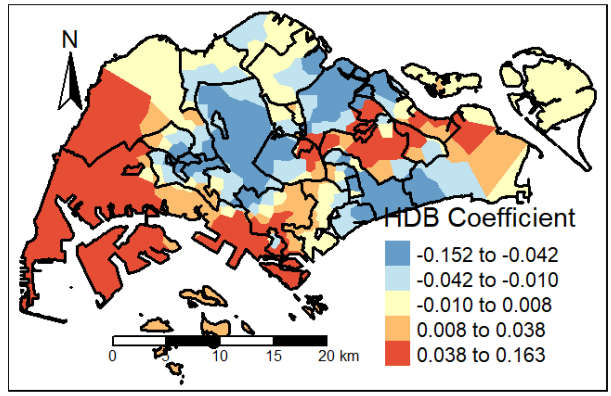

```
mapHDB.coeff <- tm_shape(gwr.HDBmap) +
  tm_fill("Poly.SP.HDB.per", n = 5, style = "quantile",
          midpoint = 0,
          title = "HDB Coefficient",
          palette = "-RdYlBu",
          border.col = "grey", border.alpha = 0.2,
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
```

```
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  ) +
  tm_shape(Electoral) +
  tm_borders(col = "black", lwd = 1.5)
mapHDB.coeff
```



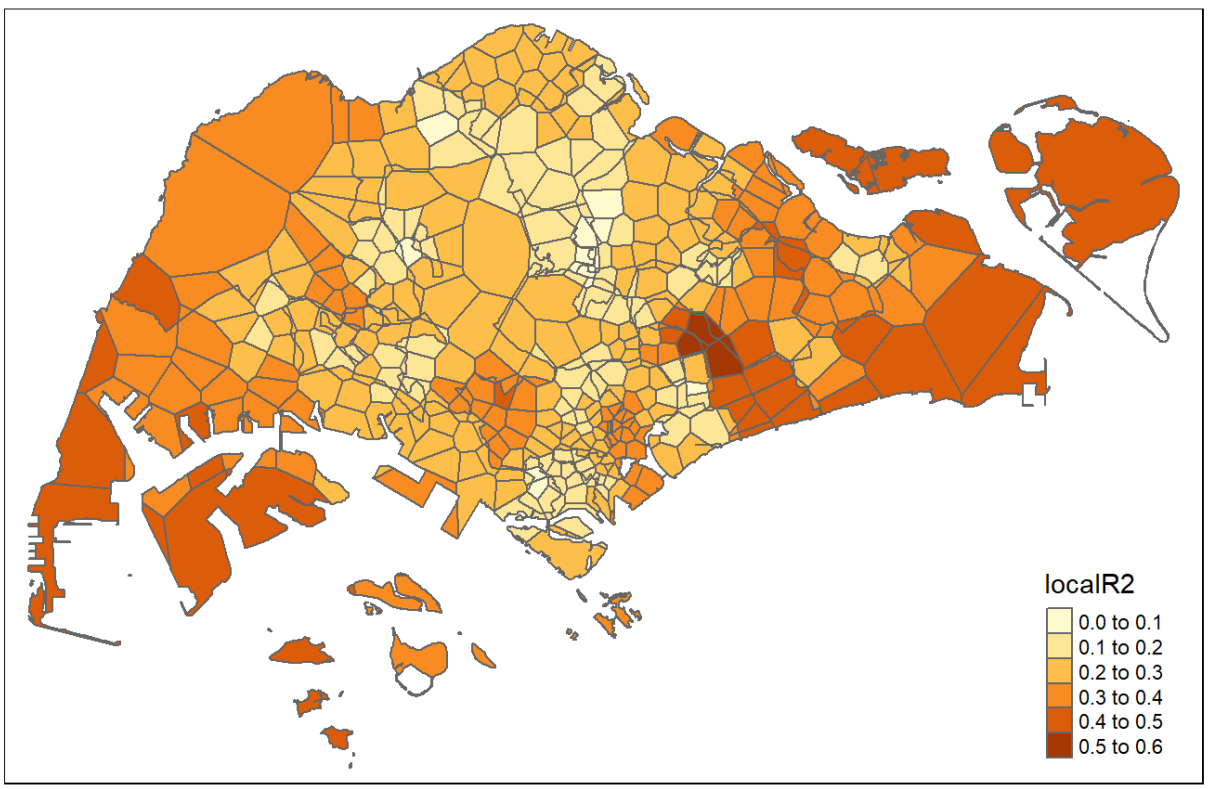

```
grid.newpage()

pushViewport(viewport(layout = grid.layout(1, 2)))

print(mapHDB, vp = viewport(layout.pos.col = 1, layout.pos.row = 1))
print(mapHDB.coeff, vp = viewport(layout.pos.col = 2, layout.pos.row = 1))
```



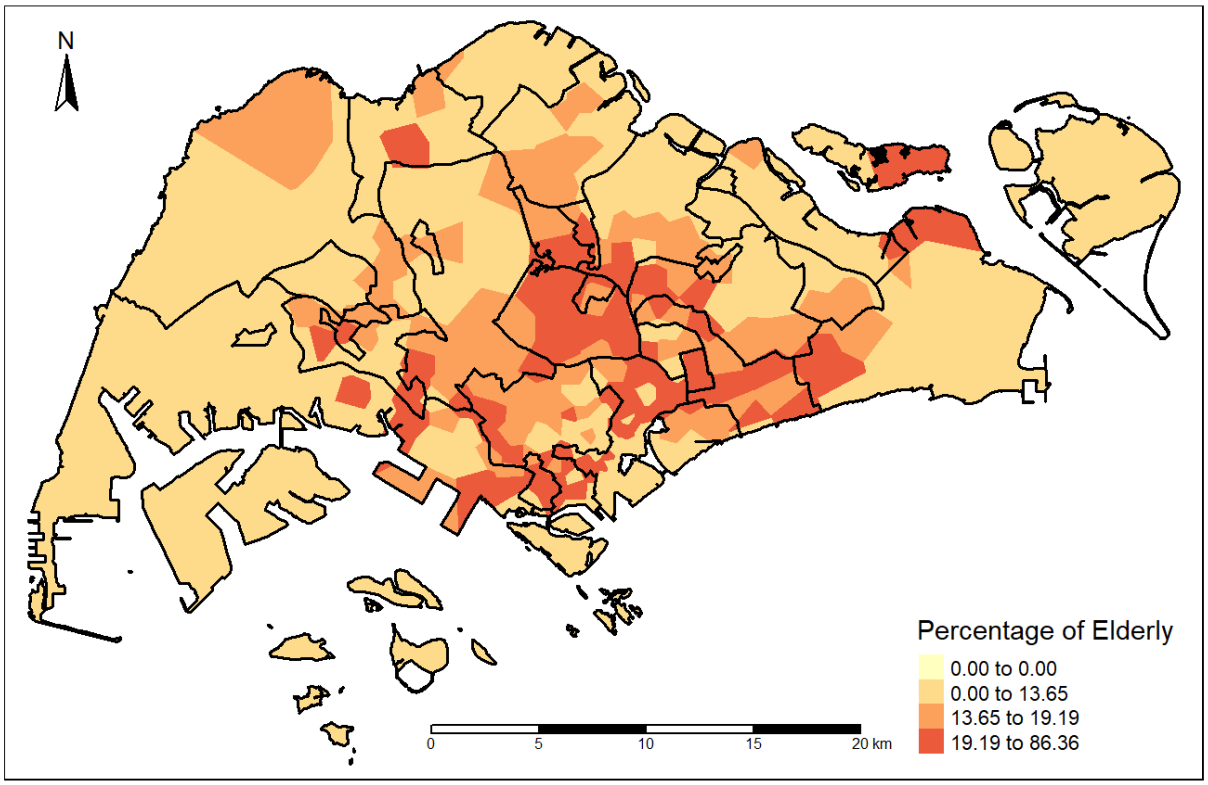



```
#elderly
formulaElderly <- Poly.SP$Margins ~ Poly.SP$Elderly
Elderlybandwidth <- gwr.sel(formulaElderly, data = Poly.SP, adapt = T)## Adaptive q:
0.381966 CV score: 182626.5
## Adaptive q: 0.618034 CV score: 186355
## Adaptive q: 0.236068 CV score: 176112.2
## Adaptive q: 0.145898 CV score: 169366.4
## Adaptive q: 0.09016994 CV score: 161952.3
## Adaptive q: 0.05572809 CV score: 153537.5
## Adaptive q: 0.03444185 CV score: 144334.5
## Adaptive q: 0.02128624 CV score: 135163.7
## Adaptive q: 0.01315562 CV score: 130357.2
## Adaptive q: 0.008130619 CV score: 131537.1
## Adaptive q: 0.01251314 CV score: 130301.3
## Adaptive q: 0.01224239 CV score: 130396.6
## Adaptive q: 0.01274396 CV score: 130240.2
## Adaptive q: 0.01278465 CV score: 130233.8
## Adaptive q: 0.01292635 CV score: 130276.9
## Adaptive q: 0.01282534 CV score: 130245.7
## Adaptive q: 0.01278465 CV score: 130233.8gwr.Elderly = gwr(Poly.SP$Margins ~
Poly.SP$Elderly, data = Poly.SP, adapt=Elderlybandwidth, hatmatrix=TRUE, se.fit=TRUE)
gwr.Elderly## Call:
## gwr(formula = Poly.SP$Margins ~ Poly.SP$Elderly, data = Poly.SP,
```

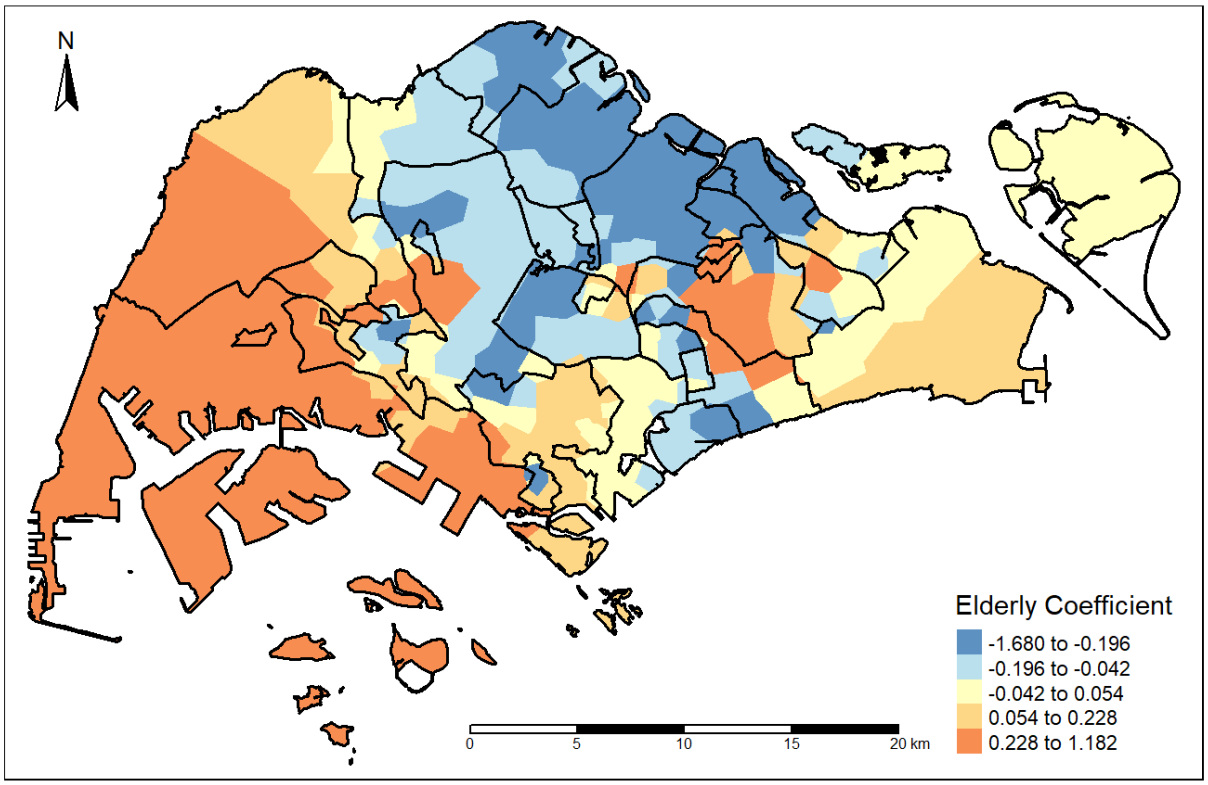
```
##     adapt = Elderlybandwidth, hatmatrix = TRUE, se.fit = TRUE)
## Kernel function: gwr.Gauss
## Adaptive quantile: 0.01278465 (about 8 of 626 data points)
## Summary of GWR coefficient estimates at data points:
##                       Min.    1st Qu.     Median    3rd Qu.       Max.  Global
## X.Intercept.    -14.294410  19.233339  26.856123  30.309978  40.757514 21.7330
## Poly.SP.Elderly  -1.680253  -0.147449   0.012322   0.173621   1.182108  0.1157
## Number of data points: 626
## Effective number of parameters (residual: 2traceS - traceS'S): 94.19296
## Effective degrees of freedom (residual: 2traceS - traceS'S): 531.807
## Sigma (residual: 2traceS - traceS'S): 13.95991
## Effective number of parameters (model: traceS): 66.29326
## Effective degrees of freedom (model: traceS): 559.7067
## Sigma (model: traceS): 13.60753
## Sigma (ML): 12.86686
## AICc (GWR p. 61, eq 2.33; p. 96, eq. 4.21): 5126.006
## AIC (GWR p. 96, eq. 4.22): 5041.232
## Residual sum of squares: 103638.1
## Quasi-global R2: 0.4525034resultsElderly <-as.data.frame(gwr.Elderly$SDF)
gwr.Elderlymap <- Area
gwr.Elderlymap <- cbind(gwr.Elderlymap, as.matrix(resultsElderly))
qtm(gwr.Elderlymap, fill = "localR2")
```



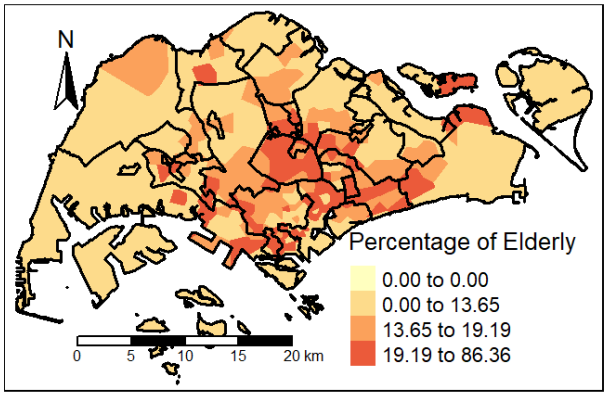
```
forElderly <- Poly.SP$Margins ~ Poly.SP$Elderly
GWRbandwidthElderly <- gwr.sel(forElderly, data = Poly.SP, adapt = T)## Adaptive q:
0.381966 CV score: 182626.5
## Adaptive q: 0.618034 CV score: 186355
## Adaptive q: 0.236068 CV score: 176112.2
## Adaptive q: 0.145898 CV score: 169366.4
## Adaptive q: 0.09016994 CV score: 161952.3
## Adaptive q: 0.05572809 CV score: 153537.5
## Adaptive q: 0.03444185 CV score: 144334.5
## Adaptive q: 0.02128624 CV score: 135163.7
## Adaptive q: 0.01315562 CV score: 130357.2
## Adaptive q: 0.008130619 CV score: 131537.1
## Adaptive q: 0.01251314 CV score: 130301.3
## Adaptive q: 0.01224239 CV score: 130396.6
## Adaptive q: 0.01274396 CV score: 130240.2
## Adaptive q: 0.01278465 CV score: 130233.8
## Adaptive q: 0.01292635 CV score: 130276.9
## Adaptive q: 0.01282534 CV score: 130245.7
## Adaptive q: 0.01278465 CV score: 130233.8mapElderly <- tm_shape(gwr.Elderlymap) +
  tm_fill("Elderly", n = 4, style = "quantile",
          midpoint = 0,
          title = "Percentage of Elderly",
          palette = "-RdYlBu",
          border.col = "grey", border.alpha = 0.2,
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  ) +
  tm_shape(Electoral) +
  tm_borders(col = "black", lwd = 1.5)
mapElderly
```

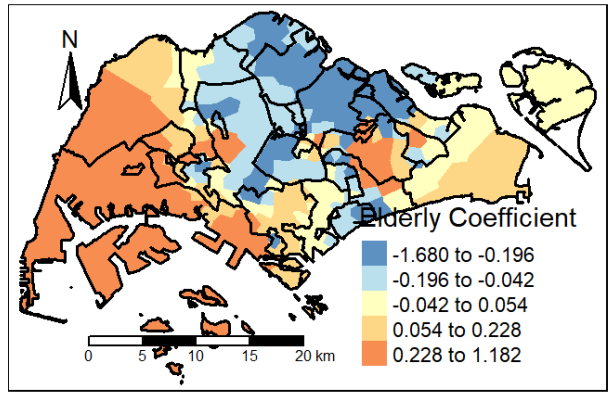



```
mapElderly.coeff <- tm_shape(gwr.Elderlymap) +
  tm_fill("Poly.SP.Elderly", n = 5, style = "quantile",
          midpoint = 0,
          title = "Elderly Coefficient",
          palette = "-RdYlBu",
          border.col = "grey", border.alpha = 0.2,
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  ) +
  tm_shape(Electoral) +
  tm_borders(col = "black", lwd = 1.5)
mapElderly.coeff
```



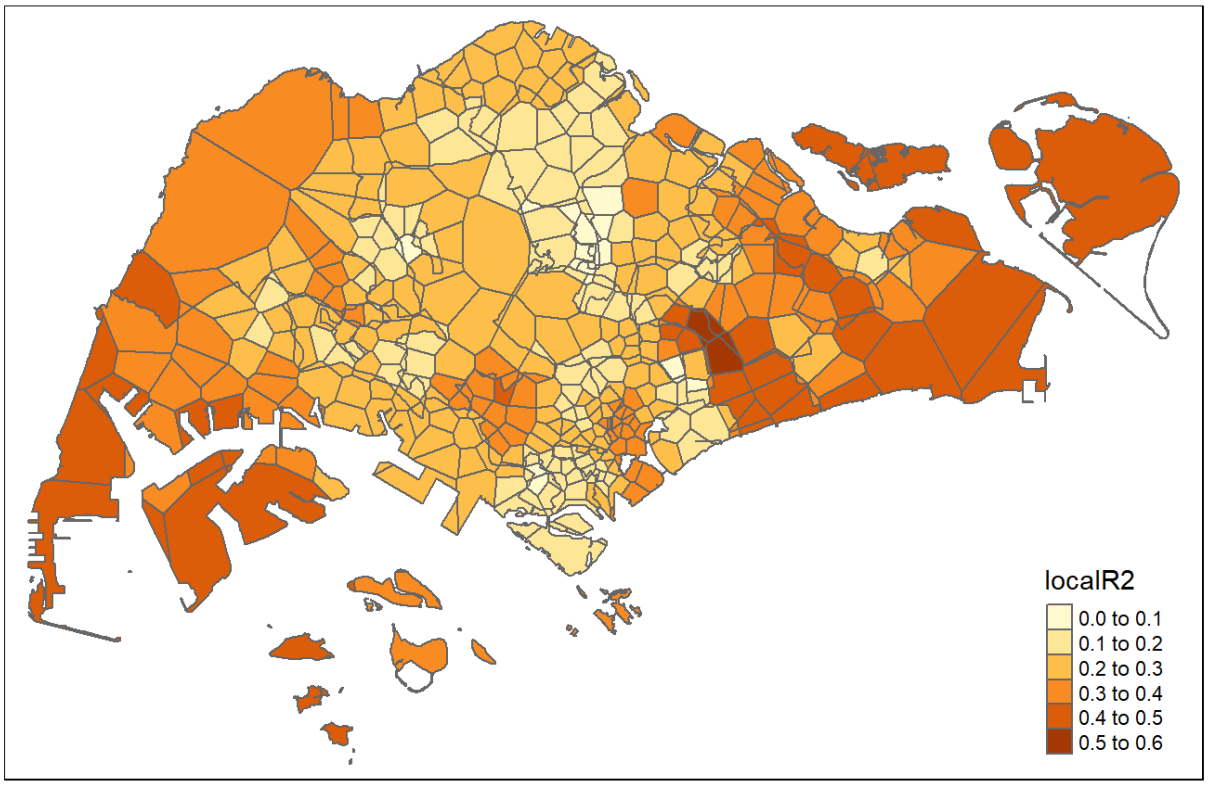

```
grid.newpage()

pushViewport(viewport(layout = grid.layout(1, 2)))

print(mapElderly, vp = viewport(layout.pos.col = 1, layout.pos.row = 1))
print(mapElderly.coeff, vp = viewport(layout.pos.col = 2, layout.pos.row = 1))
```





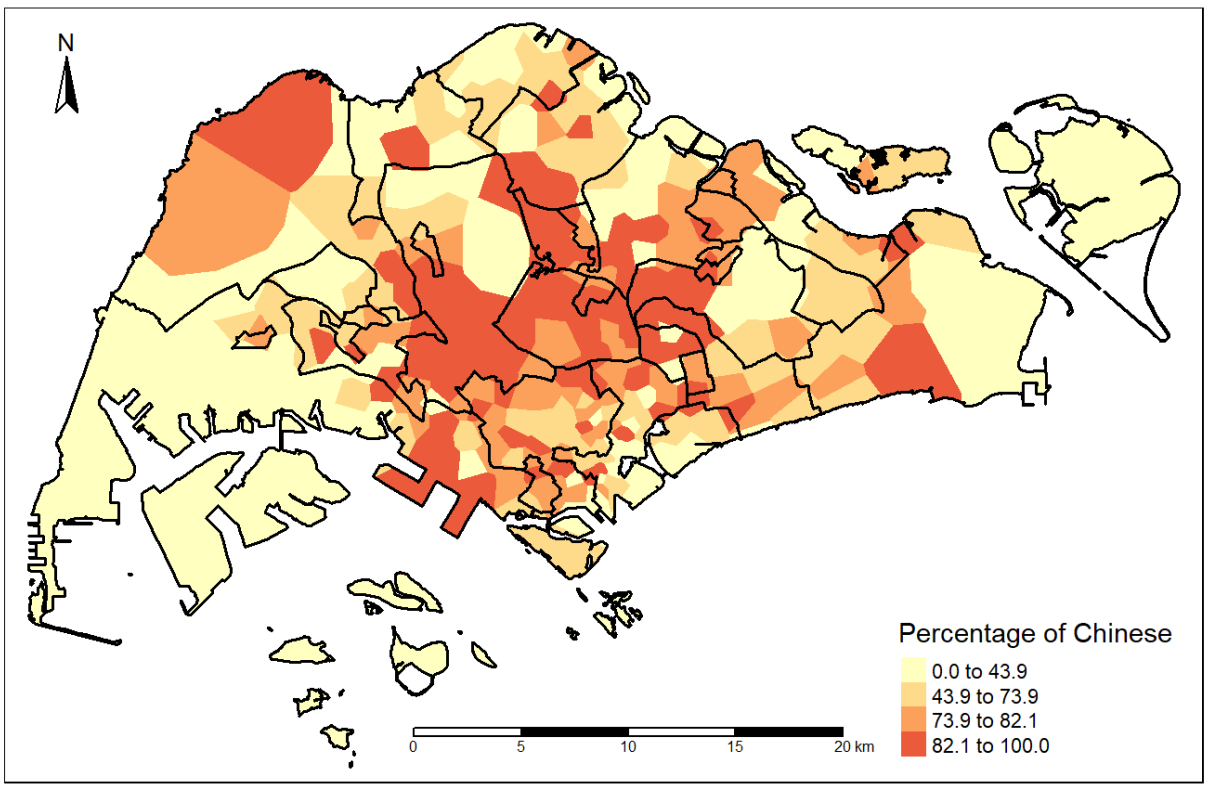

```
# Chinese
formulaChinese <- Poly.SP$Margins ~ Poly.SP$Chinese.per
Chinesebandwidth <- gwr.sel(formulaChinese, data = Poly.SP, adapt = T)## Adaptive q:
0.381966 CV score: 182725
## Adaptive q: 0.618034 CV score: 186428.6
## Adaptive q: 0.236068 CV score: 176380.2
## Adaptive q: 0.145898 CV score: 169779.1
## Adaptive q: 0.09016994 CV score: 162270.2
## Adaptive q: 0.05572809 CV score: 153640.1
## Adaptive q: 0.03444185 CV score: 143998.6
## Adaptive q: 0.02128624 CV score: 134157.5
## Adaptive q: 0.01315562 CV score: 128918.6
## Adaptive q: 0.008130619 CV score: 130521.8
## Adaptive q: 0.01282149 CV score: 128787.7
## Adaptive q: 0.0116958 CV score: 129338.5
## Adaptive q: 0.01266398 CV score: 128813.2
## Adaptive q: 0.0127808 CV score: 128773.6
## Adaptive q: 0.01274011 CV score: 128786.3
## Adaptive q: 0.0127808 CV score: 128773.6gwr.Chinese = gwr(Poly.SP$Margins ~
Poly.SP$Chinese.per, data = Poly.SP, adapt=Chinesebandwidth, hatmatrix=TRUE, se.fit=TRUE)
gwr.Chinese## Call:
## gwr(formula = Poly.SP$Margins ~ Poly.SP$Chinese.per, data = Poly.SP,
##     adapt = Chinesebandwidth, hatmatrix = TRUE, se.fit = TRUE)
```

```
## Kernel function: gwr.Gauss
## Adaptive quantile: 0.0127808 (about 8 of 626 data points)
## Summary of GWR coefficient estimates at data points:
##                            Min.     1st Qu.      Median     3rd Qu.        Max.
## X.Intercept.        -1.9834e+01  1.7548e+01  2.6869e+01  3.0121e+01  4.4393e+01
## Poly.SP.Chinese.per -2.5462e-01 -2.2973e-02  5.7919e-04  5.3316e-02  2.4305e-01
##                      Global
## X.Intercept.        21.1788
## Poly.SP.Chinese.per  0.0346
## Number of data points: 626
## Effective number of parameters (residual: 2traceS - traceS'S): 91.77926
## Effective degrees of freedom (residual: 2traceS - traceS'S): 534.2207
## Sigma (residual: 2traceS - traceS'S): 13.90748
## Effective number of parameters (model: traceS): 64.62613
## Effective degrees of freedom (model: traceS): 561.3739
## Sigma (model: traceS): 13.56696
## Sigma (ML): 12.84759
## AICc (GWR p. 61, eq 2.33; p. 96, eq. 4.21): 5119.948
## AIC (GWR p. 96, eq. 4.22): 5037.688
## Residual sum of squares: 103327.8
## Quasi-global R2: 0.4541424resultsChinese <-as.data.frame(gwr.Chinese$SDF)
gwr.Chinesemap <- Area
gwr.Chinesemap <- cbind(gwr.Chinesemap, as.matrix(resultsChinese))
qtm(gwr.Chinesemap, fill = "localR2")
```



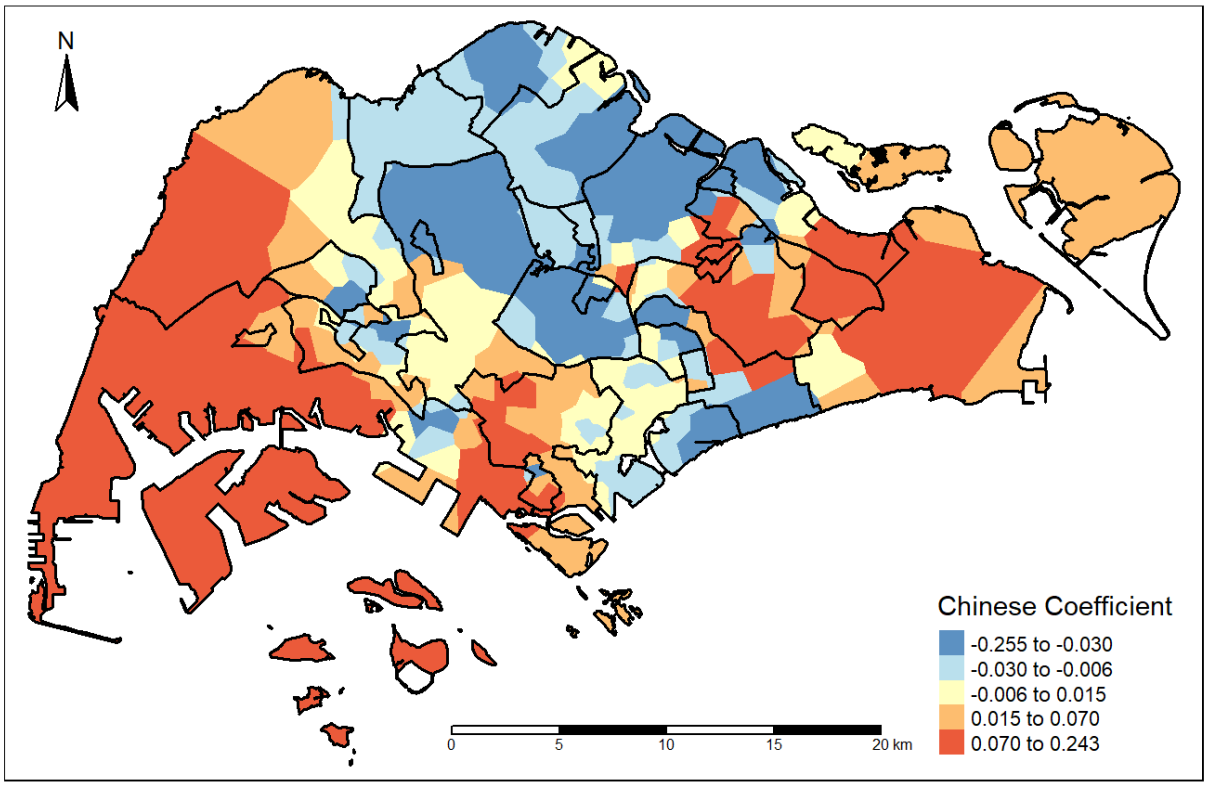

```
forChinese <- Poly.SP$Margins ~ Poly.SP$Chinese.per
GWRbandwidthChinese <- gwr.sel(forChinese, data = Poly.SP, adapt = T)## Adaptive q:
0.381966 CV score: 182725
## Adaptive q: 0.618034 CV score: 186428.6
## Adaptive q: 0.236068 CV score: 176380.2
## Adaptive q: 0.145898 CV score: 169779.1
## Adaptive q: 0.09016994 CV score: 162270.2
## Adaptive q: 0.05572809 CV score: 153640.1
## Adaptive q: 0.03444185 CV score: 143998.6
## Adaptive q: 0.02128624 CV score: 134157.5
## Adaptive q: 0.01315562 CV score: 128918.6
## Adaptive q: 0.008130619 CV score: 130521.8
## Adaptive q: 0.01282149 CV score: 128787.7
## Adaptive q: 0.0116958 CV score: 129338.5
## Adaptive q: 0.01266398 CV score: 128813.2
## Adaptive q: 0.0127808 CV score: 128773.6
## Adaptive q: 0.01274011 CV score: 128786.3
## Adaptive q: 0.0127808 CV score: 128773.6mapChinese <- tm_shape(gwr.Chinesemap) +
  tm_fill("Chinese.per", n = 4, style = "quantile",
          midpoint = 0,
          title = "Percentage of Chinese",
          palette = "-RdYlBu",
          border.col = "grey", border.alpha = 0.2,
```

```
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  ) +
  tm_shape(Electoral) +
  tm_borders(col = "black", lwd = 1.5)
mapChinese
```

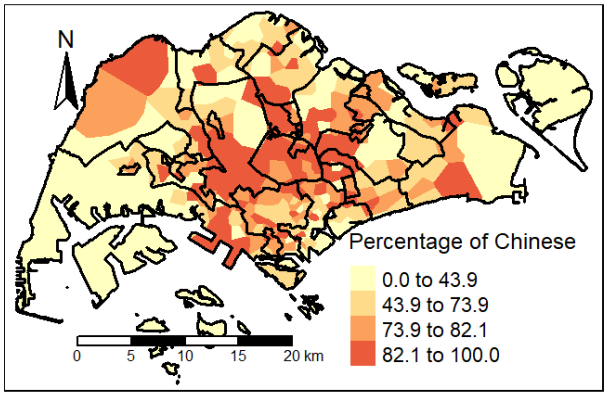


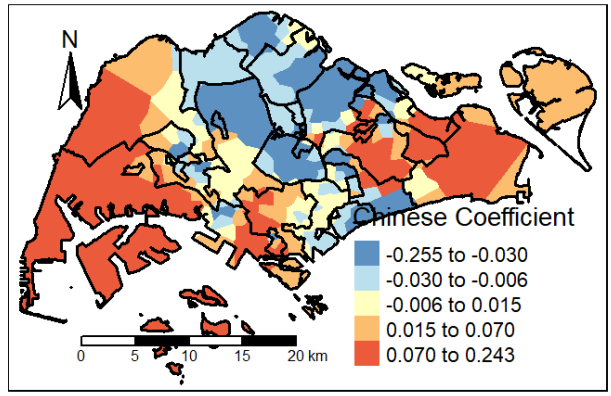

```
mapChinese.coeff <- tm_shape(gwr.Chinesemap) +
  tm_fill("Poly.SP.Chinese.per", n = 5, style = "quantile",
          midpoint = 0,
          title = "Chinese Coefficient",
          palette = "-RdYlBu",
          border.col = "grey", border.alpha = 0.2,
  ) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
```

```
  tm_scale_bar(
    breaks = c(0, 5, 10, 15, 20),
    position = c("right", "bottom")
  ) +
  tm_shape(Electoral) +
  tm_borders(col = "black", lwd = 1.5)
mapChinese.coeff
```



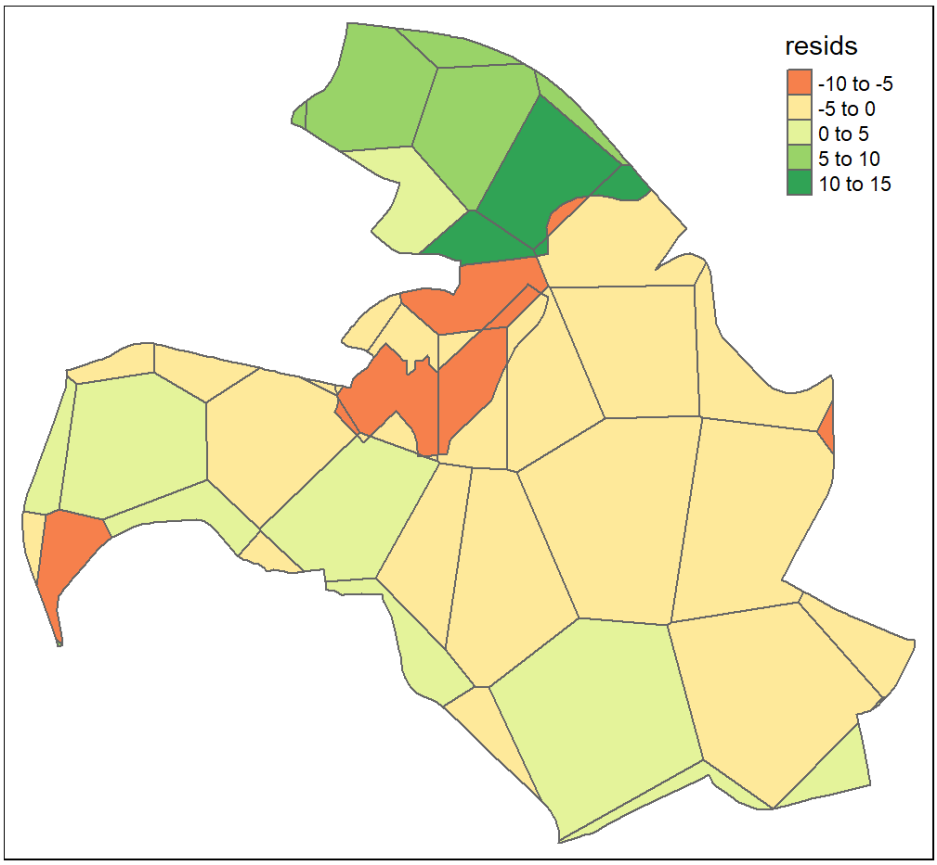

```
grid.newpage()

pushViewport(viewport(layout = grid.layout(1, 2)))

print(mapChinese, vp = viewport(layout.pos.col = 1, layout.pos.row = 1))
print(mapChinese.coeff, vp = viewport(layout.pos.col = 2, layout.pos.row = 1))
```






```
#WP analysis
WP <- Area|>
  filter(Constituency == "ALJUNIED") %>%
  bind_rows(Area %>%
              filter(Constituency == "SENGKANG")) %>%
  bind_rows(Area %>%
              filter(Constituency == "HOUGANG"))## New names:
## New names:
## New names:
## • `X0...4` -> `X0`
## • `X15...19` -> `X15`
## • `X20...24` -> `X20`
## • `X25...29` -> `X25`
## • `X30...34` -> `X30`
## • `X35...39` -> `X35`
## • `X40...44` -> `X40`
## • `X45...49` -> `X45`
## • `X50...54` -> `X50`
## • `X55...59` -> `X55`
## • `X60...64` -> `X60`
## • `X65...69` -> `X65`
## • `X70...74` -> `X70`
## • `X75...79` -> `X75`
```

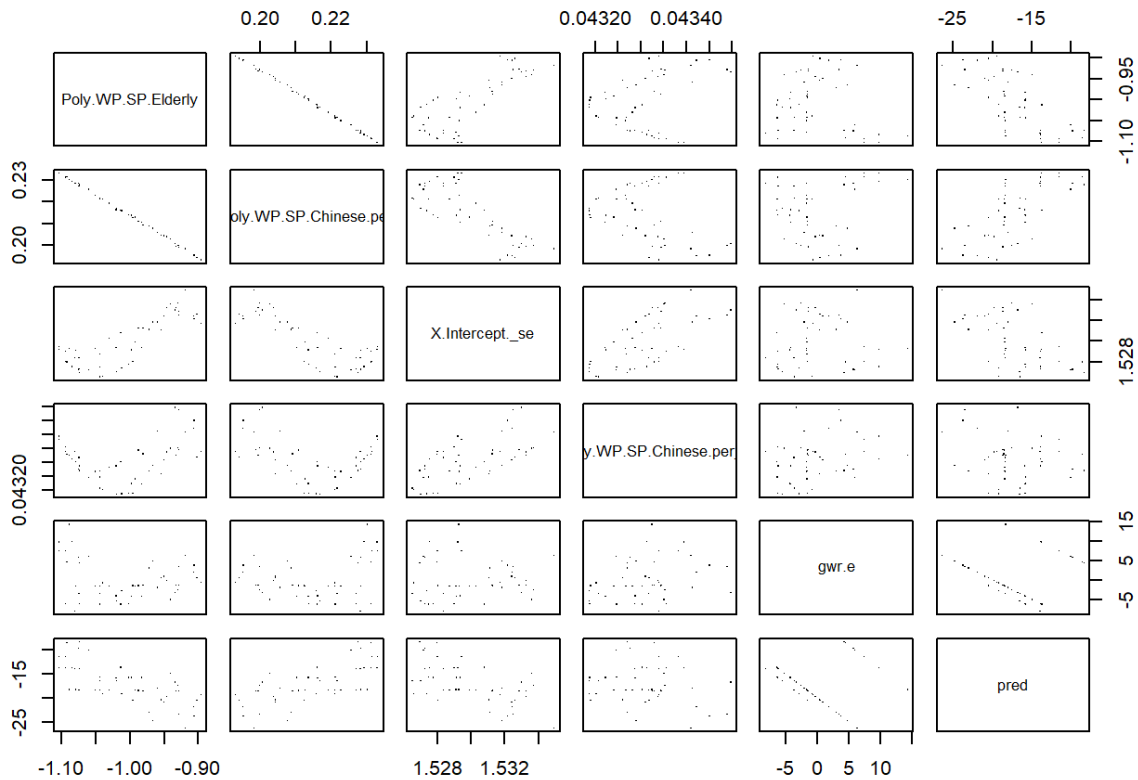
```
## • `X80...84` -> `X80`
## • `X85...89` -> `X85`WP.elec <- Electoral |>
  filter(Constituency == "ALJUNIED") %>%
  bind_rows(Electoral %>%
              filter(Constituency == "SENGKANG")) %>%
  bind_rows(Electoral %>%
              filter(Constituency == "HOUGANG"))

WP <- WP [, -c(7:47)]

#reformat
Poly.WP.SP <- as(WP, "Spatial")
# linear modelWP of margins elderly, chinese
modelWP <- lm(WP$Margins ~ WP$Elderly + WP$Chinese.per)
summary(modelWP)##
## Call:
## lm(formula = WP$Margins ~ WP$Elderly + WP$Chinese.per)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -7.975 -3.615 -1.605  3.734 14.395
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -18.39527    1.57222 -11.700 3.04e-15 ***
## WP$Elderly      -0.98549    0.19514  -5.050 7.78e-06 ***
## WP$Chinese.per   0.21184    0.04436   4.776 1.93e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.274 on 45 degrees of freedom
## Multiple R-squared:  0.3679, Adjusted R-squared:  0.3398
## F-statistic: 13.09 on 2 and 45 DF,  p-value: 3.298e-05#plot residuals
resids <- residuals(modelWP)
map.resids <- WP
map.resids <- cbind(WP, resids)
qtm(map.resids, fill = "resids")## Variable(s) "resids" contains positive and negative
values, so midpoint is set to 0. Set midpoint = NA to show the full spectrum of the color
palette.
```




```
# try to combine variables into one model
formulaWP <- Poly.WP.SP$Margins ~ Poly.WP.SP$Elderly + Poly.WP.SP$Chinese.per
WPGWRbandwidth <- gwr.sel(formulaWP, data = Poly.WP.SP, adapt = T)## Adaptive q: 0.381966
CV score: 1080.118
## Adaptive q: 0.618034 CV score: 1213.765
## Adaptive q: 0.236068 CV score: 955.8399
## Adaptive q: 0.145898 CV score: 845.3021
## Adaptive q: 0.09016994 CV score: 799.7045
## Adaptive q: 0.05572809 CV score: 817.8307
## Adaptive q: 0.09059688 CV score: 799.8217
## Adaptive q: 0.08440405 CV score: 798.2209
## Adaptive q: 0.07345081 CV score: 806.5324
## Adaptive q: 0.08022029 CV score: 799.6032
## Adaptive q: 0.08510893 CV score: 798.4114
## Adaptive q: 0.08365678 CV score: 798.0075
## Adaptive q: 0.08234416 CV score: 798.3572
## Adaptive q: 0.08349762 CV score: 797.9602
## Adaptive q: 0.08305704 CV score: 798.0272
## Adaptive q: 0.08337881 CV score: 797.9245
## Adaptive q: 0.0832559 CV score: 797.9427
## Adaptive q: 0.08333812 CV score: 797.9121
## Adaptive q: 0.08329742 CV score: 797.9255
## Adaptive q: 0.08333812 CV score: 797.9121for (i in c(1, WPGWRbandwidth)) {
```

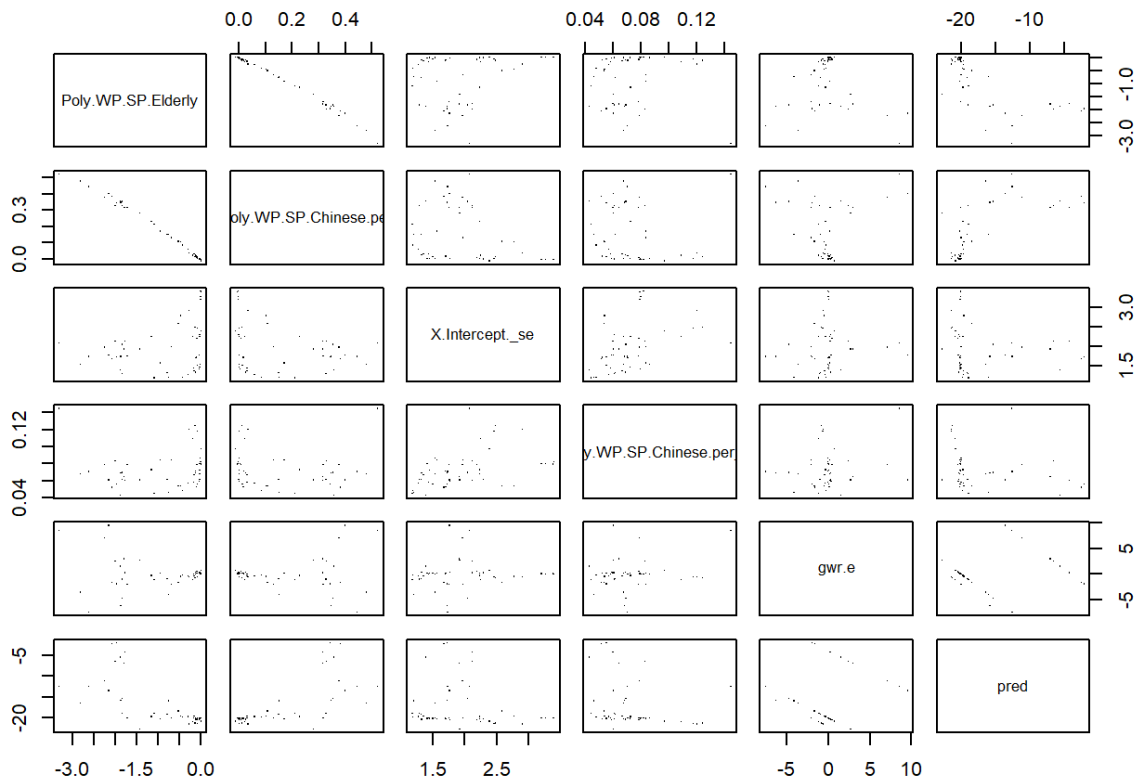
```r
  WPGWR.modelWP = gwr(formulaWP, data = Poly.WP.SP, adapt = i, hatmatrix = TRUE, se.fit =
TRUE)

  # print the results of the modelWP
  print(WPGWR.modelWP)

  results.WPGWR <- as.data.frame(WPGWR.modelWP$SDF)
  names(results.WPGWR)

  print(round(cor(results.WPGWR[, c(3, 4, 5, 7, 8, 9)], use = "complete.obs"), 2))
  pairs(results.WPGWR[, c(3, 4, 5, 7, 8, 9)], pch = ".")
}## Call:
## gwr(formula = formulaWP, data = Poly.WP.SP, adapt = i, hatmatrix = TRUE,
##     se.fit = TRUE)
## Kernel function: gwr.Gauss
## Adaptive quantile: 1 (about 48 of 48 data points)
## Summary of GWR coefficient estimates at data points:
##                              Min.   1st Qu.    Median   3rd Qu.      Max.
## X.Intercept.           -18.64856 -18.52754 -18.45352 -18.36812 -18.26339
## Poly.WP.SP.Elderly      -1.10332  -1.07346  -1.00675  -0.94921  -0.89607
## Poly.WP.SP.Chinese.per   0.19317   0.20414   0.21479   0.22705   0.23311
##                          Global
## X.Intercept.           -18.3953
## Poly.WP.SP.Elderly      -0.9855
## Poly.WP.SP.Chinese.per   0.2118
## Number of data points: 48
## Effective number of parameters (residual: 2traceS - traceS'S): 3.720298
## Effective degrees of freedom (residual: 2traceS - traceS'S): 44.2797
## Sigma (residual: 2traceS - traceS'S): 5.132714
## Effective number of parameters (model: traceS): 3.379027
## Effective degrees of freedom (model: traceS): 44.62097
## Sigma (model: traceS): 5.113049
## Sigma (ML): 4.929795
## AICc (GWR p. 61, eq 2.33; p. 96, eq. 4.21): 299.23
## AIC (GWR p. 96, eq. 4.22): 292.7457
## Residual sum of squares: 1166.538
## Quasi-global R2: 0.4108146
##                           Poly.WP.SP.Elderly Poly.WP.SP.Chinese.per
## Poly.WP.SP.Elderly                      1.00                  -1.00
## Poly.WP.SP.Chinese.per                 -1.00                   1.00
## X.Intercept._se                         0.81                  -0.81
## Poly.WP.SP.Chinese.per_se               0.32                  -0.32
## gwr.e                                  -0.17                   0.18
## pred                                   -0.69                   0.70
##                           X.Intercept._se Poly.WP.SP.Chinese.per_se gwr.e  pred
## Poly.WP.SP.Elderly                   0.81                      0.32 -0.17 -0.69
## Poly.WP.SP.Chinese.per              -0.81                     -0.32  0.18  0.70
## X.Intercept._se                      1.00                      0.64  0.04 -0.60
## Poly.WP.SP.Chinese.per_se            0.64                      1.00  0.29 -0.19
```

```
## gwr.e                                0.04                      0.29  1.00  0.02
## pred                                -0.60                     -0.19  0.02  1.00
```




```
## Call:
## gwr(formula = formulaWP, data = Poly.WP.SP, adapt = i, hatmatrix = TRUE,
##     se.fit = TRUE)
## Kernel function: gwr.Gauss
## Adaptive quantile: 0.08333812 (about 4 of 48 data points)
## Summary of GWR coefficient estimates at data points:
##                                Min.     1st Qu.      Median     3rd Qu.
## X.Intercept.           -20.6640308 -19.9992972 -19.9282223 -18.9563204
## Poly.WP.SP.Elderly      -3.3109606  -1.8040919  -0.4467807  -0.0748656
## Poly.WP.SP.Chinese.per  -0.0106895   0.0081314   0.0951985   0.3201074
##                                Max.   Global
## X.Intercept.           -11.6620729 -18.3953
## Poly.WP.SP.Elderly       0.0117505  -0.9855
## Poly.WP.SP.Chinese.per   0.5220024   0.2118
## Number of data points: 48
## Effective number of parameters (residual: 2traceS - traceS'S): 17.76953
## Effective degrees of freedom (residual: 2traceS - traceS'S): 30.23047
## Sigma (residual: 2traceS - traceS'S): 3.682204
## Effective number of parameters (model: traceS): 13.43158
## Effective degrees of freedom (model: traceS): 34.56842
## Sigma (model: traceS): 3.443424
```

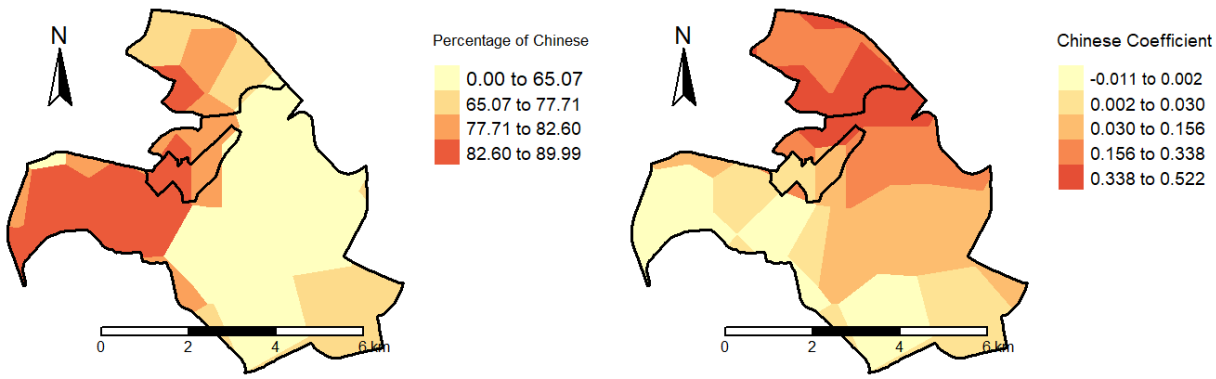
```
## Sigma (ML): 2.922198
## AICc (GWR p. 61, eq 2.33; p. 96, eq. 4.21): 281.7015
## AIC (GWR p. 96, eq. 4.22): 252.5939
## Residual sum of squares: 409.8835
## Quasi-global R2: 0.7929794
##                           Poly.WP.SP.Elderly Poly.WP.SP.Chinese.per
## Poly.WP.SP.Elderly                      1.00                  -1.00
## Poly.WP.SP.Chinese.per                 -1.00                   1.00
## X.Intercept._se                         0.25                  -0.28
## Poly.WP.SP.Chinese.per_se               0.04                  -0.11
## gwr.e                                  -0.12                   0.09
## pred                                   -0.70                   0.69
##                           X.Intercept._se Poly.WP.SP.Chinese.per_se gwr.e  pred
## Poly.WP.SP.Elderly                   0.25                      0.04 -0.12 -0.70
## Poly.WP.SP.Chinese.per              -0.28                     -0.11  0.09  0.69
## X.Intercept._se                      1.00                      0.44  0.09 -0.16
## Poly.WP.SP.Chinese.per_se            0.44                      1.00  0.22 -0.23
## gwr.e                                0.09                      0.22  1.00  0.12
## pred                                -0.16                     -0.23  0.12  1.00
```




```
#multicollinearity not present

WPGWR.map.function.WP <- WP
```

```r
# bind outputs to the WP polygon
WPGWR.map.function.WP <- cbind(WP, as.matrix(results.WPGWR))#mapping
WPChinese <- tm_shape(WPGWR.map.function.WP) +
  tm_fill("Chinese.per", n = 4, style = "quantile",
          midpoint = 0,
          title = "Percentage of Chinese",
          palette = "-RdYlBu",
          border.col = "grey", border.alpha = 0.2,
  ) +
  tm_layout(frame = FALSE, legend.outside = TRUE) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 2, 4, 6),
    position = c("right", "bottom")
  ) +
  tm_shape(WP.elec) +
  tm_borders(col = "black", lwd = 1.5)
tmap_options(check.and.fix = TRUE)

WPChineseCoeff <- tm_shape(WPGWR.map.function.WP) +
  tm_fill("Poly.WP.SP.Chinese.per", n = 5, style = "quantile",
          midpoint = 0,
          title = "Chinese Coefficient",
          palette = "-RdYlBu",
          border.col = "grey", border.alpha = 0.2,
  ) +
  tm_layout(frame = FALSE, legend.outside = TRUE) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 2, 4, 6),
    position = c("right", "bottom")
  ) +
  tm_shape(WP.elec) +
  tm_borders(col = "black", lwd = 1.5)

grid.newpage()
pushViewport(viewport(layout = grid.layout(1, 2)))

print(WPChinese, vp = viewport(layout.pos.col = 1, layout.pos.row = 1))## Legend labels
were too wide. The labels have been resized to 0.64, 0.59, 0.59, 0.59. Increase
legend.width (argument of tm_layout) to make the legend wider and therefore the labels
larger.print(WPChineseCoeff, vp = viewport(layout.pos.col = 2, layout.pos.row = 1))##
Legend labels were too wide. The labels have been resized to 0.56, 0.59, 0.59, 0.59,
```

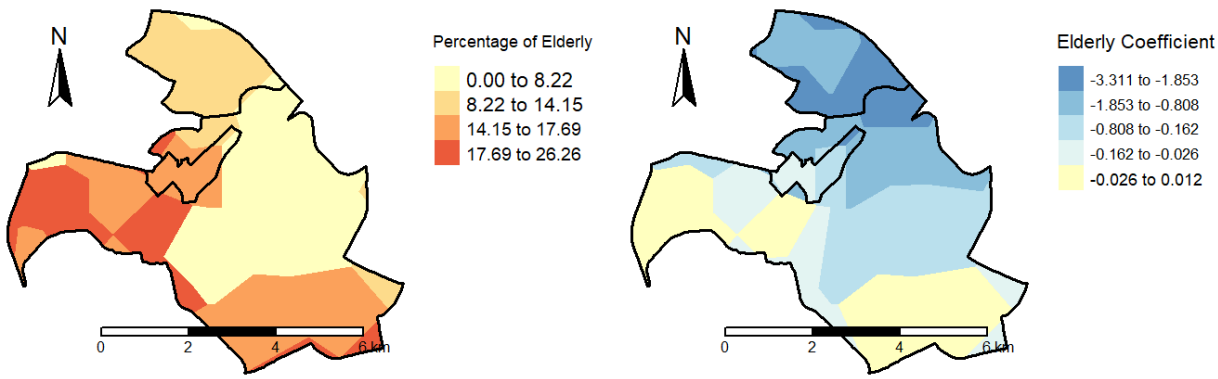
0.59. Increase legend.width (argument of tm_layout) to make the legend wider and therefore the labels larger.



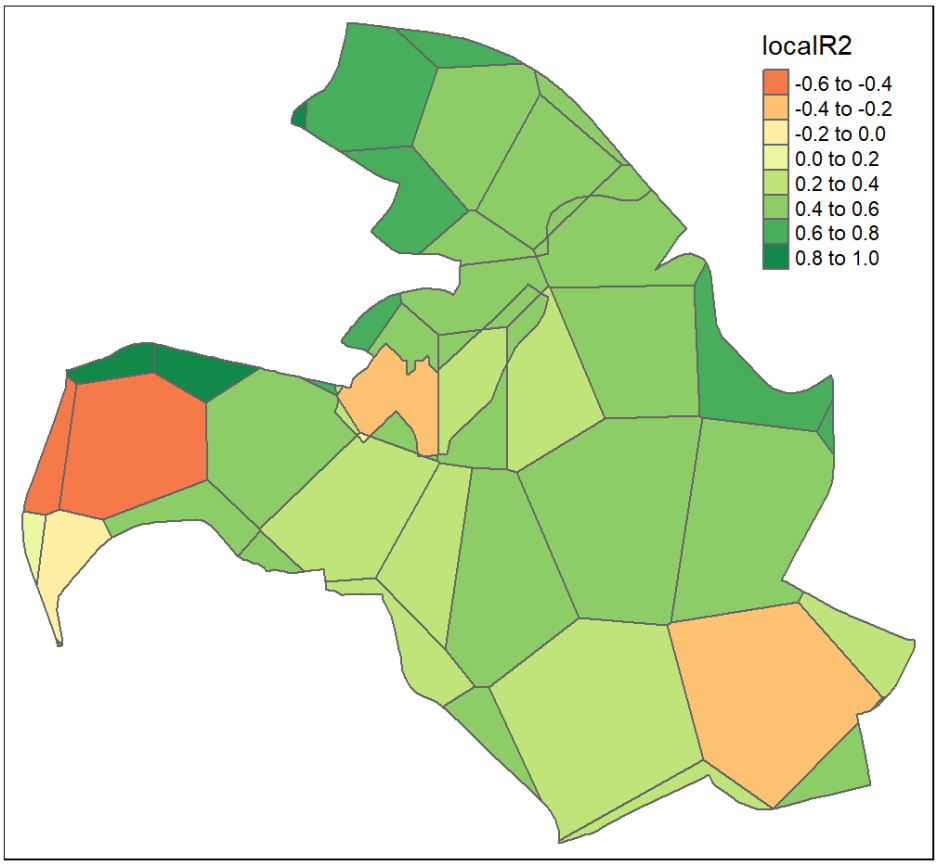

```
WPElderly <- tm_shape(WPGWR.map.function.WP) +
  tm_fill("Elderly", n = 4, style = "quantile",
          midpoint = 0,
          title = "Percentage of Elderly",
          palette = "-RdYlBu",
          border.col = "grey", border.alpha = 0.2,
  ) +
  tm_layout(frame = FALSE, legend.outside = TRUE) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 2, 4, 6),
    position = c("right", "bottom")
  ) +
  tm_shape(WP.elec) +
  tm_borders(col = "black", lwd = 1.5)

WPElderlyCoeff <- tm_shape(WPGWR.map.function.WP) +
```

```
  tm_fill("Poly.WP.SP.Elderly", n = 5, style = "quantile",
          midpoint = 0,
          title = "Elderly Coefficient",
          palette = "-RdYlBu",
          border.col = "grey", border.alpha = 0.2,
  ) +
  tm_layout(frame = FALSE, legend.outside = TRUE) +
  tm_compass(
    type = "arrow",
    position = c("left", "top")
  ) +
  tm_scale_bar(
    breaks = c(0, 2, 4, 6),
    position = c("right", "bottom")
  ) +
  tm_shape(WP.elec) +
  tm_borders(col = "black", lwd = 1.5)
```

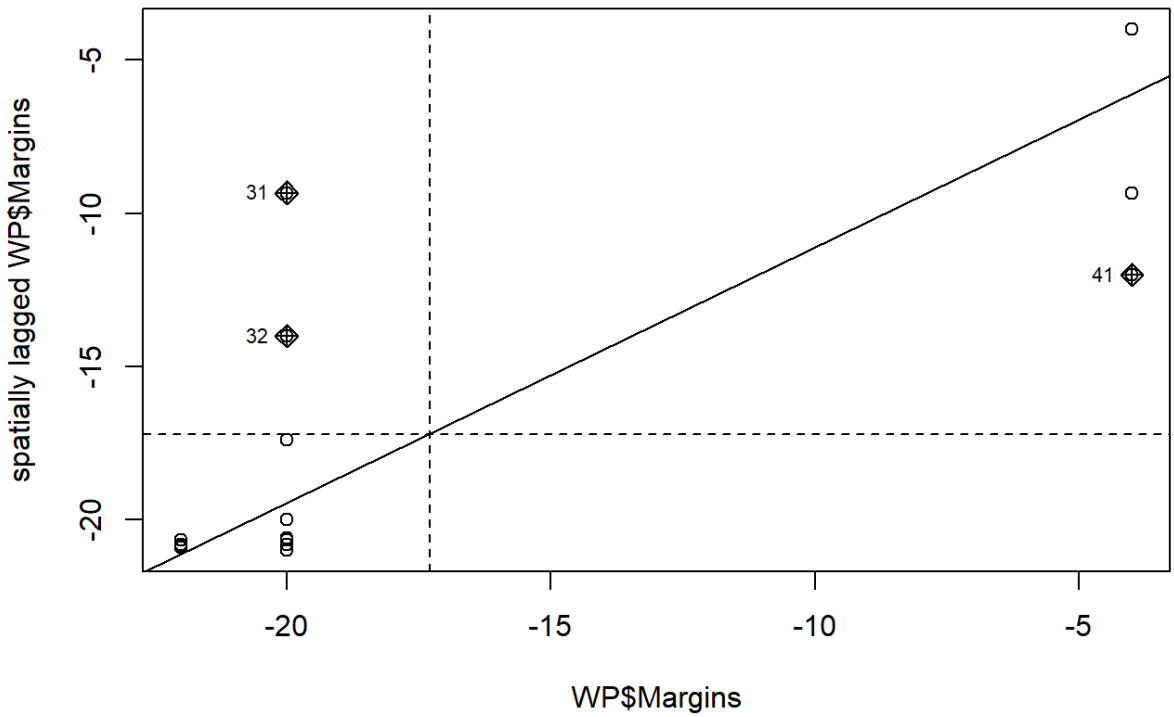
```
grid.newpage()
pushViewport(viewport(layout = grid.layout(1, 2)))

print(WPElderly, vp = viewport(layout.pos.col = 1, layout.pos.row = 1))## Some legend
labels were too wide. These labels have been resized to 0.64, 0.59, 0.59. Increase
legend.width (argument of tm_layout) to make the legend wider and therefore the labels
larger.print(WPElderlyCoeff, vp = viewport(layout.pos.col = 2, layout.pos.row = 1))##
Legend labels were too wide. The labels have been resized to 0.53, 0.53, 0.53, 0.53,
0.56. Increase legend.width (argument of tm_layout) to make the legend wider and
therefore the labels larger.
```




```
# Plot Local R2
qtm(WPGWR.map.function.WP, fill = "localR2")## Variable(s) "localR2" contains positive
and negative values, so midpoint is set to 0. Set midpoint = NA to show the full spectrum
of the color palette.
```




```
# moran's i
# global morans i
# identify neighbours
WP_queen <- WP |>
  poly2nb(queen = T)

WP_spatial_weight_queen <- WP_queen |>
  nb2listw(style = "W")

WP_centroid <- WP |>
  st_centroid()## Warning: st_centroid assumes attributes are constant over
geometries#global moran's i
moran.plot(WP$Margins, listw = WP_spatial_weight_queen)
```




```
moran_WP <- WP |>
  pull(Margins) |>
  as.vector() |>
  moran.test(WP_spatial_weight_queen)

# inspect result
moran_WP #.84 = v high (cluster)##
##  Moran I test under randomisation
##
## data:  as.vector(pull(WP, Margins))
## weights: WP_spatial_weight_queen
##
## Moran I statistic standard deviate = 9.2088, p-value < 2.2e-16
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic       Expectation          Variance
```